

A Metamodel for Ontologically Heterogeneous Social Dynamics

Part II: Modeling Six Categories of Social Facts

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Abstract

Social inquiry uses entities, relations, events, attributes, states, and processes to articulate different questions about a common situation. This paper develops their representations within the abstract action framework introduced in Part I, with ontological questions suspended. Each category receives a separate construction, derivation, critique, and philosophical interpretation. Indexed appearances support declared entity-reidentification criteria; incidence relations require invariance under participant identification; occurrence records preserve distinctions lost by idempotent effects; partial attributes require both value and definedness consistency; state descriptions depend on retained observation and memory; and process histories retain ordering, intermediate conditions, and segmentation beyond their composite effect. Established continuation semantics supplies a common state-consistency construction. Further conditional results concern the exact composition of collapse sets, coordination among category records, classification-mediated updates, changes of descriptive regime, finite probabilistic aggregation, and finite-horizon approximation error. A cooperative-agreement model is developed through its complete twelve-map transformation family. A graph model derives collective units and shows how a component description can support edge additions while failing under deletion. The philosophical analysis separates formation, description, and social uptake; predictive sufficiency and historical warrant; and behavioral equivalence and entity identity. A proposed case-study design states the evidence and revision conditions for each representation. The contribution is a coordinated architecture of explicit modeling commitments, with empirical adequacy and broader ontological interpretation remaining open.

Keywords: social facts; analytical categories; generative relational modeling; semigroup action; continuation equivalence

Research Paper Note

This paper continues the metamodel inquiry through a specified task: modeling six categories used in discussing social facts. The full series title retains the identity of Part I. The present argument concerns representational conditions, with ontological questions held open.

The term *social fact* identifies the subject matter of inquiry at a deliberately modest level. A description can concern an organization, a relation, an occurrence, a property, a situation, or a course of activity. The paper asks what information each description needs and how its representation interacts with the others. Establishing that a modeled description is warranted requires evidence appropriate to the inquiry.

The six categories constitute a selected analytical repertoire. Their uses can overlap within one investigation. A theory of their exhaustiveness, exclusivity, or metaphysical priority remains a separate undertaking.

The mathematical argument distinguishes conditional results from proposed interpretations. The finite example is constructed to expose assumptions and counterexamples. Its social labels are stipulated for illustration. A subsequent application would have to establish the relevant operations, observations, histories, and identity criteria from its own material.

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DRAFT

Notices

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Status. This expanded working draft is being developed for author review. Its formal propositions are conditional on the displayed assumptions. Category representations and the cooperative-agreement example remain revisable modeling proposals. Empirical validation, a broader originality review, and publication sequencing remain open.

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Statement on the use of language models. The author selected the research direction, clarified the six-category scope, required ontological suspension, and authorized drafting. Language-model assistance supported organization of the source discussion, mathematical reconstruction, counterexamples, website verification, and preparation in $\text{\LaTeX}$. Agent-added constructions and source provenance are recorded in the accompanying research files. This version awaits the author’s review of its formulations, arguments, and references. Authorship and decisions about circulation remain with the human author.

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1 Introduction

This paper develops a common framework for representing six categories used in discussing social facts: entity, relation, event, attribute, state, and process. Its objective is to make the modeling requirements of each category explicit and to examine their connections within a shared dynamics. The method combines conditional constructions, stepwise derivations, counterexamples, philosophical analysis, and two finite illustrations. Ontological questions remain suspended throughout.

Consider an inquiry into a cooperative agreement. Investigators may ask which organizations participate, what obligations connect them, when authorization was recorded, which resources are available, whether the agreement is operative, and how implementation developed. These questions select different aspects of the inquiry. A record of implementation may help establish an event, describe a stage of a process, and change an attributed state. The usefulness of these distinctions depends on the work they enable an account to perform.

The questions become more demanding when the records diverge. An organization can change its membership while retaining an institutional identity. An authorization can be repeated without changing the current condition. Two histories can establish the same present status through different procedures. Two arrangements can look alike under a current observation while responding differently to a later request. A completed outcome can conceal different intermediate costs or failed attempts. A common mathematical language should make these differences available for analysis.

Part I supplied an abstract framework for change and composition relative to a declared regime of distinction [3]. The continuation considered here asks how the six-category vocabulary can be represented over that framework. Its scope is intermediate between an abstract action and an empirically warranted account. It develops representational possibilities and identifies the information needed to use them.

Three levels organize the argument. A *base definition* uses the symbols and explicitly declared parameters of the underlying model. An *induced construction* builds a mathematical object, such as a quotient or a set of finite paths, from specified data. An *enrichment* supplies further information, such as an actual history, an observation rule, or a criterion for following an organization through change. Keeping these levels visible makes the commitments of a category representation available for assessment.

The distinction matters because a formalism can conceal its inputs. Writing an entity as a set of related configurations can leave the relation's justification unexamined. Writing a process as a product can conceal the history omitted by that product. Writing an attribute as a function can leave its subject, definedness, and interpretation unresolved. The paper develops each construction far enough to expose these choices and to derive the conditions under which the representation performs its intended task.

The proposed contribution is a coordinated architecture for the six categories. It includes limited positive results and discriminating counterexamples. A preserved equivalence supports quotient dynamics; a coarse current observation can fail to support them. A mechanism product determines a composite effect while omitting distinctions between possible histories. Reachability can connect configurations while leaving entity identity underdetermined. These differences matter when an inquiry moves from a mathematically possible description to a claim about a particular social situation.

The entity treatment begins with indexed appearances and a criterion of reidentification. It examines invariant descriptors, directed lineage, fission, merger, and behavioral comparison. The relation treatment distinguishes incidence among participants from historical formation and potential effects under continuations. The event treatment separates occurrence tokens from transformation types and derives the loss of occurrence counts under idempotent compression.

The attribute treatment develops predicates, value maps, and partial descriptions, including the conditions for their descent to a quotient. The state treatment examines retained memory, observation tasks, and predictive sufficiency. The process treatment follows words through their trajectories and compares the information retained by products, maps, and endpoints. Each treatment includes its own interpretation, counterconditions, and evidence requirements.

A common construction compares configurations through the observations available under all finite continuations. The resulting equivalence is preserved by the declared operations. This construction be-

longs to established behavioral semantics [7]; its function here is to show how a consistent state description can be derived once its inputs are declared. The six-category question retains further requirements, especially occurrence, historical warrant, attribute interpretation, and reidentification.

Two examples connect the abstract results to inspectable models. The cooperative-agreement example has eight configurations and three primitive operations. Its complete transformation family can be derived, allowing comparison of mechanism effects, occurrence histories, and observations. The relational-unit example forms collective units from connected components of finite graphs. Comparing edge addition with edge deletion shows how the adequacy of a reduced description depends on the operation repertoire.

The philosophical discussion asks what these constructions commit an investigator to. It separates formation of a social arrangement, description of that arrangement, and uptake of a classification into subsequent action. It also distinguishes predictive sufficiency from historical explanation and entity identity from behavioral equivalence. These distinctions permit substantive criticism while the ontological status of the categories remains open.

The extensions examine partial operations, transitions among declared descriptive regimes, and finite probabilistic updates. Each introduces additional inputs and a corresponding preservation obligation. A change of description can require information omitted by an earlier record. A probabilistic quotient requires equal transition mass into each target class. These results provide precise limits on transferring the deterministic construction to richer settings.

The paper's social claims remain conditional modeling proposals. The finite examples establish consequences of their displayed assumptions and supply counterexamples to broader identifications. Empirical adequacy requires evidence about operation meanings, histories, observations, identity criteria, and availability conditions. The research-design section specifies how a bounded application could examine those requirements and identify revisions.

Section 2 locates the method and verified antecedents. Section 3 states the abstract framework, and Section 4 develops its generative and relational interpretation. Section 5 introduces the category architecture. Section 6 establishes the common continuation construction before Sections 7–12 develop the six category representations separately.

Section 13 develops the cooperative-agreement model. The subsequent relational-unit and coordination sections examine collective formation and compatibility among category records. Section 16 develops the philosophical interpretation, Section 17 examines changes to the dynamics, Section 18 derives finite-horizon approximation bounds and their limits, and Section 19 specifies the research design and revision conditions. The conclusion draws together the construction's results and its remaining evidential obligations.

2 Methodological Commitments and Verified Antecedents

This section locates the inquiry's contribution and explains how its arguments should be evaluated. It distinguishes the supplied abstract framework, established mathematical constructions, and the proposed coordination of the six categories. It then states the role of counterexamples, finite calculations, and philosophical interpretation in the paper's method.

2.1 The Inherited Framework and the Present Construction Task

Part I supplies the abstract carrier, mechanism composition, action, and regime of distinction from which the present development begins [3]. The use of that framework creates a definite dependency: results established here assume the relevant action and preservation conditions. Their social interpretation requires an account of what the carrier, mechanisms, and distinctions represent.

The present task concerns the additional work needed to model entities, relations, events, attributes, states, and processes. A vocabulary naming these categories can suggest a unified account, yet the formal requirements differ. An event occurrence needs a way to distinguish individual happenings. An attribute

needs a subject and interpretation. An entity record needs a reidentification criterion. The argument evaluates the data required for each role.

This dependency also fixes the scope of the series continuation. The paper develops an intermediate modeling layer over Part I. It leaves the categories' ultimate ontological status suspended. A successful construction establishes that declared data support a specified representation. Further philosophical and empirical arguments would be needed to determine the broader status of that representation.

The distinction among a base definition, an induced construction, and an enrichment serves this purpose. A quotient can be constructed from an equivalence already supplied. A historically actual occurrence requires an actuality claim and relevant evidence. Both can enter one account, while their sources of warrant remain different.

2.2 Behavioral Constructions and Their Local Use

The continuation construction draws on established behavioral semantics. Rutten's account of systems and behavioral equivalence provides a verified mathematical antecedent for comparing states through their observable responses to continuations [7]. The present paper uses that machinery to derive a state description relative to declared observations and operations.

The local result is valuable because it identifies the exact information retained by the construction. Two states belong to one behavioral class when the selected continuations and readouts fail to distinguish them. Changing the observations or available operations changes the relevant comparison. Thus the quotient's meaning includes its inputs.

The six-category architecture adds questions that behavioral equivalence alone leaves open. Historically distinct occurrences can produce the same response profile. Two different organizations can occupy behaviorally equivalent positions. A formation history can answer a provenance question that a predictive state omits. These are repository applications and distinctions developed in the present argument.

The division of intellectual work should remain explicit. The mathematical existence and preservation results for behavioral descriptions have established antecedents. The proposed contribution concerns their coordination with other representational requirements in this metamodel. A broad claim of novelty for quotient-based state comparison would exceed that contribution.

2.3 Provenance Models and Historical Evidence

The W3C PROV-DM recommendation distinguishes entities, activities, agents, and relations used to represent provenance [6]. Its treatment of derivation gives an established reference point for structured historical records. The present paper uses this connection narrowly: provenance representation can organize dependence and activity records whose interpretation requires evidence.

The six categories in this paper have a different organizing purpose from the categories of that recommendation. Shared terms therefore need local definitions. An entity appearance in the present construction is an indexed subject of a proposed reidentification criterion. An occurrence is a token recorded within a declared event domain. Their relation to a provenance schema would need an explicit mapping.

This distinction helps separate documentary organization from historical warrant. A well-formed provenance record can state that one item was derived through an activity involving another. Assessing the statement requires evidence concerning the activity and its dependencies. The graph or relation structure organizes the claim; the inquiry remains responsible for establishing its application to the case.

The paper's historical examples follow the same discipline. They construct candidate histories and ask which questions a retained record can answer. A proof that several histories share a record establishes an information limit. It leaves the actual history to be identified through evidence.

2.4 Pattern Interpretation and Selective Description

Dennett's discussion of real patterns connects questions of description with compression and predictive usefulness [1]. It offers a philosophical comparison for the selection of informative descriptions. The

present paper retains its own analytic formulation and suspends the ontological conclusion that might be drawn from that comparison.

A selective description can gain explanatory value by preserving distinctions relevant to an inquiry while omitting others. The continuation quotient formalizes one such task under exact assumptions. An entity criterion, however, may also preserve historical continuity that has little effect on the selected predictions. A useful description therefore needs a stated purpose before its compression can be assessed.

The coexistence of several purposes gives the six-category architecture a philosophical role. Predictive economy, historical accountability, occurrence counting, and participant reidentification can demand different records. Their coordination makes those demands visible within one account. It also exposes tensions when a single compression is asked to serve incompatible tasks.

The position defended here is consequently methodological. An explicit representation can clarify what an account retains, what it omits, and which further claims require support. Its usefulness is assessed through the work it performs in a bounded inquiry. Ontological interpretation remains an additional question.

2.5 Conditional Proof and Interpretation

A conditional proof establishes a consequence of stated assumptions. The quotient action proof, for example, begins with an equivalence preserved by every mechanism and derives a representative-independent action on its classes. The proof can be complete while the empirical applicability of the preservation assumption remains open.

Interpretation supplies the connection between that mathematical result and a social question. If equivalence means equality of a proposed institutional status, the application must explain why equivalent cases respond equivalently to the declared operations. A counterexample to that empirical interpretation can reveal a missing variable or a poor choice of operations without affecting the abstract theorem.

The paper therefore places symbols and domains before definitions, assumptions before conclusions, and interpretation after the relevant formal construction. This order allows a reader to identify where a disagreement arises. The disagreement may concern a proof step, the meaning of a variable, the adequacy of a classification, or the evidence supporting an input.

An original formulation also carries a distinct research status. A proposed enrichment can be mathematically coherent and philosophically suggestive while awaiting empirical evaluation. The status of the construction should travel with it when it is used in another paper or applied to a case.

2.6 Counterexamples and the Repair of Tempting Identifications

Counterexamples perform a central part of the argument. They test claims that can appear plausible when expressed only in general language. A reachable configuration may seem to belong to the same entity's development. Overlapping forward closures show why that association requires an additional identity criterion. A mechanism product may seem to preserve a process. Repeated idempotent operations show how it can omit occurrence multiplicity.

Each counterexample has a specified target. The idempotence example refutes recovery of a positive occurrence count from the compressed product under the stated assumptions. It leaves open the use of a larger state or a richer record. The graph example shows that connected components can support addition-only updates while failing under a deletion operation. It leaves intact the component description for the restricted repertoire.

A good repair preserves the useful part of the original insight. An entity can remain a continuing unit through change, provided the continuity criterion is declared. A process product can summarize a composite effect, provided the historical information it omits is retained when required. The counterexample identifies the boundary of the formulation and the data needed to cross it.

This method also guards against excessive formalization. If a proposed symbol merely renames a social term, its contribution is limited. A useful formulation should define a comparison, expose an

assumption, derive a consequence, or distinguish models that would otherwise be conflated. The category sections evaluate their formulas by these functions.

2.7 Finite Calculations and Mathematical Coverage

Finite illustrations make the constructions inspectable. The cooperative-agreement model has a small carrier, a declared primitive repertoire, and a transformation family that can be derived explicitly. The relational-unit example permits comparison of component descriptions under different operation repertoires.

Enumeration checks are used to examine the finite claims and to compare independently specified constructions. Such checks can detect an omitted transformation, a mistaken partition count, or an incorrect example. The accompanying proofs explain why the claimed result follows from its assumptions.

Finite verification has a defined scope. Exhausting an eight-state example verifies claims about that example. It supplies a counterexample to a universal assertion when the assertion fails there. It supplies an illustration of a general theorem when the theorem has been proved separately. It leaves empirical correspondence to the case-study stage.

The examples are therefore analytic models. Their social labels help state the intended questions, while their transition rules are declared hypotheses of the illustration. An empirical investigation may need different states, operations, and observations.

2.8 Philosophical Discussion within the Modeling Scope

The philosophical discussion concerns the commitments made by the representations. Identity requires a rule for following a subject through change. Historical explanation requires a relation between present records and candidate histories. Potential effects require a comparison under continuations. Classification can remain descriptive or enter a response rule through actual uptake.

These questions can be developed while ontology remains suspended. Suspension sets the scope of the present argument; it leaves room for substantive claims about the adequacy, consequences, and limits of a representation. A model can be criticized for erasing a relevant history or for hiding an assumption even when its categories receive an agnostic ontological interpretation.

The same discipline governs evaluation. A greater number of reachable outcomes can serve a stated aim while making other outcomes possible as well. A more persistent relation can sustain cooperation or maintain an unwanted dependence. The mathematics identifies structural properties. Evaluation requires the aims, affected subjects, and standards relevant to the inquiry.

The paper's philosophical contribution is to articulate these commitments together with the mathematical constructions. Their coordination supports a form of criticism in which a reader can identify both the formal step and the interpretive premise on which a conclusion depends.

3 Abstract Framework for Category Modeling

This section develops the base on which the category representations depend. It explains the carrier, mechanism algebra, action, and equivalence separately; derives the quotient and the behavior of collapse under composition; and examines which additional descriptions can be defined from the base. The discussion also tests identity, reversibility, order effects, and semantic interpretation through countermodels. Its purpose is to make the assumptions perform visible work in each later derivation.

3.1 Configurations and the Scope of the Carrier

The carrier S is a nonempty set of represented configurations. A configuration can include several participants and conditions together. It can also be an abstract description whose internal structure is left unspecified. The metamodel uses this flexibility to compare forms of representation without requiring every application to begin with the same inventory of objects.

The choice of carrier determines which differences a model can express before any quotient is applied. A carrier containing only an activity flag cannot directly distinguish permission from resources. A carrier containing both conditions can represent their different roles. A history-enriched carrier can preserve formation information that a present-condition carrier omits. These are different models even if they share a verbal description of the case.

The distinction between a point of S and an entity is therefore essential. One point can describe an entire arrangement involving several organizations. The entity category later introduces appearance records to track individual subjects within and across such configurations. A state space whose points already stand for individuals makes another legitimate but stronger interpretation.

The carrier can include possible configurations that are unreachable from a selected seed. Their inclusion can support comparisons, counterfactual tests, or alternative initial conditions. A particular application should state whether S is intended to cover every relevant possibility or only a declared subset. A theorem over all of S inherits that choice of scope.

Changing the carrier changes the available questions. A richer carrier can distinguish histories or internal relations previously merged into one point. A smaller carrier can simplify calculation at the cost of information. Comparisons between such models need explicit maps or correspondences. The mere use of the symbol S in both models supplies no alignment.

3.2 Mechanisms and Associative Composition

The mechanism domain M is equipped with an associative multiplication. Its elements represent operations at the chosen level, including composites when the model treats them as mechanisms.

Assumption 3.1 (Mechanism semigroup). The nonempty set M has a product satisfying

$$\ell(nm) = (\ell n)m \quad (\ell, n, m \in M).$$

The product nm is read as applying m first and n second.

Associativity permits a finite sequence to have a composite independent of how its parentheses are placed. It leaves execution order intact. The equation equates two groupings of the same ordered sequence; commutativity would equate different orders and is a separate property.

Closure under composition also has a specific meaning. If m and n belong to the mechanism domain, their product belongs to it. This is an algebraic closure claim. Whether a particular execution is available in a social situation requires an admissibility rule. The later feasible-path construction retains that distinction at every intermediate step.

Mechanism labels can preserve distinctions that their visible effects omit. Two procedures can have the same action on the chosen variables while differing in their formation, purpose, or provenance. An abstract mechanism algebra can retain their labels, whereas a semigroup of transformations identifies them whenever their maps agree. The appropriate choice depends on the questions the model needs to answer.

The algebra also leaves temporal duration open. A product records composition, not how much time its enactment takes. Repeated applications can collapse to one algebraic element under an idempotent equation even when their durations or occurrence counts differ. The process category derives this information loss explicitly.

3.3 The Action and Its Interpretation

The action connects mechanisms to configurations. This subsection states its consistency condition and explains what that condition establishes.

Let $\alpha : M \times S \rightarrow S$ be total, and write $mx = \alpha(m, x)$.

Assumption 3.2 (Action law). For all $m, n \in M$ and $x \in S$,

$$(nm)x = n(mx).$$

The law equates the effect of the composite mechanism with successive evaluation of its factors. It makes the algebra and dynamics consistent. Once a finite word and initial configuration are selected, repeated use of the law identifies the word's endpoint with the action of its product.

Totality states that each mechanism has a modeled output at each carrier point. An application can use total maps for attempted operations, including self-loops under unmet conditions. It can instead supply an admissibility rule or use partial maps for successful executability. Those interpretations require explicit definitions because they disagree about what happens outside an operation's effective domain.

The action can be nonfaithful. Distinct $m, n \in M$ can satisfy $mx = nx$ for every $x \in S$. Their algebraic identity then retains information beyond the selected dynamics. If the action is faithful, equality of maps implies equality of mechanisms. Faithfulness is an additional assumption and is unneeded for the quotient results.

An effect can also be local. Distinct maps can agree at one initial configuration while differing elsewhere. An endpoint observation at that seed therefore gives weaker information than knowledge of the entire map. The history and process constructions will distinguish these successive reductions carefully.

3.4 Equivalence and Preserved Distinctions

The base equivalence declares which configurations the model treats alike at a selected level. This subsection explains its logical and dynamic requirements.

Let $\sim$ be an equivalence relation on S . Reflexivity, symmetry, and transitivity make its classes a partition. The additional action condition is

$$x \sim y \implies mx \sim my \quad (m \in M). \quad (1)$$

The full base assumptions are summarized as follows.

Assumption 3.3 (Base action and distinction). Let M be a semigroup acting totally on nonempty S , with $(nm)x = n(mx)$. Let $\sim$ be an equivalence preserved by every mechanism as in Equation (1). Identities and inverses are permitted without being required.

The condition concerns configurations already treated as equivalent. They must remain equivalent after the same action. Initially distinct configurations may become equivalent. The model therefore permits collapse of a represented distinction while preserving established equivalence.

The intended meaning of $\sim$ needs a domain argument. It can record a selected observational resolution, a theoretically relevant classification, or another declared equivalence. An observation-induced relation $\ker o$ qualifies only if the operations preserve it. A current measurement can fail that condition when it omits a future-relevant variable.

The equality relation always satisfies preservation under a function. The universal relation also satisfies it, because every pair is already equivalent. These extremes show that preservation alone does not select a useful descriptive level. A finer relation retains more distinctions; a coarser relation can simplify the description. The inquiry must justify which distinctions its task requires.

A value judgment about preserving a social distinction adds another requirement. The base permits some initially distinct cases to collapse. If an application needs to protect a particular difference, it must restrict the mechanisms or strengthen its conditions. Mathematical preservation of equivalence should not be conflated with a normative requirement to preserve all differences.

3.5 Symbols and the Four Analytical Roles

The base coordinates several roles in one formal structure. This subsection relates them without treating the verbal roles as interchangeable with every symbol.

Table 1 collects the notation used in the category constructions. The abstract framework continues Part I's treatment of change and composition [3], while stating the precise assumptions required for the present results.

Table 1: Symbols for the action framework and its declared enrichments

Symbol	Domain and analytical role
S	Nonempty carrier of represented configurations
$M, \cdot$	Mechanism domain with associative composition
α	Total action, written $mx = \alpha(m, x)$
$\sim$	Equivalence preserved by the action
Q, q	Quotient $S/\sim$ and projection $q : S \rightarrow Q$
B, T_b	Operation labels and their declared maps
$A(x)$	Admissible mechanisms or labels at x , with the domain specified
o, ρ	Observation and retained-description maps
I, E	Appearance indices and a declared identity equivalence

Change appears through the action. Structure appears through the composition law and any further structure supplied to the carrier. Constraint includes the requirement that equivalent configurations remain equivalent under common action. Difference appears through the selected distinction regime and the changes that operations can make to it.

These are analytical roles, not four independently measured quantities. The same mathematical component can contribute to more than one role. For example, the action both implements change and expresses a compositional consistency constraint. The collapse set defined below depends on both the action and equivalence.

Resources, permissions, values, and physical time are additional application data. They can be encoded in configurations or attached through explicit maps. Their appearance in a model requires a construction or interpretation; it does not follow merely from assigning them one of the four verbal roles.

3.6 Quotient Dynamics and Representative Independence

The quotient construction makes the preserved equivalence operational. This subsection gives both directions of its adequacy condition and derives the induced action.

Let $Q = S/\sim$ and $q(x) = [x]$. The proposed quotient update is

$$\bar{\alpha}(m, [x]) = [mx]. \quad (2)$$

The expression uses a representative x , so its value must be independent of that choice.

Proposition 3.4 (Induced quotient action). *Equation (2) is well-defined exactly when $\sim$ is preserved by every mechanism. Under Assumption 3.3, it defines an action of M on Q .*

Proof. Suppose preservation holds. If $[x] = [y]$, then $x \sim y$, so $mx \sim my$ and $[mx] = [my]$. The quotient update is therefore representative-independent.

Conversely, if the update is well-defined and $x \sim y$, the equal inputs $[x] = [y]$ give equal outputs $[mx] = [my]$. This is preservation.

For the action law, evaluate the two proposed composite updates:

$$\bar{\alpha}(nm, [x]) = [(nm)x] = [n(mx)] = \bar{\alpha}(n, \bar{\alpha}(m, [x])).$$

The middle equality follows from the base action law. Thus the quotient maps form an action. $\square$

The quotient can be viewed as a reduction of the configuration description. The projection commutes with each operation: $q(mx) = \bar{\alpha}(m, q(x))$. This equation is the precise relationship between the full and reduced dynamics.

If preservation fails, the same quotient class would need different outputs under one mechanism. An application can then refine the equivalence, retain additional state information, or adopt another type

of reduced model. The failure is informative because it identifies a specific incompatibility between description and dynamics.

Representative independence is a recurring theme in the six categories. Entity descriptors, relational incidence, attributes, and historical queries each need an analogous constancy condition when transferred to a reduced domain. The quotient proof supplies the basic pattern, while each later application has its own input and interpretive commitments.

3.7 Identity Elements and Zero-Step Continuations

An algebraic identity and an identity action are different assumptions. This subsection explains that distinction and constructs a uniform zero-step convention.

If M has an identity element e , the semigroup action law gives $\alpha(e, \alpha(e, x)) = \alpha(e, x)$. It forces the map $\alpha(e, -)$ to be idempotent. Without a unitality requirement, it need not equal id_S .

For example, take the one-element monoid $M = \{e\}$ and a carrier with at least two points. Let $\alpha(e, -)$ be a constant map. The action law holds because a constant map is idempotent, while the map differs from the identity on S . The example isolates the missing assumption.

Finite words can represent zero steps by defining $T_e = \text{id}_S$ directly. This convention does not require an identity mechanism in the original semigroup. It also avoids identifying the empty word with an old algebraic identity whose action has not been checked.

A uniform algebraic repair adjoins a fresh unit 1 disjoint from M , even if M already contains an identity. Set $1m = m1 = m$ and $1 \cdot 1 = 1$, retain all original products, and define $1x = x$. Associativity in cases involving 1 follows by deleting the unit; other cases use associativity in M .

The extended action law holds for the same reason. If a factor is 1 , both sides reduce to the unchanged action. If both factors belong to M , the original law applies. The extension therefore retains the earlier model while adding a valid zero-step action.

This distinction matters when a word is compressed to a product. A product representation that includes the empty word must specify which element or map represents it. Checking that convention prevents a zero-step history from acquiring an unintended effect.

3.8 Collapse of a Declared Distinction

An operation can identify configurations that the base initially distinguishes. This subsection defines that collapse and derives its persistence under common subsequent action.

For $m \in M$, define

$$\Delta_-(m) = \{(x, y) \in S^2 : x \not\sim y, mx \sim my\}. \quad (3)$$

The set records ordered pairs that lose their distinction under m . It is relative to the declared equivalence.

Proposition 3.5 (Persistence of common-action collapse). *Under Assumption 3.3,*

$$\Delta_-(m) \subseteq \Delta_-(nm) \quad (m, n \in M).$$

Proof. Take $(x, y) \in \Delta_-(m)$. Then $x \not\sim y$ and $mx \sim my$. Preservation under n yields $n(mx) \sim n(my)$. The action law identifies these outputs with $(nm)x$ and $(nm)y$. The initial distinction therefore also collapses under nm . $\square$

The proof holds the equivalence fixed and applies the same subsequent mechanism to both configurations. Changing either condition changes the comparison. Different later actions can give different outputs, and a finer observation can reveal differences previously identified by $\sim$.

The result also concerns represented distinctions. A collapse relative to one quotient can leave many differences intact at a finer level. Interpreting it as physical erasure, epistemic loss, or institutional homogenization requires a domain mapping and evidence. The set formula alone leaves those meanings open.

The original role of difference in the metamodel is therefore retained with a precise boundary. It identifies how an operation transforms a declared regime of distinction. It supplies no automatic measure of social importance or loss.

3.9 Exact Composition of Collapse Sets

The inclusion can be sharpened into an exact identity. This subsection derives the decomposition and explains its counting implications.

Write $\alpha_m(x) = mx$.

Proposition 3.6 (Two-stage collapse decomposition). *Under the base assumptions,*

$$\Delta_-(nm) = \Delta_-(m) \dot{\cup} (\alpha_m \times \alpha_m)^{-1}(\Delta_-(n)), \quad (4)$$

where the union is disjoint.

Proof. Take a pair in $\Delta_-(nm)$. Its inputs are initially inequivalent and its final outputs are equivalent. There are two cases.

If $mx \sim my$, the pair belongs to $\Delta_-(m)$. If $mx \not\sim my$, the intermediate pair (mx, my) belongs to $\Delta_-(n)$, so the input pair lies in the displayed inverse image.

Conversely, a pair in $\Delta_-(m)$ belongs to $\Delta_-(nm)$ by Proposition 3.5. A pair in the inverse image has $mx \not\sim my$ and $n(mx) \sim n(my)$. Its inputs must be inequivalent: $x \sim y$ would imply $mx \sim my$ by preservation. Thus it also belongs to $\Delta_-(nm)$.

The first branch requires $mx \sim my$; the second requires $mx \not\sim my$. They are disjoint, completing the proof. $\square$

The identity separates distinctions already collapsed by the first step from those surviving it and collapsing at the second. It therefore preserves the sequence of reasoning behind the final collapse set.

For finite S , cardinalities of the two branches add. The second term is the size of an inverse image, which need not equal $|\Delta_-(n)|$. Several initial pairs can map to one intermediate pair. A simple sum of the two mechanisms' raw collapse counts can therefore misrepresent the composition.

An interpreted measure would need further choices about which pairs matter and how they are weighted. The exact set identity precedes that choice and keeps its dependence on the action visible.

3.10 Order, Reversibility, and Independent Properties

Several intuitive properties of dynamics are logically distinct. This subsection uses countermodels to prevent their meanings from being merged.

On the nonnegative integers, the shift $s(k) = k + 1$ is injective and lacks an inverse within the semigroup of positive shifts. With equality as the distinction regime, its collapse set is empty. Absence of an inverse therefore does not by itself imply collapse.

On $\{1, 2, 3\}$, let a swap 1 and 2, and let b swap 2 and 3. Both are bijective and preserve every distinction. Yet $ba(1) = 3$ and $ab(1) = 2$. Order sensitivity can occur without collapse.

On $\{0, 1\}^2$, let r_1 reset the first coordinate to zero and let r_2 reset the second. They commute, because each leaves the other's coordinate unchanged. Each also collapses pairs differing only in its reset coordinate. Collapse can therefore occur without an order effect between these operations.

These examples distinguish three questions: whether an operation can be reversed within the chosen algebra, whether it identifies a represented pair, and whether its order with another operation changes an outcome. A historical interpretation can involve any combination of the three.

History itself is another question. A sequence can have a history record even when its composite is bijective or returns a seed to its initial point. The record of what occurred belongs to the event and process representation. It is not determined by the presence of a collapse set.

3.11 Definitions, Constructions, and Enrichments

The category account uses several forms of mathematical development. This subsection states their differences so that a representation's contribution can be assessed.

The two-sorted base language is

$$L_0 = \{S, M, \cdot, \alpha, \sim\}.$$

An explicitly defined predicate uses a stated formula in that language, with every parameter disclosed. The order-effect predicate is one example because it refers only to mechanism products, actions, and equivalence.

A quotient is an induced set construction from a specified equivalence. A word space is constructed from a specified alphabet. These constructions are available in the mathematical setting, but treating their elements as objects of the original first-order language requires an encoding or additional sorts.

An enrichment supplies further information. An actual history, observation map, availability set, or identity criterion can be supplied in this way. The enrichment can be justified and useful while remaining distinct from a consequence of the bare base data.

An underdetermination test holds the entire base fixed and varies the proposed addition. If two permitted enrichments disagree, the fixed base does not uniquely determine that addition. This does not exclude a particular definition under stronger premises. It identifies the information that those premises must supply.

The distinction also clarifies philosophical claims of emergence. A mathematical construction can derive a pattern from specified relational data. Whether that construction explains the emergence of a social category depends on the interpretation and evidence for those data. Calling the construction generative cannot replace that argument.

3.12 Symmetry and Parameter-Free Definability

A symmetry test can expose omitted distinctions. This subsection gives the logical reasoning and its interpretive boundary.

A base automorphism consists of bijections on the two sorts that preserve multiplication, action, and equivalence. Every parameter-free first-order predicate defined in L_0 has the same truth value on a tuple and its image under such an automorphism.

The proof proceeds by induction on expressions. Variables transform by the assigned bijections. Preservation of function symbols makes term evaluation commute with the automorphism. Atomic equalities and the relation $\sim$ therefore preserve truth. Boolean connectives retain that invariance.

For an existential quantifier, a witness is sent to a witness by the automorphism. Its inverse gives the converse. Universal quantification follows by the same bijective correspondence. The induction establishes truth invariance for every formula.

Consider a carrier $\{0, 1\}$ with only identity action and ordinary equality. Swapping the two states preserves the whole base. A parameter-free unary predicate therefore cannot select just one state. A named parameter or supplied subset can make that selection, adding information that breaks the symmetry.

The symmetry test is necessary, not a general converse. Passing it does not display a defining formula. It is useful because a failed test shows that a proposed semantic distinction requires additional structure or parameters.

The same point applies to social meanings. If two interpreted cases receive identical base representations, a base-only calculation cannot recover the omitted difference. The appropriate response is an explicit encoding and justification, not a stronger verbal claim about the unchanged formula.

3.13 The Base and the Category Construction Task

The preceding results establish a common formal discipline for the rest of the paper. This subsection states how that discipline guides the six categories.

The base provides configurations, composition, action, and a preserved distinction regime. It supports quotient dynamics, order comparisons, and collapse calculations. The category constructions use those resources while declaring their additional domains and criteria.

Entities require reidentification across appearances. Relations require participant incidence, formation records, and potential-effect comparisons. Events require occurrence indices. Attributes require coding rules at a declared level. States require sufficient retained information for the chosen task. Processes require words, trajectories, and any grouping or time records.

The contribution is therefore a coordinated development under explicit assumptions. Some components are derived from the base, some are induced constructions, and some are justified enrichments. Each can be examined through proofs, counterexamples, and evidence without turning the category repertoire into an asserted ontology.

4 Interpretation of Generative Relational Dynamics

This section explains the generative and relational vocabulary through historical formation, future possibilities, and the conditions of realization. It constructs mechanism fibres, derives the relation between support and outcome images, and examines prefix feasibility and reachability. The philosophical discussion identifies which comparison can justify a claim of generation, expansion, or reproduction. The aim is to connect the abstract action to a research question without assigning more meaning to the algebra than its inputs support.

4.1 Historical Formation and Further Activity

A working relational interpretation concerns a relatively reidentifiable dependence with a formation history and possible effects on later activity. A relation can arise historically and participate in producing further history. The two directions connect provenance with prospective dynamics while retaining distinct information requirements.

The first direction asks how a pattern came to obtain. Its answer can involve a sequence of events, a negotiation, a maintained practice, or another specified process. The second direction asks what later activity depends on that pattern. Its answer can involve an available operation, a changed outcome, or a route whose feasibility requires the pattern to persist.

A current state can encode enough information for a selected future prediction while omitting the formation sequence. That sufficiency leaves the historical question open. Conversely, a detailed history can remain insufficient for prediction when the model lacks the relevant transition rules or current conditions. Retaining records and specifying dynamics are complementary tasks.

The potential effect can remain unrealized. An authorization can make a continuation possible even during a period in which it is never used. The model must specify the conditions and test under which the effect would appear. An observed sequence that bypasses the continuation supplies different information from a complete test of the proposed possibility.

The terms *generative* and *relational* therefore orient an inquiry. They motivate questions about formation and dependence. Their use in a title or variable name does not supply the missing history, interpretation, or evidence. Those components must be developed in the model and argument.

4.2 A Base-Definable Interaction Witness

Order sensitivity gives one interaction predicate within the abstract language. This subsection states its use before the broader relation category develops its limits.

Define

$$R_{\text{ord}}(m, n, x) \iff (nm)x \not\sim (mn)x. \quad (5)$$

The predicate compares the two orders from one seed under the same distinction regime. It requires no additional relation symbol because every component belongs to the base.

The calculation can reveal that one operation changes the conditions under which another operates. In the cooperative example, authorization followed by attempted activation has a different effect from the reverse order. The interpretation depends on the specified maps and conditions.

Commuting operations can also contribute jointly to an outcome. Two operations that set separate prerequisite bits commute, yet a later activity can require both bits. An order predicate therefore captures one kind of interaction and leaves other forms of dependence open.

The distinction matters to the ambition of deriving relations. A base-definable predicate establishes that one structural pattern is available in the language. A social relation requires an account of its participants, formation, and meaning. The later relation section keeps those additional commitments explicit.

4.3 Mechanism Fibres and Possible Outcomes

The generative interpretation needs a way to describe possibilities before their realization. This subsection constructs their witnesses and separates them from the image of outcomes.

Let $A(x) \subseteq M$ be a declared set of admissible composite mechanisms at x . For a quotient outcome $v \in Q$, define

$$\Pi_A(x, v) = \{m \in A(x) : q(mx) = v\}, \quad (6)$$

$$P_A(x) = \{v \in Q : \Pi_A(x, v) \neq \emptyset\}, \quad (7)$$

$$G_A(x) = P_A(x) \setminus \{q(x)\}. \quad (8)$$

Definition 4.1 (Support of a possible outcome). The fibre $\Pi_A(x, v)$ retains the admissible mechanism witnesses for outcome v . Its nonemptiness establishes possibility in the declared model. The image $P_A(x)$ retains outcomes with witnesses, and $G_A(x)$ restricts that image to changed quotient outcomes.

The fibre and image answer different questions. The image states which outcomes are possible. The fibre states which admissible mechanisms support one outcome. Several witnesses can share an outcome while differing in cost, history, duration, or enabling conditions.

The definition of $G_A(x)$ makes another selection. It excludes the present quotient state. This is appropriate when the research question concerns a changed outcome at that resolution. An activity that reproduces the current state can still require effort, sustain a relation, or have historical significance. Those roles remain visible through the fibre or occurrence record.

The set $A(x)$ is additional information unless a definition derives it from represented conditions. A semigroup action alone does not determine which mechanisms are available at a particular situation. The same action can be paired with different availability rules, producing different support sets.

4.4 Support Enlargement and Outcome Enlargement

Adding mechanisms can increase routes without adding outcomes. This subsection derives the exact condition for strict image expansion.

Fix the seed and quotient, and suppose $A_1(x) \subseteq A_2(x)$. Every mechanism in $\Pi_{A_1}(x, v)$ then belongs to $\Pi_{A_2}(x, v)$. Consequently

$$P_{A_1}(x) \subseteq P_{A_2}(x).$$

Proposition 4.2 (Strict outcome expansion). *The inclusion of outcome images is strict exactly when some $m \in A_2(x)$ satisfies $q(mx) \notin P_{A_1}(x)$.*

Proof. If such a mechanism exists, its image belongs to the larger outcome set and lies outside the smaller one, proving strictness.

Conversely, a member of $P_{A_2}(x) \setminus P_{A_1}(x)$ has a witnessing mechanism in $A_2(x)$ by definition. That mechanism satisfies the stated condition. $\square$

An added witness for an old outcome leaves the image unchanged. It can nevertheless matter if its enabling conditions differ from those of earlier witnesses. The outcome may become available under circumstances that previously blocked every route. To analyze that change, the model must retain the route conditions.

Duplicating a label can also enlarge a syntactic catalog while adding neither an effect nor a distinct enabling structure. Raw mechanism counts therefore depend on representation. A substantive claim about expanded opportunity needs a comparison that identifies what the new witness actually changes.

For infinite sets, strict inclusion need not increase cardinality. Even in a finite model, cardinality alone omits the meaning of outcomes and the relations among routes. Set-based comparison is therefore the starting point; a numerical summary requires further choices.

The result keeps several possible claims separate: more labels, more mechanism witnesses, more feasible paths, more supported outcomes, and more valued opportunities. Each transition between them requires its own premises.

4.5 Primitive Words and Prefix Feasibility

A composite mechanism can conceal the stages needed for its enactment. This subsection retains those stages and states the feasibility condition.

Choose an operation alphabet B whose labels specify mechanisms and their maps. A word $w = b_1 \cdots b_k$ gives

$$x_0 = x, \quad x_i = T_{b_i} x_{i-1}.$$

Let $A_{\text{prim}}(x) \subseteq B$ be the admissible primitive labels at x .

Definition 4.3 (Prefix-feasible word). The word is feasible from x when

$$b_i \in A_{\text{prim}}(x_{i-1}) \quad (1 \leq i \leq k). \quad (9)$$

The empty word is feasible and leaves the seed unchanged.

Every test is performed at the configuration reached by the earlier prefix. The total action can calculate the hypothetical endpoint of a word whose intermediate execution is unavailable. Prefix feasibility identifies that difference.

Primitive admissibility and composite admissibility have different domains. The former concerns the next label. The latter concerns a whole mechanism. A composite can be called feasible when at least one of its factorizations is prefix-feasible. Retaining the witness matters when the process must actually be followed.

Other factorizations of the same product can remain infeasible. An existential composite feasibility claim therefore does not establish that every word with that product can be executed. The feasible-path paper develops this factorization issue through an exact descent criterion.

The distinction is especially important when prerequisites can be consumed, withdrawn, or changed by an earlier step. An endpoint image does not record those intermediate conditions. A resource or permission model must specify their update rules.

4.6 Partial Maps and Domain-Sensitive Composition

Partial operations provide an alternative representation of availability. This subsection explains the domain law and its relation to the base algebra.

Suppose m is executable on D_m and n on D_n . Their partial composition is defined on

$$D_{n \circ m} = \{x \in D_m : mx \in D_n\}.$$

The first condition permits the initial step; the second permits the next step at its result.

Three successive partial maps have the same prefix conditions under either parenthesization. This establishes associativity of partial-map composition. The domain and effect together determine the partial transformation.

Arbitrary domains attached to elements of an existing total semigroup can fail to preserve its equations. A total flip can satisfy $f^2 = \text{id}$, while restricting its domain to one state makes its two-step partial composition empty. The restricted product has changed meaning.

An application can retain free words until their domain compatibility is established. It can instead use the semigroup of partial transformations. Either approach records the additional structure openly and avoids silently transferring identities that no longer hold.

A blocked-state totalization offers another convention. It can make failed execution observable, but the marker and its later behavior require interpretation. The state and event sections distinguish modeled unavailability from an actual failed attempt.

4.7 Realization, Selection, and Valuation

A possible route and an actual occurrence have different statuses. This subsection separates the additional commitments needed for realization and appraisal.

Nonempty support means that a witness exists under the model's rules. It leaves open whether anyone selects the witness, whether execution occurs, and whether the modeled outcome is observed. Each claim needs an occurrence or selection account.

A probability distribution is another enrichment. The set of witnesses does not supply a canonical measure, and a uniform distribution over labels can change when labels are duplicated. A probability model must identify its sample space and justify its weights.

Intention and reasons remain distinct from a selection law. A statistical account of which route is chosen can omit the participant's understanding of the choice. A philosophical account of agency may need those reasons and the conditions under which they are formed.

Valuation adds further structure. A larger outcome set can include undesirable possibilities. Greater route robustness can sustain an arrangement that participants wish to change. A generative or reproductive effect therefore has no fixed evaluative sign in the formalism.

The model can accommodate values, preferences, or constraints when they are supplied explicitly. It should then distinguish the formal consequence of those inputs from an argument that the inputs are justified. This preserves the open relation between generation, reproduction, and evaluation.

4.8 Unrestricted Reachability in a Fixed Semigroup

A fixed semigroup already contains every finite composite of its mechanisms. This subsection derives the consequence for unrestricted future reachability.

Define the reflexive reachable set

$$\text{Reach}^*(x) = \{x\} \cup \{mx : m \in M\}.$$

Proposition 4.4 (Reachability after a represented action). *For every $c \in M$,*

$$\text{Reach}^*(cx) \subseteq \text{Reach}^*(x). \quad (10)$$

Proof. A member of the left side is either cx or $m(cx)$ for some $m \in M$. The first is reachable from x by c . The second equals $(mc)x$, and closure puts mc in M . Both therefore belong to the right side. $\square$

The result can appear to conflict with a claim that an operation opens possibilities. Its scope resolves the apparent tension. The reachable set already includes all composites, without a finite horizon, resource budget, or changed availability rule.

An enabling action can make a later primitive locally effective while its composite route was already included in the unrestricted set. The cooperative example illustrates that distinction. Recording authorization changes what an immediate activation can do, even though the authorization-plus-activation word was already a represented composite.

The inclusion also leaves strict contraction possible. After an irreversible action, some earlier alternatives may no longer be reachable. Equality requires additional conditions, such as a suitable route back. The theorem itself guarantees only inclusion.

A claim of generative expansion therefore needs to identify a comparison different from this unrestricted fixed-semigroup one, or explain how the model's carrier or repertoire changes.

4.9 Horizon, Availability, and Comparison Design

Finite resources and time horizons produce different possibility comparisons. This subsection explains their relation to the unrestricted theorem.

Let $\text{Reach}_h^A(x)$ contain endpoints of prefix-feasible words of length at most h under a fixed primitive admissibility rule. If c is feasible at x , then

$$\text{Reach}_h^A(T_c x) \subseteq \text{Reach}_{h+1}^A(x).$$

A witness word from $T_c x$ can be prefixed by c , adding one feasible step and preserving its later prefix conditions.

Comparing the same remaining horizon h before and after the enabling step resets the budget. It is therefore different from comparing two plans under one fixed total budget. A statement that more outcomes are available after the step can depend on which comparison is used.

Changing admissibility while holding the seed fixed is another comparison. If a rule makes additional primitive operations available, new feasible words can appear. Their endpoint images may or may not expand, depending on whether the new words reach new outcomes.

Changing the operation repertoire or carrier is stronger still. The old and new models need an alignment before their possibility sets can be compared. A novel label alone supplies no evidence of a novel effect.

These distinctions make the generative claim more precise. A researcher can state whether an intervention changes immediate activity, bounded routes, availability scenarios, represented outcomes, or the model's own vocabulary. The proof obligations follow that choice.

4.10 Interpretive Role within the Category Architecture

The generative-relational interpretation connects the base to the six categories. This subsection states that connection and the remaining philosophical task.

Entities can be formed or reidentified through changing relations. Relations can condition later activity. Events record actual applications. Attributes describe conditions and outcomes. States retain the distinctions relevant to the selected task. Processes preserve ordered formation and enactment.

The formalism supplies routes, images, and compatibility tests. Historical and social meaning enters through declared records, comparisons, and interpretations. A useful model makes those additions inspectable and demonstrates why they serve the inquiry.

This preserves the central insight that history can form the conditions of further history. It also exposes the separate questions that a comprehensive account must answer: what was formed, how it was formed, which possibilities depend on it, and what evidence warrants the interpretation.

5 Representations of Six Social-Fact Categories

This section gives entity, relation, event, attribute, state, and process distinct representational jobs within a single inquiry. After the common consistency construction, six category sections develop each con-

Table 2: Representational roles and additional commitments for six categories

Category	Representational role	Additional commitment
Entity	Reidentification across indexed appearances	Identity criterion or directed lineage
Relation	A dependence pattern with formation and possible effects	Relata, history, comparison, and domain meaning
Event	An indexed occurrence of a transition or attempt	Occurrence evidence and relevant circumstances
Attribute	A predicate or value assigned at a declared descriptive level	Coding rule, scale, and evidential interpretation
State	A configuration class under a selected distinction regime	Coverage and observation adequacy
Process	An ordered history with a trajectory	Step records, grouping, duration where relevant

struction, its derivation, and the information needed to interpret it. Table 2 summarizes these dependencies. The repertoire is a working selection for this research program; its completeness remains an open question.

Sections 7–12 treat the categories individually. Each develops its mathematical representation alongside the philosophical question that motivates it. Their formal roles overlap in one inquiry, while the evidence needed for a particular claim remains specific to the category and question.

The distinction between a representation and its interpretation governs this architecture. An equivalence class can support an entity label, but its identity criterion needs justification. A transition map can support an event description, but occurrence requires a record and warrant. An attribute can be mathematically well-defined while its classification rule remains debatable. These are explicit dependencies to investigate through the detailed constructions.

6 Consistency of State Descriptions under Continuation

This section establishes a common construction used by the category treatments that follow. Given operations and an observation, it derives the coarsest equivalence that retains the observation and supports deterministic updates. The discussion gives the full proof, examines the relation to an inherited distinction regime, derives finite refinement, and treats blocked operations. It then identifies the construction’s interpretive and computational limits.

6.1 The State-Description Problem

A current observation can merge configurations that respond differently to a later operation. This subsection states the resulting problem and the information needed to repair it.

Let $o : S \rightarrow Y$ report an aspect of a configuration. Equality of o values defines an equivalence on S . That relation is sufficient for reporting the current observation. It can fail to support an update rule on observation values.

If $o(x) = o(y)$ while $o(T_b x) \neq o(T_b y)$, the same current observation would need two different successor observations under b . A deterministic update using only $o(x)$ is therefore impossible. The failure identifies information omitted by the observation.

A repair can retain the original configuration, but a smaller record may suffice. The desired relation should merge configurations only when every declared continuation leaves their observations equal. That criterion retains exactly the distinctions needed for the selected task.

The mathematical construction has an established antecedent in deterministic behavior semantics. Rutten’s final-system example uses observations indexed by finite input words [7, p. 44, Example 10.2(5)]. The present application uses that construction to audit state descriptions within the six-category framework.

6.2 Symbols and Continuation Signatures

The construction requires explicit operations and an observation. This subsection fixes their domains and the execution convention before stating the theorem.

Assumption 6.1 (Declared observation system). Let S be nonempty, let B be an operation alphabet, and let $T_b : S \rightarrow S$ be a total deterministic map for every $b \in B$. Let $o : S \rightarrow Y$ be an observation. Words execute left to right, with $T_\epsilon = \text{id}_S$ and $T_{bw} = T_w \circ T_b$.

Definition 6.2 (Continuation signature). For $x \in S$, define

$$\sigma_x : B^* \rightarrow Y, \quad \sigma_x(w) = o(T_w x), \quad x \equiv_o y \iff \sigma_x = \sigma_y. \quad (11)$$

An equivalence E is compatible with the task when $E \subseteq \ker o$ and $x E y$ implies $T_b x E T_b y$ for every $b \in B$.

The signature includes the empty word, so it retains the current observation. Positive words record possible later observations. Equality of signatures concerns the declared test system, rather than actual execution of every test.

The alphabet is also an input. Different labels can share a map, while an omitted operation can reveal a difference that the selected repertoire leaves invisible. The scope of the resulting equivalence must therefore name the operations used.

6.3 The Greatest Compatible Equivalence

The theorem derives the desired relation through four separate properties. This subsection includes the quotient maps so that the construction's dynamic use is explicit.

Proposition 6.3 (Greatest compatible equivalence). *The relation $\equiv_o$ is the greatest compatible equivalence under inclusion. Its quotient supports well-defined updates $[x] \mapsto [T_b x]$ and observation $[x] \mapsto o(x)$.*

Proof. First, equality of functions is reflexive, symmetric, and transitive, so $\equiv_o$ is an equivalence. Taking the empty word in signature equality gives $o(x) = o(y)$ whenever $x \equiv_o y$. Thus $\equiv_o \subseteq \ker o$.

Second, suppose $x \equiv_o y$ and fix $b \in B$. For every word w ,

$$o(T_w T_b x) = o(T_{bw} x) = o(T_{bw} y) = o(T_w T_b y).$$

Hence $T_b x \equiv_o T_b y$, proving update preservation.

Third, let E be any compatible equivalence and suppose $x E y$. Induction on word length gives $T_w x E T_w y$ for every word. The empty case is the assumed pair. The induction step uses preservation by the next operation. Since $E \subseteq \ker o$, the updated observations agree. Therefore $x \equiv_o y$, proving $E \subseteq \equiv_o$.

Finally, define the quotient update by $[x] \mapsto [T_b x]$. Preservation makes its value independent of the representative. Define the quotient observation by $[x] \mapsto o(x)$. Containment in $\ker o$ makes that value representative-independent as well. $\square$

Greatest under inclusion means coarsest as a partition. The theorem merges the largest set of pairs consistent with the specified responsibilities. It does not seek the most informative possible description.

A finer equivalence can retain additional history or identity information. A coarser one must lose either the current observation or compatibility with some declared continuation. The theorem identifies that precise boundary.

The result also supplies a positive response to underdetermination. The base alone leaves the observation and repertoire open, but once they are declared, the compatible quotient is determined by them. Its interpretive adequacy remains a separate question.

6.4 Inherited Distinctions and Generator Scope

A new observation task can coexist with an earlier distinction regime. This subsection derives the appropriate intersection and limits the scope of generator-based preservation.

Suppose the original equivalence $\sim$ is preserved by the same operations on the same carrier. Define

$$E^\dagger = \sim \cap \equiv_o. \quad (12)$$

The intersection is an equivalence because both factors are. Each operation preserves both factors and therefore their intersection. The relation also retains the current observation.

Any compatible equivalence contained in $\sim$ lies in $\equiv_o$ by Proposition 6.3. It therefore lies in the intersection. Thus $E^\dagger$ is the greatest compatible equivalence that also respects the inherited distinctions.

The construction uses the same carrier and operation system. An earlier equivalence need not remain preserved after new availability restrictions or new operations are introduced. Its compatibility must be checked in the revised model.

Preservation under B extends by induction to every word over B . To claim preservation under all of an ambient semigroup M , the declared maps must generate its action, or the remaining maps must be checked separately.

For a counterexample, take $S = \{0, 1, 2\}$ with $o(0) = o(1) = 0$ and $o(2) = 1$. If B contains only the identity, the continuation classes are $\{0, 1\}$ and $\{2\}$. An added map sending zero to zero and one and two to two splits the first class. The old quotient was valid for its old repertoire.

6.5 Finite Horizons and the Refinement Recursion

A finite horizon gives a bounded approximation to full continuation equality. This subsection derives its recursive form and the relation between horizons.

For $h \geq 0$, define

$$x \equiv_o^{(h)} y \iff o(T_w x) = o(T_w y) \text{ for every } |w| \leq h. \quad (13)$$

At horizon zero, this is $\ker o$. At the next horizon,

$$x \equiv_o^{(h+1)} y \iff o(x) = o(y) \text{ and } T_b x \equiv_o^{(h)} T_b y \text{ for every } b \in B. \quad (14)$$

To derive the recursion, separate the empty word from a nonempty word bw . The empty word tests current observation. The suffix w has length at most h after the first update. These conditions cover exactly the words of length at most $h + 1$.

The relations decrease under inclusion because a longer horizon imposes all the shorter horizon's tests. Their intersection over all finite horizons equals $\equiv_o$, since every finite word belongs to some bounded horizon.

A bounded relation need not itself be update-preserved. One initial operation can expose a distinction requiring the full remaining horizon. The stable relation, rather than an arbitrary bounded one, supplies the exact unbounded state quotient.

6.6 Finite Stabilization and Its Certificate

On a finite carrier, the refinement process reaches the full relation after a bounded number of strict splits. This subsection proves the bound and distinguishes attainment from certification.

Proposition 6.4 (Finite refinement bound). *If $|S| = n < \infty$ and $\ker o$ has r_0 classes, some $h \leq n - r_0$ satisfies $\equiv_o^{(h)} = \equiv_o$. Equality of consecutive refinement relations certifies stabilization.*

Proof. Every strict refinement of an equivalence on S splits at least one class. It therefore increases the number of classes by at least one. Starting from r_0 classes, at most $n - r_0$ strict increases are possible.

If consecutive relations agree, Equation (14) applies the same operator to the same input, so every later relation also agrees. The stable relation equals the intersection of all horizons, which is $\equiv_o$. A stable value must therefore be attained by the stated depth. $\square$

An implementation may compare the attained partition with one further stage to certify that it has stopped changing. The bound concerns the depth of attainment, while the additional comparison supplies the certificate.

A finite alphabet makes each refinement directly enumerable. The mathematical class-count bound can hold with an infinite alphabet, but an effective computation then needs a way to check the operation family.

The result concerns a specified model. Observing a finite number of traces from an unknown social system does not establish that its unmodeled transitions agree with the assumed maps. Model identification requires additional evidence.

6.7 Blocked Operations and Totalization

Partial availability can be represented through a fresh blocked state and observation. This subsection states that convention and examines the information it preserves.

Let T_b initially be partial with domain $D_b \subseteq S$. Adjoin a fresh state $\dagger$ and define

$$\widehat{T}_b(x) = \begin{cases} T_b(x), & x \in D_b, \\ \dagger, & x \in S \setminus D_b, \end{cases} \quad \widehat{T}_b(\dagger) = \dagger.$$

Extend the observation by $\widehat{o}(\dagger) = \perp$, where $\perp$ is outside the ordinary observation range.

The totaled system satisfies the theorem's map assumptions. A word executable from one configuration and blocked from another can now distinguish them through the fresh observation. Availability becomes part of the test output.

The absorbing convention records failure of the modeled execution. It need not describe a physical state from which recovery is impossible. A repairable failure requires a different transition or a richer failure record.

The construction uses free operation words. Equations between products in an earlier total semigroup may fail after domains are restricted. Their validity must be checked before the totaled word action is compressed through that presentation.

Observable blockage, silent failure, and an unattempted operation also carry different meanings. The observation should reflect the distinctions the inquiry can justify. A convenient marker is an explicit modeling choice.

6.8 Interpretive Limits and Category Use

The continuation quotient supplies a consistent state description for a selected task. This final subsection identifies how the six categories can use it and where their requirements remain independent.

An entity can persist while its continuation profile changes. Different entities can have the same profile under a coarse observation. The quotient therefore cannot replace a separate reidentification criterion.

Event occurrences and process histories can also be merged by behavioral equality. A repeated idempotent operation can leave the same future state while adding an occurrence. A historical query can require that distinction even when the predictive task does not.

Attributes can supply the observation or contribute to a family of observations. The quotient must preserve their values if those values belong to the task. Partial attributes also require a definedness criterion, developed in the attribute section.

The construction thus provides one shared tool within the category architecture. It derives dynamic consistency from declared observations and operations, while leaving identity, occurrence, history, evidence, and evaluation visible as additional commitments. The following sections develop those commitments for each category.

7 Entity Reidentification and Historical Continuity

This section develops the entity category through the problem of reidentifying a participant, organization, agreement, document, or other focal object across changing appearances. It begins with configurations and appearances, constructs an identity quotient under a declared criterion, and derives consequences for attributes and lineage. It then examines orbit-based alternatives, branching, behavioral similarity, relational individuation, and evidential warrant. The objective is an entity representation whose mathematical conditions and philosophical commitments remain separately assessable.

7.1 The Reidentification Problem

An entity description lets an inquiry connect claims made at different moments or in different settings. A researcher may ask whether the group that entered an agreement is the group that later exercised its permission, whether a revised document continues an earlier document, or whether a newly configured organization carries forward an earlier commitment. These questions require a way to join appearances while allowing some of their properties to change. A catalogue of currently visible features cannot settle the question by itself, because the significance of a changed feature depends on the criterion of continuity being used.

Consider a stipulated association that changes its meeting place, replaces several participants, and modifies its procedures. A study of its historical activity might follow it through all three changes. A study of the membership cohort could treat replacement of participants as a change of its focal object. A study of the meeting practice might group several otherwise different associations together because they enact the same procedure. These descriptions can be internally coherent while answering different questions. The inquiry needs to state what is being tracked and what a claim of sameness is intended to preserve.

The word *entity* marks a representational role in this paper. It provides a subject to which claims across appearances can be attached. This role is compatible with several positions about the metaphysical status of the represented subject. Ontological suspension keeps those positions available for later argument. It also prevents a mathematical equivalence class from receiving more philosophical authority than its construction supplies.

The distinction matters for the carrier S , which represents configurations of an inquiry. A configuration can describe many participants and conditions together. Conversely, one participant can appear in many configurations. Taking every $x \in S$ to be an individual would identify the overall situation with each object discussed within it. Taking every trajectory to be an individual would import another assumption about continuity. The appearance construction provides a separate domain on which the entity question can be stated explicitly.

7.2 Appearances, Contexts, and Identification

The construction introduces the records needed for the question before assigning identity classes. This order makes additional commitments visible and permits comparison between competing criteria.

Let I be a set of indexed appearances. An appearance is a position in the representation at which a focal object is presented or referred to. It can include a time, context, document reference, or observation source when the inquiry requires those data. The index distinguishes records that would otherwise have the same description. A map

$$p : I \longrightarrow S$$

associates an appearance with a represented configuration. Its mathematical role is modest: it connects entity records to the action framework. Several appearances may have the same image, and the image may contain information about several focal objects.

The appearance domain is itself a modeling choice. It can be constructed from positions in a documented history, versions in a repository, or a declared set of possible manifestations. Each source has a different epistemic status. A possible appearance supplies a place in a model; a recorded appearance

supplies documentary material; an evidentially warranted appearance supports a claim about the case. The mathematics can represent each, provided the interpretation of I remains clear.

Definition 7.1 (Declared entity criterion). Let E be an equivalence relation on I . The represented entity domain is $\mathcal{E} = I/E$, with projection $e : I \rightarrow \mathcal{E}$. The assertion iEj records that the inquiry reidentifies appearances i and j under its stated criterion.

Reflexivity assigns an appearance to its own class. Symmetry permits the same identification to be read from either appearance. Transitivity lets a chain of warranted identifications support a single class. These properties make a partition possible: classes either coincide or are disjoint. They also constrain interpretation. A weaker relation of influence or succession may fail symmetry or allow branching that the partition would obscure.

An equivalence can be supplied in several ways. A specified identifier may induce iEj when identifiers agree. A documentary reconstruction may establish pairwise links and assess whether they justify an equivalence closure. A formal model may stipulate classes for an illustration. The resulting partitions can have the same abstract form even when their evidential bases differ. Writing the relation as E records the criterion; its justification belongs to the accompanying argument.

A possible circularity becomes visible at this point. If the input is defined to contain all appearances of an already identified entity, identity has already been used to select the domain. The construction can organize those appearances, but presupposes their initial identification. A stronger reconstruction begins with indexed records whose grouping remains open, states admissible evidence for joining them, and examines the consequences. The framework accommodates either use and requires disclosure of which one is intended.

7.3 Invariant Descriptors and Changing Properties

The quotient provides a subject for invariant descriptions. The following result identifies the appearance-level values that can be assigned to it without dependence on the chosen representative.

Proposition 7.2 (Descent of an entity descriptor). *Let $d : I \rightarrow V$ be a descriptor. A map $\bar{d} : \mathcal{E} \rightarrow V$ satisfying $d = \bar{d} \circ e$ exists exactly when*

$$iEj \implies d(i) = d(j). \quad (15)$$

When it exists, it is unique.

Proof. Suppose first that $d = \bar{d} \circ e$. If iEj , then $e(i) = e(j)$. Applying $\bar{d}$ to the equal arguments gives $d(i) = d(j)$.

Conversely, suppose Equation (15) holds. For $C \in \mathcal{E}$, define $\bar{d}(C)$ to be the common d -value of its members. The class is nonempty, and the condition ensures that any two members give the same value. This proves existence and representative independence. Since $e(i)$ contains i , $\bar{d}(e(i)) = d(i)$.

Every class is the image of an appearance. Any map satisfying the factorization must send that class to its members' descriptor value. Two such maps therefore agree on all of $\mathcal{E}$, proving uniqueness. $\square$

A descriptor that changes within an identity class cannot be an invariant function of the class alone. It can remain an appropriate attribute of the entity at a specified time or context. A change of membership, location, or resources therefore need not defeat an entity criterion. It may show that the attribute's domain needs an extra coordinate.

Let $c : I \rightarrow C$ record context and consider $(e, c) : I \rightarrow \mathcal{E} \times C$. An attribute $a : I \rightarrow V$ factors through the image of this joint map precisely when

$$e(i) = e(j) \text{ and } c(i) = c(j) \implies a(i) = a(j).$$

The proof repeats the two directions above with (e, c) in place of e . The resulting value is assigned on represented entity/context pairs. Extending it to every conceivable pair requires further information. If

records disagree even at one such pair, the model can retain conflicting evidence, uncertainty, or a need for finer context.

This distinction blocks an indiscriminate inference from changed properties to changed identity. Such an inference needs a premise making the property identity constitutive. Conversely, an invariant identifier can continue while the object relevant to another inquiry has changed substantially. Identity and descriptive attributes constrain one another through explicit decisions. A convenient database key settles only the identity question built into that database.

Structural type and entity token also differ. A descriptor may assign the same procedural form to several entity classes. Equality of its value groups similar entities. Recovering token identity from the descriptor would require injectivity of $\bar{d}$, which the proposition never assumes. A classification can describe many objects efficiently while remaining insufficient to reidentify an individual object.

7.4 Forward Reachability and Orbit Proposals

The action framework supplies a natural family of reachable configurations. This subsection examines its contribution to continuity and the additional conditions needed for identity.

For a declared subsemigroup $N \subseteq M$, write

$$\text{Orb}_N(x) = \{nx : n \in N\}, \quad \text{Orb}_N^*(x) = \{x\} \cup \text{Orb}_N(x).$$

The first set collects results of at least one mechanism in N . The second retains the seed. If an identity action is available, the seed may already belong to the first set; otherwise its inclusion requires an explicit convention.

The associated reflexive reachability relation is a preorder. A zero-step continuation gives reflexivity. If $y = mx$ and $z = ny$, then $z = (nm)x$, so composition gives transitivity. These properties track where a configuration can lead. Symmetry requires a path back, which a semigroup action need not supply.

Example 7.3 (Overlapping forward closures). Let $S = \{u, v, w\}$ and let k send every point to v . Then $k^2 = k$, but

$$\text{Orb}_{\{k\}}^*(u) = \{u, v\}, \quad \text{Orb}_{\{k\}}^*(w) = \{w, v\}.$$

The closures intersect at v while remaining distinct.

Applying k once to either seed reaches v . Further applications stay there. The reflexive closures therefore contain their respective seeds and the common destination. Since u belongs only to the first and w only to the second, the closures differ. Equivalence classes, by contrast, coincide whenever they intersect. These forward closures fail to provide an identity partition.

Two organizations can be represented as reaching a common later configuration because the model retains only a shared outcome. Their common destination leaves several interpretations open: a merger, a coarse observation that discarded their distinction, or an actual identity claim supported by other evidence. The reachability pattern alone leaves the choice unresolved. An entity criterion needs information appropriate to its intended interpretation.

Calling N identity preserving can make the orbit a useful description of transformations undergone by an already identified entity. That description presupposes the identity judgment it might appear to derive. The construction is conditional: given a criterion and mechanisms that respect it, the orbit records reachable configurations. It can contribute to persistence analysis while leaving the criterion's justification visible.

Absence of inverses also leaves the identity question open. An injective forward shift can lack an inverse in the selected semigroup while retaining every difference between starting points. An idempotent operation can collapse distinct beginnings immediately. An entity's history can involve either behavior. Historical continuity, injectivity, reversibility, and identity preservation require separate conditions.

7.5 Directed Lineage, Fission, and Merger

Lineage records directional continuation between appearances and permits branching to remain visible. This subsection derives the constraints that arise when such continuity is identified with equivalence.

Let $L \subseteq I \times I$ record immediate continuation. Its reflexive transitive closure L^* connects appearances by finite lineage paths. An edge may represent organizational succession, document derivation, or another declared continuation. Its meaning remains narrower than a general identity claim unless identity is independently justified.

Proposition 7.4 (Branching consequence of equivalence). *If an equivalence E satisfies iEj and iEk , then jEk . Consequently, keeping two successors in distinct identity classes prevents both from belonging to the predecessor's identity class.*

Proof. Symmetry gives jEi . Transitivity with iEk yields jEk . If the successors occupy distinct classes, at least one proposed identification with the predecessor must be revised. $\square$

The result imposes a structural choice without prescribing a philosophical resolution of fission. An inquiry can treat a predecessor as ending and both successors as new. It can identify one successor as continuing under a privileged criterion. It can preserve shared descent in a lineage description while leaving identity undecided. Its statements must agree with the relation type chosen to represent them.

For example, take

$$I = \{r, u, v, u', v'\}, \quad L = \{(r, u), (r, v), (u, u'), (v, v')\}.$$

The partition $\{\{r\}, \{u, u'\}, \{v, v'\}\}$ treats the predecessor as ending and two successors as continuing separately. The partition $\{\{r, u, u'\}, \{v, v'\}\}$ privileges one continuing branch. Both accompany the same lineage. Choosing between them requires a criterion beyond the directed graph.

Automatic closure operations supply further alternatives. Mutual reachability under L^* is an equivalence: reflexivity and symmetry are immediate, and concatenating forward and reverse paths proves transitivity. In this acyclic example, however, distinct appearances lack paths back to one another, so the equivalence has five singleton classes. Symmetrizing the edges and taking transitive closure gives one connected component containing all five appearances. The first construction can be too fine for continuity, the second too coarse for distinguishing branches. Mathematical availability alone leaves their adequacy unsettled.

Merger produces a corresponding issue. If two earlier entities are both identified with a common successor, equivalence identifies them with each other. An inquiry that distinguishes them before merger needs to differentiate lineage from identity. It can record the successor's multiple origins without erasing the earlier distinction. The substantive question is what continues: organization, membership, resources, commitments, or a declared combination.

Temporal indexing makes the disagreement articulable. Investigators can agree that a successor inherits something from both predecessors while disagreeing about whether either predecessor continues numerically. Lineage records inheritance directions. Identity records the stronger claim of sameness. Their coordination preserves the disagreement for argument and evidence.

7.6 Behavioral Equivalence and Entity Tokens

Continuation signatures characterize agreement under specified tests. This subsection separates that result from reidentification across appearances and explains how the two tasks can be coordinated.

Section 6 identifies configurations when every declared continuation produces the same observation. Equal signatures justify shared treatment for that prospective observation task. They can support a behavioral type or compressed predictive state. Whether the records concern one entity remains an additional question.

Two stipulated devices can have identical update rules and conditions while remaining separately indexed. Two document copies can have matching content and editing possibilities. Two associations

can occupy the same position in a coarse procedural model. A token-sensitive inquiry can preserve their differences through appearance or provenance records. Behavioral agreement supplies no rule for discarding those records.

The converse is equally significant. An entity can continue under a criterion while its signature changes. An association can gain permission, acquire resources, or lose facility access. If these alterations affect a continuation observation, the signature changes, even when the chosen identity criterion permits the changes. Requiring signature invariance would impose a special persistence criterion under which such changes terminate the entity.

Use separate maps to coordinate the questions. The identity projection e groups appearances by continuity. A behavior map β sends an appearance to the signature of its configuration $p(i)$. The pair (e, β) retains the entity label and behavior at each appearance. It can represent a continuing entity with changed behavior, or different entities with matching behavior, without requiring one map to determine the other.

Dennett's discussion of patterns connects descriptive usefulness with compression and prediction [1]. This comparison motivates attention to what a representation preserves. The present construction adds an independent reidentification task and leaves the metaphysical conclusions of that discussion open.

A stable pattern can contribute evidence for continuity. Making it sufficient requires explaining why this pattern, at this resolution and over this interval, settles the entity question. An account must also address equally predictive descriptions that support different token groupings. The distinction between identity and behavior exposes that obligation while allowing pattern-based theories to supply a further argument.

7.7 Relational Individuation and Boundaries

An entity can be tracked through relations to other objects and conditions. This subsection evaluates that strategy without concealing its dependence on jointly specified records.

A relational description can identify an appearance through participation, commitment, dependence, or derivation. A document's relation to an earlier version can distinguish it from a matching document with another history. An association's commitments and recognition by participants can support continuity. These are proposed forms of evidence whose relevance depends on the target.

Such descriptions encounter mutual dependence. Identifying an entity through its relations requires identifying the relations and their other participants. Identifying a relation through its participants requires tracking those participants. A joint representation can constrain these components together. The distinction is between a coherent joint specification and a claim that one component is uniquely derived from data that already presuppose it.

One approach begins with provisional appearance indices and relational records, then assesses candidate assignments of identity classes and incidence relations. A candidate fails when a required predicate becomes representative-dependent or when its grouping collapses evidentially required distinctions. Several candidates may remain. The result is then a family of admissible reconstructions. Later evidence can narrow the family without pretending that the initial formulation already settled identity.

Boundary selection changes the task. The focal object can be a participant, a group, a working arrangement, or a system including resources and procedures. An alteration internal to one boundary becomes an interaction across another. The action model can represent either choice through a suitable carrier and update maps. Explanatory relevance and evidence must supply reasons for preferring a boundary.

This permits a disciplined plurality of descriptions. Different questions can use different boundaries while each description states what its identity claims preserve. Grounds for criticism remain concrete: descriptor inconsistency, unsupported continuity links, loss of a required distinction, or failure to represent the subject of the inquiry. The availability of several representations therefore leaves substantial room for disagreement and correction.

7.8 Continuity, Potential Efficacy, and Explanation

The motivating intuition combines reidentifiable continuity with the capacity to matter in subsequent activity. This subsection evaluates these components separately before discussing their conjunction.

Continuity relates appearances. Potential efficacy concerns a difference under a continuation or comparison. A stable identity record can be irrelevant to the chosen future observations. A configuration difference can affect those observations while giving little guidance about which entity persisted. The two conditions therefore need distinct representations and warrants.

For a proposed efficacy test, take a defensible comparison (x, y) and a nonempty continuation w . The inequality $o(T_w x) \neq o(T_w y)$ shows that the model distinguishes the configurations under that test. Attributing the contrast to an entity requires an account of how the comparison isolates that entity or a relevant condition. Other differences in resources, rules, or participants leave the source of the effect underdetermined.

Dormant efficacy remains meaningful under an existential test. A permission-bearing organization can have a modeled route to activity that is never exercised. Its absence from an observed sequence leaves the untested potential unresolved. An interpretation depends on admissible continuations, persistence of enabling conditions, and the comparison's empirical warrant. The relation category and the companion historical paper develop those requirements in more detail.

The source intuition also links continuing appearance with withholding a commitment to independent self-subsistence. Here it functions as a research proposal. Its mathematical content concerns tracking continuity and effects through explicit records. A conclusion about the ultimate constitution of the object requires a philosophical argument beyond those records and their algebraic properties.

Historical explanation introduces further information. Explaining a present capacity requires evidence about its formation and transformation. A current identity label or signature can omit those histories. Two organizations with the same present model may have acquired capacity through different sequences. The difference can matter to a historical, causal, or normative question. Keeping current efficacy separate from formation history preserves the possibility of asking those questions.

7.9 Evidence, Revision, and Disagreement

A usable entity representation includes a way to assess its identification claims. This subsection identifies evidence problems left open by the formal construction.

An appearance record can accurately report a local feature while misidentifying its subject. Matching names need not identify one organization; different names can belong to a continuing organization. The evidence task therefore concerns both record reliability and the inference linking records. A naming convention can support that inference when its reliability is justified in the case.

Conflicting records illustrate the distinction between identifier and warrant. Attach sources to appearances and preserve competing identity assignments. The inquiry can then distinguish conclusions shared by all admissible assignments from conclusions dependent on one reconstruction. A definitive identifier assigned too early can conceal disagreement and make downstream calculations appear more certain than their premises.

Revision also differentiates a claim from a record of that claim. Evidence may justify regrouping appearances while the fact that a different grouping was previously accepted remains historically relevant. Preserving both supports inquiry into changing recognition without asserting that the earlier appearances were rewritten. Occurrence, recording, and later interpretation retain their separate roles.

An operational assessment can ask what would justify joining two classes, what would require splitting one, how the criterion handles a changed name or a gap in records, and whether it distinguishes succession from continuation when needed. It should also ask whether the criterion forces attributes to remain invariant that the inquiry needs to examine as changing. Each answer requires domain-specific decisions and a way to retain unresolved cases.

Disagreement then becomes more precise. An investigator can challenge the evidence, the criterion, the appearance domain, or the fit between the criterion and the explanatory purpose. These objections call

for different revisions. An unqualified claim to have derived entity identity would collapse distinctions that the formal result leaves open.

7.10 Coordination with the Other Categories

The entity construction supplies one part of the six-category framework. This subsection states how that part connects to the others and what a successful application would establish.

Entity classes can support context-sensitive attribute descriptions. Appearances can participate in relational incidence records. Event tokens can index occurrences involving those appearances. Process histories can connect successive configurations, while a state description retains conditions relevant to a declared prediction. These links let an inquiry ask several questions about a continuing subject without forcing every answer to follow from one quotient.

The formal tasks remain distinguishable. Equivalence supplies a partition. Descriptor descent identifies invariant class-level information. Lineage preserves direction and branching. Signatures retain agreement under chosen tests. Evidence supports or challenges their interpretation. A successful application coordinates those tasks and demonstrates why its criterion serves a concrete question.

Stronger philosophical theories of identity, persistence, and selfhood can propose privileged criteria or criticize this separation. Their conclusions enter through argument. The present constructions expose their consequences and counterexamples while keeping the original inquiry visible.

8 Historically Formed Relations and Potential Effects

This section develops the relation category through three connected questions: how participants are represented, how a dependence is historically formed, and how it can matter for later activity. It first constructs contextual incidence, then examines the base-definable order predicate and a comparative potential-effect test. The philosophical discussion concerns relational individuation, dormancy, persistence, and the evidence needed for causal interpretation. The proposed account remains a focal research criterion rather than a definition of every mathematical or social relation.

8.1 The Focal Relational Question

A mathematical relation can be any subset of a product of domains. That broad usage permits precise representation of many patterns, including similarity, order, participation, and dependence. The research question here is narrower. It concerns a relatively reidentifiable dependence that has a formation history and can affect subsequent activity.

The motivating insight joins two directions of inquiry. Relations are historically formed, and they can participate in further historical formation. The first direction asks how an arrangement came to obtain. The second asks which later possibilities depend on its presence, alteration, or removal. The two directions can interact while requiring different evidence and formal records.

A stipulated agreement illustrates the distinction. Its participants can be represented through appearance indices. Its formation can be represented by a history of proposals and authorization. Its present role can be represented by a permission condition. Its potential effect can be tested through a continuation whose availability or result depends on that condition. None of these components alone supplies the entire relational account.

The category also leaves evaluation open. A dependence can enable cooperation, sustain an unwanted arrangement, constrain a participant, or preserve an existing practice. Potential efficacy identifies a role in the declared dynamics. An appraisal requires further premises about purposes, interests, and justification. Generative and reproductive roles can each have several evaluations depending on the case.

8.2 Participants and Contextual Incidence

A social relation needs a representation of its participants or conditions. This subsection constructs incidence and derives the requirement for describing it at an entity level.

Let I be the appearance domain and let E be a declared identity equivalence, with projection $e : I \rightarrow \mathcal{E}$. For a fixed finite arity n , let

$$R \subseteq I^n$$

record an incidence predicate. The tuple positions can have different roles, such as issuer, recipient, or object of an authorization. Binary notation is a special case; the framework imposes no requirement that every relation have two participants.

Suppose the inquiry wants a predicate on entity classes, $\bar{R} \subseteq \mathcal{E}^n$. The intended exact descent condition is

$$R = (e^n)^{-1}(\bar{R}).$$

This says that the appearance predicate can be recovered from the entity-level predicate without losing its truth conditions.

Proposition 8.1 (Incidence descent). *Such a predicate $\bar{R}$ exists exactly when coordinatewise replacement by E -equivalent appearances preserves membership in R .*

Proof. If R is the displayed inverse image, coordinatewise equivalent tuples have equal e^n projections and therefore equal membership truth.

Conversely, assume replacement invariance. Define $\bar{R}$ on a tuple of entity classes by the R -value of any representative tuple. The assumption makes the value independent of representatives. Pulling the predicate back then recovers R . $\square$

The condition can fail for historically changing participation. One appearance of an entity may belong to an agreement and a later appearance may follow its withdrawal. An entity-only predicate would assign one value to both appearances. Retaining a time or context argument permits the relation to change while the entity criterion remains fixed.

Existential projection is another option: an entity tuple is related when some representative tuple belongs to R . This always supplies a quotient-level predicate, but its pullback can be larger than R . It changes the statement from participation at this appearance to participation at some appearance. The difference matters to claims about present authorization or historical membership.

Incidence therefore performs one clear task: it identifies participation at a declared level. Historical formation and potential effect require additional structures. An incidence table can record who stands in a relation while leaving how it arose and what it enables unresolved.

8.3 Relation Tokens and Relational Types

A relation can persist, be renewed, or recur between the same participants. This subsection distinguishes a general relational type from a particular historically tracked arrangement.

A predicate such as participation in an agreement can classify many arrangements. Two agreements between the same parties may have different formation histories, scopes, or durations. Conversely, one continuing arrangement may change some terms while remaining reidentified under a chosen criterion.

An occurrence or relation-record index can retain those distinctions. Let Λ index proposed relational arrangements and attach participant, context, and formation records to each index. A classification map can send several indices to the same relational type. The map's image retains the type while its domain retains the separate arrangements.

The indexing does not establish their identity by itself. Reidentifying one arrangement through renewal requires a criterion, just as reidentifying an entity does. The criterion might concern continuing commitments, a declared institutional record, or another warranted feature. The model can represent the criterion and test its consequences without treating it as self-evident.

A relation can also change arity or participant roles in an extended model. Fixed-arity incidence then captures only one stage or one selected description. A more general record can retain a role-indexed participant assignment. That extension adds flexibility while introducing new questions about comparison across stages.

These distinctions prevent a relational type from absorbing its historical tokens. Two arrangements can share an extensional participant pattern and differ in provenance. A present type description can support one prediction while leaving a historical question unanswered. The historical-relations paper develops that record-sufficiency problem directly.

8.4 Order Effects in the Abstract Base

The base action supplies an exact predicate for order-sensitive effects. This subsection derives its scope and tests a broader relational reading.

For $m, n \in M$ and $x \in S$, define

$$R_{\text{ord}}(m, n, x) \iff (nm)x \not\sim (mn)x. \quad (16)$$

The predicate compares two orders from one seed under the same equivalence. Its symbols belong to the base language, so it is explicitly definable there.

A simple numerical example uses $a(x) = x + 1$ and $b(x) = 2x$ with ordinary equality. Then $ba(x) = 2x + 2$ and $ab(x) = 2x + 1$. The order predicate holds at every input. The calculation establishes order dependence without assigning a particular social meaning to either map.

Commuting operations give a complementary countercondition. The translations $a(x) = x + 1$ and $c(x) = x + 2$ both affect the same coordinate, yet $ca(x) = ac(x) = x + 3$. A shared target of action therefore need not produce noncommutativity. Likewise, two prerequisite-setting operations can commute while jointly enabling a later activity.

The predicate is thus neither a necessary nor a sufficient criterion for a particular social relation. It can be a useful witness when the interpretation of the maps and outcome distinction is independently supplied. It leaves friendship, authorization, dependence, or another substantive meaning open.

The locality in x also matters. Two operations can commute at one configuration and differ at another. A global statement quantifies over the carrier; a local witness concerns one selected seed. An empirical comparison must identify which configurations are relevant and accessible rather than treat a local result as a universal law.

8.5 Historical Formation and Present Representation

A relation's formation history can be compressed into present conditions for some tasks. This subsection explains why that possibility preserves the historical question rather than settling it.

Let a word $w = b_1 \cdots b_k$ represent a candidate formation sequence from a seed. Its endpoint can contain a record that the relation currently obtains. Another word can produce the same endpoint. The present state can therefore be sufficient for a future operational prediction while failing to recover which formation sequence occurred.

Consider two labels n and i with identical maps that set an authorization coordinate to 1. They can be interpreted in a stipulated example as negotiated and imposed formation. Their effects agree on the chosen carrier while the labels retain different histories. A calculation using only the common endpoint cannot recover the label.

This is an underdetermination argument. Keep the base data and endpoint fixed while varying the formation record. Any function of the fixed data returns the same value in both cases. A target query that distinguishes the histories therefore requires richer input. The result identifies a representational limit; it does not claim that formation history is always inaccessible.

Historical formation also differs from dependence on information omitted by the present state. A sufficiently rich current state can encode every condition needed for the selected future task. The ar-

rangement still has a history. A statement that it was historically formed therefore does not imply failure of a present state description.

The distinction supports two legitimate research directions. One can ask which historical information must be retained for future updates. One can ask what happened in the formation of an arrangement even when that information is redundant for the selected prediction. A comprehensive relational model keeps both directions available.

8.6 Potential Effects under Declared Continuations

Potential efficacy concerns a comparison of possible futures. This subsection states the relevant domains and explains the positive-length condition.

Fix total operation maps T_b , an observation $o : S \rightarrow Y$, and a set $\mathcal{C} \subseteq S^2$ of admissible comparison pairs. For a word w , write $\sigma_x(w) = o(T_w x)$.

Definition 8.2 (Comparative future-effect witness). For $(x, y) \in \mathcal{C}$, a nonempty word $w \in B^+$ is a future-effect witness when

$$o(T_w x) \neq o(T_w y). \quad (17)$$

Potential efficacy under the declared tests requires at least one such witness.

The positive-length requirement separates a present distinction from a future effect. Let the observation distinguish two current states, while the only operation resets both to one common state. The present observations differ, but every nonempty continuation agrees. A definition using only current inequality would count a different property.

The condition is existential. It leaves open whether the witness is ever enacted. It also permits other continuations to erase or bypass the distinction. Requiring different outcomes under every continuation would impose a stronger criterion and exclude many dependence patterns that can matter under some relevant circumstances.

The comparison set $\mathcal{C}$ carries substantive information. It identifies which alternative configurations can be compared meaningfully. A pair that differs in many respects can witness a predictive difference without isolating the relation under study. A causal interpretation needs a justified design for the contrast.

The operation repertoire and observation likewise constrain the result. A test can fail to reveal a difference because its observation is too coarse, its repertoire omits the relevant continuation, or the relation has no effect under those declared conditions. The formal witness separates these possibilities only when the model retains the relevant distinctions.

8.7 Dormancy, Erasure, and Availability

A relation can affect access to a route or remain unused throughout an observed history. This subsection examines those cases through the declared comparison.

A permission can be represented by a condition that enables an activity. Comparing otherwise specified configurations with and without that condition can yield a distinguishing continuation. The permission's effect remains potential if the activity is never attempted. The absence of an observed enactment leaves that potential unresolved.

An erasing operation can remove the condition before the distinguishing test. The resulting actual sequence may therefore show no difference. This is compatible with an earlier potential witness. The interpretation must specify the time at which the comparison is made and the conditions that persist during the test.

Availability can itself be the relevant difference. For partial operations, let $W(x)$ contain the words whose every prefix is executable from x . Use a fresh marker $\perp$ and define

$$\hat{\sigma}_x(w) = \begin{cases} o(T_w x), & w \in W(x), \\ \perp, & w \notin W(x). \end{cases}$$

A word executable in one configuration and blocked in another gives a signature difference even when every commonly executable word has matching observations.

The marker records a testing convention. It can represent modeled unavailability, not an ordinary social outcome. A physical failed attempt or observable refusal requires the corresponding transition or occurrence record. Comparing these interpretations is part of the model's adequacy audit.

An inquiry that considers only the intersection $W(x) \cap W(y)$ omits availability differences. That restriction can be appropriate for comparing outcomes under shared opportunities, but it changes the question. A relational account of access needs to preserve the difference between available and unavailable continuations explicitly.

8.8 Relational Persistence and Change

A historically formed relation can persist while some of its participants, terms, or effects change. This subsection identifies the criteria needed to describe that continuity.

Persistence is a reidentification claim. It asks whether appearances of an arrangement belong to the same tracked relation under a declared criterion. Repeating an incidence pattern can support that claim while leaving open whether the arrangement was renewed, replaced, or independently recreated.

A relation's current extension can also change while its institutional or practical continuity is retained. A group can admit a new participant or revise a permission. Whether the result is the same relation, a modified relation, or a new arrangement depends on the criterion of tracking.

The model can keep these decisions visible through relation-record indices, contextual incidence, and histories. A single invariant predicate may be too coarse when the inquiry needs changing terms. A fresh relation index after every update may be too fine when continuity is central. The appropriate representation needs an argument about the kind of persistence at issue.

Potential effects can persist, vanish, or become observable under a changed repertoire. These are different claims from relation identity. One continuing agreement can lose an operational effect; two distinct agreements can have matching effects. A comprehensive account therefore tracks relation continuity and prospective behavior through separate components.

8.9 Causal Interpretation and Evidence

An observational difference is a formal witness. This subsection explains the further work needed to interpret it causally or historically.

The comparison in Equation (17) shows that the model distinguishes two configurations after the same word. Attributing the difference to one relation requires assumptions that isolate its role. Other differences between the configurations can account for the observation.

The historical question likewise needs evidence. A present authorization record may be compatible with several formation histories. Naming a history in the model makes it available for analysis, but the claim that it occurred depends on records, testimony, or other appropriate sources.

These obligations can interact. A formation history may explain why a present condition has the efficacy assigned to it. A present effect can motivate investigation of the history. Neither direction licenses an automatic inference. The account must state which premises support the transition between historical and prospective claims.

Counterfactual tests can also depend on the stability of other conditions. Removing one relation may alter participants, rules, or resources rather than leave everything else fixed. A proposed comparison must explain why its alternative configurations are admissible for the causal question. The set $\mathcal{C}$ records the choice; evidence and theory justify it.

The formalism remains useful under this restraint because it exposes the required comparison. It distinguishes a mathematical difference, a historical reconstruction, and a causal attribution. An application can then identify which parts are established and which remain hypotheses.

8.10 Relational Explanation and Philosophical Scope

The relational category gives dependence a central explanatory role while retaining the conditions of that explanation. This subsection examines its philosophical implications.

A relation can help explain why a participant has a capacity in one context and lacks it in another. The explanation locates the difference in a structured arrangement rather than treating the capacity as an invariant isolated attribute. The model supports that move when its variables and comparisons represent the arrangement adequately.

The explanatory direction need not deny that participants have other properties. The same configuration can include individual conditions and relational conditions, with their interaction specified by the update rules. The question is which distinctions are required for the phenomenon being studied.

Relational individuation can also shape entity descriptions. An organization's commitments and dependencies can contribute to its reidentification. The entity and relation criteria may then be jointly specified. Coherence of the joint specification is a mathematical and conceptual requirement; uniqueness and empirical warrant require further argument.

The present account suspends the claim that relations are metaphysically prior to all relata. It permits a research program to investigate how represented units emerge or become identifiable through activity and dependence. Such an investigation must show its constructions and evidence rather than assume the desired priority in its choice of primitives.

The distinction preserves the original insight about historical formation while making it testable. A relation can be investigated through its formation record, continuity, incidence, and possible effects. Each component can be challenged, refined, or found inadequate without forcing a decision about the entire ontology.

8.11 Relation Use in the Coordinated Framework

The relation category connects participants and conditions across historical and prospective questions. This subsection states what a successful application needs to coordinate.

Entity appearances provide participant references. Events identify formation, exercise, renewal, or withdrawal occurrences. Attributes describe terms and conditions. State descriptions retain present dependencies. Process histories record how the arrangement developed. The relation account joins these records to a declared potential-effect comparison.

A successful model must specify its incidence domain, relation identity criterion, formation query, observations, operation repertoire, and admissible comparisons. It must explain which parts are given, which are constructed, and which are evidenced. Where current effects and historical identity diverge, it must preserve both questions.

This makes the relational account comprehensive at the level of its intended task. It combines a formation question with a future dependence question while retaining their distinct formal and evidential requirements.

9 Event Occurrences and Transition Descriptions

This section develops the event category by distinguishing a possible transition, an occurrence, and a record of that occurrence. It constructs the required maps, derives the loss of multiplicity under effect compression, and examines how exact counting changes an idempotent model. The later discussion concerns attempts, event boundaries, temporal resolution, and evidential revision. The objective is to represent a happening without allowing a formula for its possible effect to substitute for the claim that it occurred.

9.1 Happenings within an Action Framework

An action model states what a mechanism would do at a represented configuration. Given $x \in S$ and a mechanism $m \in M$, the expression mx identifies the modeled result. The triple (x, m, mx) therefore describes a candidate transition. The same base is compatible with different occurrence histories: the mechanism may be applied once, repeatedly, or remain unused. The distinction between those histories requires information beyond the action map.

Social inquiry often needs precisely that additional information. A permission may be issued, renewed, refused, or merely proposed. Several meetings may enact the same procedure. Two attempts may have matching outcomes while differing in timing, participants, and significance. A model that retains only the input and final condition can answer an effect question while leaving those occurrence questions open.

The occurrence distinction remains relevant even in a wholly stipulated example. A simulation can declare a sequence of executed labels and count its positions. A counterfactual analysis can declare several possible sequences while selecting none as actual. An empirical study can reconstruct a sequence from records. The same word notation can appear in all three settings, so its epistemic role must be stated.

An event category accordingly requires a criterion of occurrence and a criterion of individuation. The first concerns whether the happening belongs to the selected history. The second concerns whether two descriptions refer to one happening or to separate happenings. A transition formula contributes a consistency condition for the modeled effect. It leaves occurrence and individuation available for their own evidence and argument.

9.2 Occurrence Indices and Record Maps

The following construction places occurrence information alongside the base dynamics. Its domain carries token distinctions while its image records the transition description.

Fix an operation alphabet B with maps $T_b : S \rightarrow S$. Let Ω be an index set of represented occurrences. Write

$$d : \Omega \longrightarrow S \times B \times S, \quad d(\omega) = (x_\omega, b_\omega, x'_\omega).$$

The middle coordinate identifies the represented operation. The first and third identify the initial and resulting configurations.

Definition 9.1 (Consistent execution record). An occurrence record d is execution-consistent with the declared total maps when

$$x'_\omega = T_{b_\omega} x_\omega \quad (\omega \in \Omega). \quad (18)$$

An occurrence criterion separately specifies what membership in Ω means for the inquiry.

Equation (18) can be checked by substitution. It says that the recorded transition agrees with the action model. Agreement establishes internal consistency. Establishing an actual event requires a warranted interpretation of the occurrence index and its source records. A model can consistently describe a sequence that was never enacted.

The map d need not be injective. Distinct occurrence indices can map to the same triple. This allows repeated executions to remain distinct even when their recorded conditions and effects match. Conversely, one occurrence can have several records. To represent that distinction, introduce a record domain $\mathcal{R}$ and, when warranted, a map

$$r : \mathcal{R} \longrightarrow \Omega.$$

Several documents can then refer to one occurrence. Their multiplicity belongs to $\mathcal{R}$; event multiplicity belongs to Ω .

The map r may itself be uncertain. A practical inquiry can retain candidate assignments or a relation between records and possible occurrences. Forcing a total definitive assignment would assert that each record has already been identified with a particular happening. A representation of uncertain evidence should expose that assumption or use a suitably partial or set-valued relation.

Times, participants, locations, and source references can be attached to ω through further maps. Each supplies a possible basis for distinguishing occurrences, but none is universally sufficient. Matching timestamps can accompany simultaneous distinct events. Different timestamps can record different stages or reports of one event. A participant list can remain unchanged across repeated enactments. The occurrence index provides a formal place to retain the intended distinction; evidence must justify its use.

9.3 Transition Types and Event Multiplicity

The image of the occurrence map classifies represented transitions. This subsection derives the information lost when only that image is retained.

Suppose a finite occurrence domain contains two distinct indices ω_1, ω_2 with

$$d(\omega_1) = d(\omega_2) = (x, b, x').$$

The image contains one transition description, while the domain contains two occurrences. A second record system containing only ω_1 has the same image. The two systems therefore agree about which transition types are represented and disagree about how often the transition occurred.

Proposition 9.2 (Multiplicity loss under image reduction). *The set $d(\Omega)$ alone generally fails to determine the cardinality of a finite occurrence domain Ω or the multiplicity of a transition description within it.*

Proof. Choose any consistent transition triple t . A one-element domain and a two-element domain can each map constantly to t . Both images equal $\{t\}$, whereas their domain cardinalities and the multiplicity of t differ. Any function of the image alone would have to assign the same answer to both systems. It therefore cannot recover both counts. $\square$

A multiset representation can preserve more information than an ordinary set image. For a finite domain, define

$$n(t) = |\{\omega \in \Omega : d(\omega) = t\}|.$$

The multiplicity function retains the count of each represented type, while still discarding the identity and other attributes of its individual occurrences. If timing or responsibility matters, the occurrence records remain necessary. There is consequently a sequence of increasingly coarse descriptions: indexed records, type multiplicities, and the set of represented types.

These reductions suit different questions. A catalogue may need only which procedures have ever been used. A workload study may need how often each procedure was used. A historical analysis may need who participated in particular enactments and in what order. The appropriate representation follows from the question. Calling each of them an event description without specifying its retained information would make their answers appear interchangeable.

The proof also shows why an elaborate notation can conceal a simple loss. Adding a symbol for the set of events to an endpoint model does little if the symbol is defined only as a set of transition types. The occurrence domain has to carry the distinctions that later questions require. Those distinctions may be supplied through recorded sequence positions, independent identifiers, or another justified indexing rule.

9.4 Repeated Applications of an Idempotent Operation

Idempotence provides a particularly sharp test because it identifies repeated effects while leaving repeated applications conceptually available. This subsection derives the incompatibility and constructs an explicit repair.

Suppose $T_b^2 = T_b$. Applying b twice has the same final effect as applying it once. A flag-setting operation is a simple example: $T_b(z) = 1$ on a Boolean carrier. One application and two applications both leave the flag at 1.

Let words retain application positions, with b^k denoting k applications. The count $c(b^k) = k$ distinguishes b and bb . Their effect representation identifies them:

$$T_b = T_{bb}, \quad c(b) = 1, \quad c(bb) = 2.$$

A count function of the total effect alone would therefore receive equal arguments and need to return different values.

Proposition 9.3 (Counting and idempotent compression). *If a product or effect representation identifies b with bb , exact application count fails to factor through that representation. On $S \times \mathbb{N}$, the update*

$$\tilde{T}_b(x, n) = (T_b x, n + 1) \tag{19}$$

retains application count and projects to the old effect, but it fails the idempotent equation.

Proof. For the first statement, a factorization $c = \bar{c} \circ e$ would imply $c(b) = c(bb)$ whenever $e(b) = e(bb)$. The counts 1 and 2 contradict that implication.

For the extension, apply Equation (19) twice:

$$\tilde{T}_b^2(x, n) = (T_b^2 x, n + 2) = (T_b x, n + 2).$$

One application gives $(T_b x, n + 1)$. The natural-number coordinates differ, so $\tilde{T}_b^2 \neq \tilde{T}_b$. Finally, projection pr_S satisfies

$$\text{pr}_S \circ \tilde{T}_b = T_b \circ \text{pr}_S,$$

showing that the visible effect on S is preserved. $\square$

The repair changes the information and the algebra together. Freely composed words act consistently on the enlarged carrier because each application increments the counter. Imposing the old equation $b^2 = b$ on the enlarged transformations would erase that count. The model therefore needs to distinguish an effect semigroup from a history-sensitive action. Each is coherent, and their connection is the projection that forgets count.

This example gives a general methodological lesson about enrichment. Adding a variable is mathematically consequential when the variable records distinctions previously identified by an algebraic law. A proposed enrichment should be checked against the old equations, rather than assumed to preserve them automatically. The same issue arises for duration and intermediate-state records in the process category.

The counter also remains interpretation-relative. It can count attempts, successful completions, or documented reports, depending on which updates increment it. An attempt counter and a completion counter need different update rules when an action can fail. Repeated reports can inflate a documentary counter without changing an occurrence counter. The equation records exactly what the increment rule specifies and leaves its empirical warrant open.

9.5 Quotient-Silent Events

A state description can treat an occurrence as silent when it leaves the selected observation unchanged. This subsection explains why that classification is a property of the description and event criterion.

For a quotient map $q : S \rightarrow Q$, a transition is silent at that resolution when $q(T_b x) = q(x)$. Silence means that this quotient records the same class before and after the operation. Another observation can distinguish the configurations, and an occurrence index can distinguish the application even when the carrier point itself remains fixed.

A stipulated renewal illustrates the issue. Let a Boolean flag indicate whether a permission is currently recorded. A renewal applied while the flag is 1 leaves it at 1. If the inquiry concerns current permission alone, the transition is silent. If it concerns how often the agreement was renewed, who endorsed the renewal, or when its evidential record was updated, the occurrence can be central. Those questions require additional observations or occurrence records.

A failed attempt can likewise matter despite an unchanged outcome. The model may record that a participant tried to activate a mechanism, that a prerequisite was absent, and that the selected outcome remained unchanged. The attempt, refusal, and unchanged state have distinct descriptive roles. A definition that counts only changed quotient outcomes selects a narrower event category. It should state that selection explicitly.

The converse also holds: a change in recorded state can reflect a revised observation rather than a newly occurring event in the target history. If an investigator corrects an earlier record, the research database changes. The target event may have occurred earlier. A model of the investigation and a model of its subject then have different carriers and update meanings. Their alignment requires explicit maps or provenance relations.

This distinction is philosophically important because event significance often depends on the question under which a happening is described. The mathematics can show that an update is silent relative to a map. It cannot establish that the occurrence is insignificant in every respect. Equally, adding a counter can make every application visible, but visibility alone supplies no criterion of social importance.

9.6 Attempts, Blocking, and Completion

An event representation needs a clear convention when an operation is unavailable or its attempt fails to achieve the target. This subsection compares three modeling choices.

A total map can encode a failed attempt by leaving the configuration unchanged. This is appropriate when the intended rule specifies a self-loop under unmet prerequisites. The self-loop remains an executed application in the model, even though the selected state is unchanged. An occurrence record or attempt counter can retain it.

A partial map can encode successful executability through its domain. For x outside that domain, the operation has no modeled result. This convention expresses unavailability, and it leaves the effect of attempting the unavailable operation unspecified. An application that needs to discuss refusal, discovery of blockage, or sanctions for an attempt must supply an additional transition or record type.

A totalized partial map introduces a special blocked marker. That marker can distinguish availability failure from an ordinary outcome. Its interpretation requires care: it may record a failed test, a terminal analysis state, or an observable refusal. Each meaning places different demands on subsequent dynamics. An absorbing marker, for example, says that later modeled operations remain blocked after the failure. That is a substantive testing convention whose adequacy must be justified for the question.

These representations can agree about successful routes while differing about failed attempts. A comparison between models should therefore identify which event class is being compared. Counting successful applications in a partial model and counting every attempted label in a totalized model can produce different totals without any mathematical inconsistency. The counts answer different questions.

A comprehensive event record can declare an outcome type such as completion, refusal, interruption, or unresolved attempt. These labels remain proposed coding choices. Their relation to observed evidence, institutional categories, and the modeled state transition must be specified separately. Adding a label for success does not prove that the intended target was achieved.

9.7 Event Boundaries and Temporal Resolution

The beginning, end, and granularity of an event determine its occurrence domain. This subsection examines how changing those boundaries affects counting and interpretation.

A meeting can be represented as one event, a sequence of agenda events, or a process containing many individual contributions. The choice depends on what the inquiry needs to distinguish. A process model retains intermediate states; an event-level compression may retain only its overall transition. The two can be coordinated by a segmentation map that groups finer occurrences into coarser units.

Such grouping adds information. An ordered sequence alone can have several segmentations, each with the same total effect. A grouping criterion might concern a common purpose, an institutional procedure, a temporal interval, or a continuous activity. Each criterion can be debated. Associativity

guarantees consistency of composite effects when grouping changes, while leaving the intended event boundaries undetermined.

Temporal resolution can also alter the appearance of simultaneity. Events that receive one coarse timestamp may be sequential at a finer resolution. Events known to overlap may resist a faithful representation by a single total order. A word model can impose an order for analysis, but its empirical significance requires justification when operations fail to commute.

For two operations a, b , equality $T_b T_a = T_a T_b$ can make the endpoint insensitive to their ordering. It leaves the occurrence order unresolved. The same final effect can arise from two different sequences, and the difference may matter to a historical explanation. When the operations fail to commute, uncertainty about order can imply uncertainty about the predicted endpoint as well.

A partial-order event record offers another possible extension. It can preserve known precedence constraints while leaving unrelated events unordered. To derive an endpoint from it, the model must explain how linear orders consistent with those constraints are evaluated and whether they agree. Agreement can follow from suitable commutation assumptions, but those assumptions need proof for the specified maps. This paper retains the sequential base and records the partial-order extension as an explicit modeling obligation.

9.8 Occurrence Evidence and Reconstruction

An event formula becomes part of an empirical inquiry through evidence. This subsection identifies the gap between internal consistency and warranted reconstruction.

A log entry can support the claim that a system recorded an operation. Its relation to a real-world completion depends on how the log is generated. A proposal document can support an intention without proving execution. A retrospective account can describe an event while requiring corroboration about timing or participants. These are generic evidence problems; an application must examine the particular source mechanism.

One can formalize the reconstruction task without pretending to have solved it. Let H be a declared space of candidate histories and let $\rho : H \rightarrow Z$ record the information retained by the investigation. An event-count query $g : H \rightarrow \mathbb{N}$ is exactly recoverable from ρ when it is constant on each record fibre. Different histories with the same record and different counts witness failure of exact recovery. The condition identifies the information requirement while leaving evidence about the actual history open.

A reconstruction can remain useful when exact recovery fails. The set of counts compatible with a record expresses the unresolved alternatives. Additional independent records can narrow that set when their interpretation is justified. Assigning probabilities would add another layer of assumptions; the set of alternatives alone supplies no frequency or likelihood.

The same distinction applies to event identity. Two reports may be compatible with either one jointly reported event or two matching events. A model that retains both possibilities makes the uncertainty visible. A single merged event index suppresses it. The appropriate choice depends on the evidence and the inferential purpose, not on the convenience of a compact event table.

Revision of an event reconstruction should preserve the earlier record when that record has historical relevance. Correcting a count or participant assignment changes the current representation. It also creates a new occurrence in the inquiry's own history: a correction was made on the basis of particular evidence. Keeping those levels separate supports a clear account of what changed and why.

9.9 Philosophical Role of the Event Category

The formal distinctions support a philosophical account of occurrence without settling a general metaphysics of events. This subsection states what the construction contributes and where further argument would be required.

The event category permits reference to a happening that has a place in a history. Its identity can depend on a declared boundary, a continuity criterion, and a relevant description. The action framework specifies the transformations associated with a possible happening; the occurrence domain speci-

fies which tokens the inquiry considers. Their combination supports claims about repetition, order, and participation that an endpoint alone cannot express.

A single occurrence can receive several descriptions. An enactment may be described through its physical procedure, its authorization role, or its effect on a record. The model can attach several attributes to one occurrence index while retaining the possibility that some descriptions refer to distinct subevents. Whether they should be joined or separated depends on the event criterion and evidence.

This creates a relation between event and process categories. An event can be analyzed internally as a process. A process can be compressed into one event for a coarser inquiry. The mathematical consistency of the compression concerns its retained effects and records. Its philosophical adequacy concerns which changes, reasons, and meanings the inquiry needs to explain. A useful event boundary is justified through those purposes and can remain revisable.

The category also clarifies the status of actuality. A possible transition, an intended action, a documented attempt, and a completed occurrence occupy different places in an argument. Moving between them requires premises. The framework exposes these transitions by giving each a distinct representational component and avoiding an automatic inference from a map's existence to its enactment.

9.10 Event Use in the Coordinated Framework

The event category contributes indexed happenings to the six-category account. This subsection connects that contribution to the rest of the model and identifies the required adequacy checks.

Entities can participate in occurrences through appearance indices. Relations can be formed, exercised, or modified through recorded events. Attributes can classify event types or report conditions at their occurrence. State descriptions specify before-and-after distinctions. Processes preserve sequences and intermediate configurations. The event domain supplies the token distinctions needed to coordinate these uses.

A successful application must declare what counts as an event, how occurrence indices are warranted, which repeated applications remain distinct, and how blocked or silent attempts are treated. It must check that any enriched variables respect the algebra it continues to use. It must also explain what information is lost when records are reduced to types, counts, or effects.

These requirements make the event category more than a transition triple. The triple remains essential for the action model, while occurrence, individuation, and evidence determine how that possible transition enters an account of social history.

10 Attribute Descriptions and Classification Commitments

This section develops attributes through the mathematical question of assigning a predicate or value at a declared descriptive level. It derives the condition under which an attribute descends to a quotient, extends the result to partial information, and examines overlapping classifications and numerical scales. The philosophical discussion concerns the difference between a coding rule, the constitution of a status, and the evidence for an assignment. The objective is to make the information supplied by an attribute explicit rather than conceal it in the notation.

10.1 Subjects and Descriptive Levels

An attribute says something about a subject under a description. That subject can be a configuration, an entity at a time, an event occurrence, or a process record. Its domain is therefore the first modeling decision. A value assigned to a current configuration can change while an entity continues. A value assigned to an event can remain historically fixed while later evidence about it changes. A value assigned to a process can depend on intermediate steps that its endpoint omits.

The carrier S supplies one available domain. A quotient $Q = S/\sim$ supplies a coarser domain when the inquiry treats equivalent configurations alike. An appearance domain supplies context-sensitive sub-

jects. The choice matters because a well-defined function must assign one value to each input. If the input merges cases that require distinct values, the attribute needs a finer domain or a different representation.

Consider a stipulated permission classification. A configuration may include whether a permission is recorded. A current-state attribute can read that coordinate. An entity-only attribute would additionally require the permission value to remain invariant across all appearances of the entity. An entity/time attribute can retain its variation. These representations can use similar verbal labels while making different claims about their subjects.

The six-category framework therefore treats an attribute's domain as part of its meaning. A statement that a group is authorized differs from a statement that a particular record classifies the group as authorized. The two can coincide under appropriate rules and evidence. Their coincidence is a further claim, and a model should retain enough structure to state it rather than assume it through a shared variable name.

10.2 Boolean Predicates and Value Maps

The first construction represents a declared classification on the quotient. Its simplicity makes the added information particularly clear.

For $U \subseteq Q$, define

$$a_U : S \longrightarrow \{0, 1\}, \quad a_U(x) = \mathbf{1}_U(q(x)). \quad (20)$$

The map assigns 1 to configurations whose quotient class belongs to U , and 0 to the others. It is constant on every $\sim$ -class because equivalent inputs have the same quotient image.

A value-valued attribute uses a map $f : Q \rightarrow V$ and defines $a = f \circ q$. The codomain can contain names, categories, numbers, or structured values. Calling the values numbers adds arithmetic notation, but the meaningful operations on them depend on the intended scale. A set of numbered categories can still carry only a nominal distinction.

The subset U and the map f supply information. The quotient alone does not select which of its classes count as authorized, isolated, resilient, or relevant to another substantive property. A particular classification may be defined through the base language and declared parameters. A freely supplied classification is a legitimate enrichment when its origin and interpretation are stated.

This distinction affects claims of derivation. Showing that an attribute can be represented by an indicator proves a representational possibility. It leaves the membership criterion for U open. Deriving a substantive attribute requires an argument that determines its membership from specified data or rules. The indicator formula then expresses that criterion's result; it cannot replace the argument selecting the criterion.

10.3 Descent from Configurations to Quotient Classes

A proposed attribute may initially be defined on the full carrier. This subsection derives the exact condition for retaining it at the quotient level.

Proposition 10.1 (Attribute factorization). *Let $a : S \rightarrow V$. There exists a map $\bar{a} : Q \rightarrow V$ satisfying $a = \bar{a} \circ q$ exactly when*

$$x \sim y \implies a(x) = a(y). \quad (21)$$

The map $\bar{a}$ is unique when it exists.

Proof. If $a = \bar{a} \circ q$ and $x \sim y$, then $q(x) = q(y)$. Applying $\bar{a}$ gives $a(x) = a(y)$, proving necessity.

For sufficiency, define $\bar{a}([x]) = a(x)$. If $[x] = [y]$, Equation (21) gives $a(x) = a(y)$. The definition therefore does not depend on the representative and satisfies $\bar{a}(q(x)) = a(x)$.

Every quotient class has a representative. Any factorization must take that class to its representative's a -value. This determines $\bar{a}$ uniquely. $\square$

For a Boolean attribute represented on S by a subset A , the condition says that A is a union of $\sim$ -classes. If one member of a class belongs to A , every equivalent member must belong. This saturation requirement makes explicit what a quotient classification assumes.

Failure of the condition has several possible responses. The inquiry can refine the equivalence so that distinct attribute values remain separate. It can change the attribute's target, perhaps by reporting a set of possible values for a coarse class. It can retain the attribute on the full carrier alongside a coarser state description. These choices preserve different information and should be assessed against the research question.

A set-valued report can describe uncertainty introduced by aggregation:

$$A([x]) = \{a(y) : y \sim x\}.$$

The definition is representative-independent because equivalent inputs have the same class. A singleton set permits exact recovery of the attribute at that class. Several values expose the ambiguity. Replacing the set by one selected value requires an additional selection or approximation rule. The construction is useful precisely because it displays what the coarse representation leaves unresolved.

The quotient condition also disciplines changes of scale. A finer state description can support an attribute that the coarser description cannot recover. Calling both descriptions states does not make their attribute languages interchangeable. When an inquiry moves between them, it should identify which values are preserved, lost, or newly supplied.

10.4 Partial Attributes and Definedness

A classification can be incomplete. This subsection extends the descent argument to show that its domain of definedness carries information alongside its reported values.

Let $a : D \rightarrow V$ be a partial attribute, where $D \subseteq S$. An input outside D lacks a value under the declared coding. The reason may be missing evidence, an unresolved classification, or inapplicability. Those reasons can require different representations, even when a preliminary formal treatment uses one partial domain.

Proposition 10.2 (Partial-attribute descent). *A partial map $\bar{a} : \bar{D} \rightarrow V$ on Q can satisfy*

$$D = q^{-1}(\bar{D}), \quad a = \bar{a} \circ q \text{ on } D$$

exactly when D is a union of $\sim$ -classes and a is constant on each class contained in D .

Proof. Suppose the partial factorization exists. If $x \sim y$, then $q(x) = q(y)$. Membership of that image in $\bar{D}$ is the same for both, so $x \in D$ exactly when $y \in D$. Thus D is saturated. For members of D , applying $\bar{a}$ to equal projections gives equal values.

Conversely, assume both conditions. Set $\bar{D} = q(D)$. If $q(x) \in q(D)$, then some $y \in D$ satisfies $q(x) = q(y)$. Saturation puts x in D , so $q^{-1}(\bar{D}) = D$. Assign $\bar{a}([x])$ the common value of the class's members. Value constancy makes this well-defined and gives the required factorization. $\square$

A two-point counterexample isolates the domain condition. Let $S = \{x, y\}$ with $x \sim y$, and let $D = \{x\}$. Even if a has only one reported value, it cannot preserve definedness on the one-point quotient. The single class would need to be both inside and outside $\bar{D}$. Checking only equality of reported values would miss this failure.

A fresh marker can totalize the partial map:

$$\hat{a}(x) = \begin{cases} a(x), & x \in D, \\ \perp, & x \notin D, \end{cases}$$

where $\perp$ is distinct from every value in V . Constancy of $\hat{a}$ on quotient classes combines the two requirements in the proposition. The marker makes missingness visible; its meaning remains an interpretive choice.

For a Boolean attribute, the codes 0, 1, and $\perp$ can represent false, true, and unknown to the investigation. This coding does not by itself assert that the underlying proposition has three truth values. Unknown can concern evidence rather than the world's state. Likewise, inapplicability can differ from both falsehood and missing evidence. A careful application separates these meanings before aggregating or comparing observations.

10.5 Overlapping Attributes and Classification Partitions

Several attributes can apply to the same subject. This subsection distinguishes that possibility from the stronger structure of a partition.

For a family $(U_j)_{j \in J}$ of subsets of Q , the profile

$$a(x) = (\mathbf{1}_{U_j}(q(x)))_{j \in J}$$

records simultaneous Boolean classifications. A configuration may belong to several U_j , or to none. The formula imposes neither mutual exclusivity nor exhaustive coverage.

A partition requires both. Its cells must be pairwise disjoint and their union must cover Q . A single-valued map $f : Q \rightarrow V$ induces a partition by its nonempty fibres. Conversely, a partition supplies a single-valued cell assignment. The structure is suitable when the classification task explicitly requires one exhaustive alternative.

A cultural or institutional classification may need overlapping predicates. A participant can hold several roles. A practice can belong to several traditions under the inquiry's declared criteria. A configuration can be both authorized and resource-constrained. Turning these predicates into one list of mutually exclusive categories changes the task unless the list retains every relevant combination.

One can create a partition from the full Boolean profile, grouping subjects that agree on all attributes. This preserves overlap information at the cost of a potentially larger classification space. For a finite family of k Boolean attributes there are at most 2^k profiles, though some may be absent. The resulting cells represent combinations of predicates, not a discovery that the original predicates were exclusive.

The philosophical point concerns what the classification claims. Overlapping predicates organize several aspects of one subject. A partition organizes alternatives at a particular level. Both can be useful, but they impose different commitments about how descriptions fit together. The six social-fact categories are used here as overlapping analytical roles and therefore require no partition of all social facts into six disjoint kinds.

10.6 Coding Rules and Constitutive Conditions

A classification can record a pattern, implement a rule, or represent an investigator's judgment. This subsection separates those functions while permitting their interaction.

An observational coding rule maps available material to a reported value. A constitutive rule specifies conditions under which a status obtains within a stipulated practice. An interpretive judgment connects records to a substantive claim. These functions can coincide in a particular model, but their coincidence requires premises about how the practice and its records work.

For a hypothetical registry, let $c(x)$ be the current registered classification and let $g(x)$ be the substantive target of the inquiry. If the target is precisely the registry's classification, then $g = c$ by definition. If the target concerns an independently specified condition, equality is a further claim. The registry can be evidence for that condition while remaining fallible with respect to it. The formal distinction protects the investigator from changing the target silently when evidence is convenient.

A constitutive classification can also influence later dynamics. If access depends on $c(x)$, two otherwise similar configurations with different classifications may admit different operations. That effect is a modeled consequence of the rule. It leaves open whether the classification accurately describes another feature, whether its implementation is stable, and whether the rule is justified.

The formation of a category and the classification of an instance are therefore different research problems. A community may develop a classification practice, after which an investigator asks which

instances satisfy it. A revision of the practice can change the classification rule itself. Modeling that revision requires a rule index or a dynamic state containing the relevant rule. A fixed subset U represents one selected rule at a time.

This distinction matters for philosophical accounts of social facts. Some attributes concern statuses constituted through practices; others concern patterns that an institution attempts to record. The same function notation can represent both. Their difference lies in the interpretation and the arguments connecting the code to the subject, not in the bare mathematical form of the map.

10.7 Numerical Values and Measurement Commitments

A numerical attribute invites arithmetic operations. This subsection shows why those operations require more structure than a numerical label alone.

A nominal classification can be recoded by a bijection of labels without changing its intended distinctions. An ordinal classification can be recoded by a strictly increasing function while preserving order. A quantitative scale may permit a narrower class of transformations when differences or ratios carry meaning. Which transformations are admissible is a modeling and measurement commitment.

For an ordinal attribute a and a strictly increasing ϕ ,

$$a(x) < a(y) \iff \phi(a(x)) < \phi(a(y)).$$

The equivalence follows directly from strict monotonicity on the scale. It establishes preservation of individual comparisons. It leaves aggregate operations such as addition open.

Consider two stipulated profiles $(0, 3)$ and $(2, 2)$. Their numerical sums satisfy $3 < 4$. The increasing recoding of $\{0, 1, 2, 3\}$ to $\{0, 1, 2, 100\}$ changes the sums to $100 > 4$. Individual rank information is preserved, while the aggregate ranking reverses. A sum-based comparison therefore uses more than the ordinal information.

This counterexample does not prescribe a social ranking. It identifies an assumption that a scalar model must justify. If a score represents authorization categories, degrees of recognition, or a researcher-defined index, meaningful addition cannot be inferred from its numerical appearance. The intended interpretation of the scale must explain the operation.

Aggregation can also conceal heterogeneity. Two configurations can have the same scalar score while differing in which conditions contribute to it. Those differences may affect feasible continuations. If they do, equality of the score can fail the state-preservation condition. A scalar summary can remain useful for reporting while being inadequate as the sole predictive state variable.

10.8 Aggregate Attributes and Predictive Closure

A scalar attribute can accurately summarize a configuration while leaving its later value unrecoverable from the scalar alone. This subsection derives a complete characterization for a simple aggregation model. The example gives the earlier warning about heterogeneity a precise mathematical content: the suitability of a summary depends jointly on the aggregation map and on the update family.

Let $S = \mathbb{R}^2$, where the two coordinates represent a stipulated pair of numerical attributes, and define

$$q(x, y) = \frac{x + y}{2}.$$

The map retains the arithmetic mean. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous, and apply the same update independently to both coordinates:

$$T_f(x, y) = (f(x), f(y)).$$

A closed mean description requires a map $\bar{T} : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $qT_f = \bar{T}q$. This is the action-preservation question for the equivalence relation induced by the mean.

The assumptions are deliberately narrow. The coordinate values lie on a scale supporting arithmetic means. Membership and weights stay fixed. Each component receives the same deterministic update,

and the update depends only on that component's value. Interaction between components, changing membership, different local update rules, or a restricted carrier would change the problem. The narrow formulation allows the consequences of the chosen summary to be derived completely.

Proposition 10.3 (Mean closure for a continuous component update). *Under the stated assumptions, a reduced map satisfying $qT_f = \bar{T}q$ exists exactly when f is affine: $f(t) = at + b$ for constants $a, b \in \mathbb{R}$. The reduced map then equals f .*

Proof. For each t , the diagonal point (t, t) has mean t , and its updated mean is $f(t)$. Any reduced map therefore satisfies $\bar{T}(t) = f(t)$.

For arbitrary x, y , put $m = (x + y)/2$. The points (x, y) and (m, m) have the same mean, so descent implies

$$\frac{f(x) + f(y)}{2} = f\left(\frac{x + y}{2}\right).$$

Define $g(t) = f(t) - f(0)$. Then $g(0) = 0$, and g satisfies the same midpoint equality. Substitution of $2t$ and 0 gives $g(2t) = 2g(t)$. Consequently,

$$g(x + y) = 2g\left(\frac{x + y}{2}\right) = g(x) + g(y).$$

Repeated addition yields $g(n) = ng(1)$ for nonnegative integers. Since $g(n) + g(-n) = g(0) = 0$, the same expression holds for negative integers. For an integer p and positive integer q ,

$$qg(p/q) = g(p) = pg(1),$$

so $g(p/q) = (p/q)g(1)$.

Given a real number t , choose a sequence of rationals tending to t . Continuity of g implies $g(t) = tg(1)$. Thus $f(t) = at + b$, with $a = g(1)$ and $b = f(0)$.

Conversely, for $f(t) = at + b$,

$$qT_f(x, y) = \frac{ax + b + ay + b}{2} = aq(x, y) + b = f(q(x, y)).$$

Taking $\bar{T} = f$ supplies the required factorization. □

Continuity enters at a specific step: it transfers rational linearity to all real arguments. The proof should therefore preserve that assumption even when the substantive interpretation makes smooth changes seem natural. Its conclusion concerns the displayed carrier and update rule. A finite selected set of attribute pairs can permit additional reduced maps because fewer equal-mean pairs need to be compared.

For a direct failure, take $f(t) = t^2$. The configurations $(0, 2)$ and $(1, 1)$ both have mean 1. Their updated configurations are $(0, 4)$ and $(1, 1)$, whose means are 2 and 1. Any proposed mean-only update would need to assign two different next values to the same input 1. The disagreement establishes failed closure without relying on an estimation error or an imperfect observation.

The missing information in this example can be displayed by expanding the formula:

$$\frac{x^2 + y^2}{2} = \left(\frac{x + y}{2}\right)^2 + \frac{(x - y)^2}{4}.$$

The second term records heterogeneity omitted by the mean. It vanishes on the diagonal and is positive for unequal coordinates. Retaining the mean and this additional term would recover the next mean for the squaring update. Whether the enlarged record supports its own future updates requires another closure check.

That last qualification matters. Adding a variable that repairs one prediction can leave later predictions unresolved. The construction of an adequate state description therefore proceeds relative to an

update family and an intended horizon. A reporting statistic answers a present descriptive question. A state variable used for repeated prediction must additionally support the continuation requirements of the inquiry.

The mathematical distinction remains useful even where an arithmetic mean has a clear interpretation. Average resources, average reported confidence, or an average numerical performance measure may summarize the selected subjects accurately. Interpreting the average as an autonomous dynamic subject requires evidence that the omitted distributional features are irrelevant to the modeled updates. Proposition 10.3 supplies one exact test in a restricted model and demonstrates why that additional requirement can be demanding.

The result also places a limit on claims that an aggregate creates a new entity. The map q defines equivalence classes of numerical configurations. Those classes can support a closed update under the proposition's conditions. Reidentifying a collective entity across time requires the appearance and continuity records developed in Section 7. A successful mean quotient establishes a descriptive and dynamic achievement, with entity identification remaining a separately stated task.

10.9 Changing Composition and Aggregate Direction

A second aggregation problem arises when the subjects entering a summary change their relative weight. This subsection derives an exact decomposition and a numerical counterexample. The objective is to separate a change within the classified groups from a change in how those groups contribute to the reported aggregate.

For a finite set of strata indexed by j , let r_j be an earlier rate and r'_j a later rate. Let w_j and w'_j be the corresponding nonnegative weights, each family summing to one. Write

$$A = \sum_j w_j r_j, \quad A' = \sum_j w'_j r'_j.$$

The strata, rates, denominators, and weighting rule are declared components of the representation. A stratum can be a reporting category without being an entity, and membership in it can change between observations.

Proposition 10.4 (Rate and composition decomposition). *The change in the aggregate satisfies*

$$A' - A = \sum_j w'_j (r'_j - r_j) + \sum_j (w'_j - w_j) r_j. \quad (22)$$

For fixed weights, weak improvement in every stratum gives weak improvement in the aggregate.

Proof. Add and subtract $\sum_j w'_j r_j$ in the expression for $A' - A$. The first difference is $\sum_j w'_j (r'_j - r_j)$; the second is $\sum_j (w'_j - w_j) r_j$. This proves the identity. When $w'_j = w_j$ for every j , the second sum vanishes. If every rate change is nonnegative, each term in the first sum is nonnegative, proving the direction statement. $\square$

The first term evaluates within-stratum rate changes at the later composition. The second evaluates the weight change at the earlier rates. Reversing these baseline choices gives another exact decomposition. The allocation of an interaction between changing rates and weights depends on the stated convention. The total difference remains the same.

Table 3 gives stipulated counts that make the distinction visible. In the earlier observation the high-rate stratum records 90 successes out of 100 cases, while the low-rate stratum records 1 out of 10. In the later observation the high-rate stratum records 10 out of 10, and the low-rate stratum records 20 out of 100. These counts define a mathematical illustration; they report no empirical population.

Both subgroup rates rise by $1/10$. Yet the pooled rate falls from $91/110$ to $30/110$, a change of $-61/110$. The earlier weights are $(10/11, 1/11)$ and the later weights are $(1/11, 10/11)$. The within-stratum term in (22) is $1/10 = 11/110$. The composition term is

$$-\frac{9}{11} \frac{9}{10} + \frac{9}{11} \frac{1}{10} = -\frac{72}{110}.$$

Table 3: Stipulated subgroup rates and changing composition in two observations. Each subgroup rate increases while the pooled rate falls.

Observation	High-rate stratum	Low-rate stratum	Pooled rate
Earlier	$90/100 = 0.9$	$1/10 = 0.1$	$91/110$
Later	$10/10 = 1$	$20/100 = 0.2$	$30/110$

Adding the terms gives $-61/110$, matching the pooled calculation.

The apparent directional conflict disappears once the two questions are stated explicitly. The subgroup comparison asks how each rate changed. The pooled comparison asks for the success fraction among the cases represented in each observation, under that observation's composition. The numerical answers can move in opposite directions because the later pool contains a much larger share of the lower-rate stratum.

A fixed-weight standardized comparison answers another declared question. Holding the weights constant removes the composition term. It can make the within-stratum change easier to compare across observations. The choice of standard weights still requires a reason: earlier composition, later composition, equal weights, and a separately chosen reference population can all answer different comparative questions. The formula cannot select the relevant population by itself.

The example also exposes a connection to entity and process records. If the research question concerns the same individuals over time, aggregate counts from changing pools are insufficient by themselves. An appearance-level link or another identification method is needed to establish which subjects persist, enter, or leave. If the question concerns a changing institution's current composition, the movement of subjects between categories may itself be part of the process under study. The required record depends on that distinction.

Interpreting a pooled decline as evidence of deteriorating practice would add a causal claim. The displayed example supplies one countercondition: a change of composition can outweigh improvement within every displayed stratum. Conversely, favorable standardized rates can coexist with a practically significant redistribution of cases toward difficult conditions. Choosing one statistic to settle the overall assessment requires an evaluative argument about which changes matter and to whom.

A comprehensive attribute record should consequently preserve the numerator, denominator, inclusion rule, stratum definitions, and weighting convention wherever those components affect interpretation. A scalar can remain the principal reported value while its construction remains recoverable. The retained detail permits criticism of the formula, comparison of alternative descriptions, and correction when the original question was expressed too coarsely.

10.10 Attribute Change and Explanatory Change

An attribute can change for several reasons. This subsection distinguishes a changed subject, a changed rule, and changed knowledge about the subject.

With a fixed map $a : S \rightarrow V$, a transition from x to mx can change the value because the represented configuration changes. With a fixed configuration, replacing a by a new map a' can change the reported value because the coding rule changes. A record can also be revised while both the target configuration and intended rule remain fixed, reflecting new evidence or correction.

These are different histories. A time-indexed report alone may combine them. To interpret the report, an inquiry can retain the configuration, rule version, and evidential record as separate components. A change in one component then has an explicit location in the model.

For example, a hypothetical practice can broaden its membership criterion. Some appearances receive a new category even when their previously recorded features remain fixed. A study of recognition may treat the reclassification itself as a social event. A study of changes in the underlying features would need to distinguish it from such changes. Both inquiries can use the same archive if its rule history is retained.

The metamodel can accommodate rule change through an expanded carrier that includes the rule state. It can also compare separate models with an explicit alignment. In either case, a formula developed for one fixed quotient or classification needs renewed checking when the regime changes. Equality of labels across regimes supplies little evidence of equality of meaning.

10.11 Evidence and Interpretive Disagreement

Attribute assignments require evidence appropriate to their meaning. This subsection explains how formal consistency and substantive warrant can diverge.

A coding procedure can assign one value to every record consistently and still misrepresent the target. Internal consistency concerns whether the rule is applied as specified. Warrant concerns whether the rule, evidence, and interpretation justify the substantive claim. A reliable procedure for reading a database field does not establish that the field accurately represents an independently defined social condition.

Disagreement can concern different components. Investigators may agree on the records but disagree on the category. They may agree on the category but disagree on how the records support an assignment. They may disagree about the subject's identity or context, changing the domain to which the value belongs. A representation that retains these components permits a more precise diagnosis than a single disputed number.

Partial and set-valued classifications are useful when they express this unresolved state faithfully. They should not be adopted merely to avoid difficult decisions. A study can state which evidence would reduce uncertainty, which coding changes would alter the target, and which disagreements remain conceptual rather than empirical.

A further issue concerns reflexivity. Publicly assigning an attribute can affect the subject's later opportunities or behavior in a stipulated model. The reported classification then participates in the dynamics being studied. The investigator needs to distinguish evidence about an earlier condition from effects of the public classification. That distinction can be represented through separate event and state variables when the application warrants it.

10.12 Philosophical Role of Attribute Representation

The attribute construction clarifies how a description can be precise while remaining dependent on conceptual choices. This subsection develops that implication for the broader six-category account.

An attribute map is exact relative to its domain, codomain, and rule. Exactness does not make the rule inevitable. Different rules can classify the same carrier differently, and the appropriate comparison concerns their explanatory purpose, evidential basis, and consequences. Precision makes those choices inspectable; it leaves them available for criticism.

This also clarifies the difference between discovering a pattern and supplying a useful classification. A researcher can propose a category to organize inquiry, then investigate whether its instances share relevant dynamics. Conversely, a regularity may motivate a later coding scheme. The direction of inquiry matters to claims about emergence. A supplied subset becomes a representation of an observed regularity only through an argument linking it to evidence.

The framework leaves room for disagreement about which attributes matter. Formal compatibility can show that a classification survives a quotient or supports an update. It cannot establish that the classification deserves priority in every explanatory or normative inquiry. A social model therefore needs reasons for its attribute choices in addition to proofs of their internal consistency.

Within the six-category architecture, attributes supply descriptive predicates and values. Entities supply subjects across appearances; events and processes supply historical occurrences; relations supply dependencies; states supply the current descriptive level. An attribute can apply at each of these levels when its domain and rule are explicit. This flexibility is the construction's practical strength and the reason its commitments require careful disclosure.

11 State Descriptions, Observation, and Retained Memory

This section develops the state category by separating a carrier configuration, an observation, and a description sufficient for a declared dynamical task. It derives the condition for deterministic updates on a reduced description, examines hidden memory and several observation tasks, and discusses actual occupancy and historical interpretation. Section 6 subsequently supplies the full continuation construction. The present section explains why that construction is needed and what its result means.

11.1 The Descriptive Role of a State

A state description records the conditions under which the model evaluates subsequent operations. In the abstract base, a point $x \in S$ has that role. The point can encode a large configuration containing participants, records, resources, and rules. It need not correspond to a single entity or to one directly observable snapshot.

The quotient point $q(x)$ supplies another descriptive level. It groups configurations according to the declared equivalence $\sim$. The action-preservation assumption ensures that equivalent configurations remain equivalent after the same mechanism, allowing the quotient to carry its own deterministic action. This is a mathematical condition on the description. Its empirical interpretation depends on what the carrier and quotient represent.

An observation map $o : S \rightarrow Y$ supplies a third object. It records the features visible or relevant to a chosen inquiry. Two configurations may have the same observation while producing different later observations under one operation. In that case, the present observation is insufficient to determine the next observation. Treating Y as a deterministic state space would require a repair or a different model.

These three objects can coincide in a simple application, but their roles remain distinct. The carrier supports the full declared dynamics. The quotient supports an agreed level of identification. The observation expresses a measurement or reporting task. A model should justify each identification between them rather than derive it from the shared word *state*.

11.2 Deterministic Updates on Reduced Descriptions

A reduced description is useful when its future values can be updated from its present value. This subsection derives the condition that makes such an update well-defined.

Let $\rho : S \rightarrow Z$ be a description and let $T_b : S \rightarrow S$ be a total operation. The question is whether a function $F_b : \rho(S) \rightarrow \rho(S)$ can satisfy

$$\rho(T_b x) = F_b(\rho(x)) \quad (x \in S). \quad (23)$$

The function is defined on the image of ρ , because unrepresented values of Z need no arbitrary transition.

Proposition 11.1 (Reduced-update criterion). *A function satisfying Equation (23) exists exactly when*

$$\rho(x) = \rho(y) \implies \rho(T_b x) = \rho(T_b y). \quad (24)$$

If the condition holds for every operation label, the description supports the corresponding deterministic updates.

Proof. If F_b exists, equal current descriptions are equal arguments to the same function and give equal next descriptions. This proves necessity.

Conversely, assign to a represented value $z = \rho(x)$ the value $F_b(z) = \rho(T_b x)$. Whenever another representative y gives the same z , the stated condition gives the same proposed output. The assignment is therefore well-defined and satisfies the equation. $\square$

The condition says that $\ker \rho$ is preserved by the operation. It is precisely the representative-independence needed for a quotient update. The argument applies to one operation at a time, so a repertoire of operations requires preservation under every member.

The result distinguishes an accurate current report from a sufficient dynamic description. A report can correctly record the present value and still omit information needed for its next value. Accuracy at one moment therefore does not establish state sufficiency. Conversely, a mathematically sufficient description may be difficult to observe or may encode variables whose interpretation remains uncertain.

11.3 Hidden Memory and a Minimal Counterexample

A finite example makes the failure and repair explicit. Its purpose is to show why the omitted information matters under a specified operation.

Let $S = \{0, 1\}^2$, write $x = (v, h)$, and observe $o(v, h) = v$. Interpret v as a current reported flag and h as a retained condition. Define

$$T_r(v, h) = (h, h).$$

The configurations $(0, 0)$ and $(0, 1)$ have the same current observation, but

$$o(T_r(0, 0)) = 0, \quad o(T_r(0, 1)) = 1.$$

Thus no deterministic function of the current flag alone can reproduce both next observations.

Retaining the pair (v, h) repairs this failure because the transition rule is already defined on that pair. The repair is sufficient for this stipulated system. Its adequacy in an application would require evidence that h represents an available condition and that the update captures the relevant dependence.

The variable can be called memory when its interpretation records a historically formed condition. The mathematical role, however, concerns future update dependence. A latent condition can perform that role without being a person's recollection. A document archive, an authorization record, or a retained resource state can each function as memory in a model if the interpretation and update rule justify the term.

The example also shows that a failed reduction does not uniquely identify the missing variable. Another sufficient encoding can be related to (v, h) by a change of coordinates, or a richer record can retain more detail. The failure proves that the chosen current observation is too coarse for the declared update. Identifying a particular hidden cause or historical mechanism requires additional evidence.

A stochastic reduced model offers a different response. It can assign a distribution to the next observation conditional on the current report. That move requires a probability model over the omitted distinctions or transitions. The deterministic failure alone supplies no canonical probabilities. Choosing a richer state and choosing a stochastic reduction are distinct modeling decisions.

11.4 Predictive Sufficiency and Historical Sufficiency

A description can be sufficient for one purpose and insufficient for another. This subsection distinguishes future observation from historical explanation.

Suppose two configurations have identical continuation signatures for the selected observations and operations. Their distinction can be omitted without affecting that predictive task. They may nevertheless correspond to different formation histories. If the historical question asks which sequence produced the present condition, the predictive quotient can discard the answer.

A stipulated permission illustrates the distinction. Two histories can produce the same current permission and all the same modeled future opportunities. One history may involve negotiation and another an imposed rule. If the current dynamics has no variable distinguishing those histories, prediction of the chosen outcome can remain identical. The historical query still differs, provided the formation labels have been meaningfully specified and evidenced.

The same issue appears in records of completed events. A state flag recording whether an activity has ever occurred can be sufficient for a future rule that depends only on that flag. It omits the count, timing, and participants of earlier occurrences. A historical inquiry needs those records even if they are redundant for the chosen transition rule.

The appropriate repair is task-specific. A joint description can retain both predictive information and the historical query. If the query must remain updatable after future operations, preserving only its current value can be insufficient. The historical-relations paper develops the joint continuation construction in detail. Here the general lesson is that sufficiency belongs to a declared family of questions.

This limitation has philosophical force. A compressed state can be the right object for predicting a selected variable without being a complete account of the social situation. Explanations of formation, responsibility, recognition, and significance can require distinctions absent from that prediction. The model should state its task so that predictive success receives the appropriate evidential scope.

11.5 Coordination of Observation Tasks

A six-category inquiry can require several observations at once. This subsection derives their coordination and clarifies the meaning of the resulting refinement.

Let $o_j : S \rightarrow Y_j$ be observations indexed by $j \in J$. The product observation is

$$o_G(x) = (o_j(x))_{j \in J}.$$

For each j , let E_j identify configurations whose o_j observations agree after every finite word. Define

$$E_G = \bigcap_{j \in J} E_j. \quad (25)$$

Proposition 11.2 (Joint-task equivalence). *The relation E_G is the continuation equivalence for o_G . It is the greatest update-preserved equivalence contained in every $\ker o_j$.*

Proof. Equality of product observations under a word means equality of each coordinate under that word. Requiring this for every word gives precisely the intersection in Equation (25).

Each E_j is an update-preserved equivalence by the continuation construction proved in Section 6. Intersecting those relations retains equivalence and update preservation.

If another update-preserved equivalence retains every current observation, the greatest-property argument for each individual task places it inside each E_j . It therefore lies in their intersection. $\square$

Adding an observation coordinate can only refine the relation, since it adds an equality requirement. Adding operation labels with the earlier maps unchanged can likewise expose more distinctions: every old word remains among the new tests. Neither change alone establishes that new underlying entities or configurations have appeared.

This distinction matters when a revised study reports more categories of state than an earlier one. The increase can reflect additional measurements or a richer repertoire of hypothetical tests. It can also reflect a changed underlying model, but that is another comparison requiring explicit alignment. A count of descriptive classes cannot distinguish the possibilities by itself.

The product construction can retain too much for a practical task. An infinite family can give an infinite record; even a finite family can be costly to measure. A researcher can therefore compare exact sufficiency, finite approximations, and selective omission. Each choice should state the errors or questions it leaves unresolved.

11.6 Actual Occupancy and Epistemic State

The action framework represents possible configurations. This subsection distinguishes those configurations from what an investigator knows about the current case.

An empirical occupancy assertion identifies the configuration or class occupied by the subject at a specified time. Such an assertion needs a mapping from evidence to the model. The existence of a state space and transition maps alone leaves the initial condition unspecified.

Incomplete evidence can be represented by a set $K \subseteq S$ of configurations compatible with the available information. This set is an epistemic record. Its elements are candidate world configurations.

Confusing K with a single world configuration would combine uncertainty about the case with the case's own modeled conditions.

Under a known deterministic operation T_b , the predicted compatible set is

$$T_b(K) = \{T_b x : x \in K\}.$$

A subsequent observation y can restrict it to $T_b(K) \cap o^{-1}(\{y\})$, assuming the observation model is exact. This update follows by retaining precisely the candidates that satisfy both the dynamics and the observation. Noise or uncertain execution requires further rules.

The set update can be informative without selecting one actual state. If every surviving candidate gives the same answer to a query, that answer is identified relative to the model and evidence assumptions. If candidates disagree, uncertainty remains. Assigning a probability distribution would add a measure whose justification is separate from set compatibility.

A social inquiry can also include participants' beliefs or records inside S . Those variables belong to the modeled situation, whereas the investigator's compatible set concerns knowledge of that situation. Both are legitimate, and their relation can matter dynamically. A clear model distinguishes the participant's information, the public record, and the investigator's uncertainty before drawing conclusions.

11.7 State Revision and Changes of Description

A state report can change because the world changes, because evidence changes, or because the description changes. This subsection separates those transitions.

Under a fixed carrier, observation, and operation system, a transition $x \mapsto T_b x$ is a modeled change in the subject. Under a fixed subject configuration, revising a compatible set changes the inquiry's information. Replacing the observation or equivalence changes the descriptive regime. The resulting reports can look similar while having different meanings.

For equivalences $E' \subseteq E$ on the same carrier, the finer quotient S/E' maps to the coarser quotient S/E by $[x]_{E'} \mapsto [x]_E$. Representative independence follows from the inclusion. This map gives a precise alignment between the two descriptions. It permits questions about which distinctions the coarser quotient forgets.

Arbitrary quotients need not be ordered by refinement. Two observations can preserve different features. Their joint observation provides a common refinement, but recovering one from the other requires a factorization condition. Shared numerical labels or equal numbers of classes establish little about their semantic agreement.

A change of carrier introduces another level of difficulty. An enlarged model can include newly represented participants or mechanisms. Comparing states across the models then needs an explicit embedding, projection, correspondence, or other alignment. The original quotient theorem applies within its fixed domain and does not automatically settle that comparison.

These distinctions prevent an inference from greater descriptive detail to greater generative capacity. A new classification can reveal a possibility that the old description omitted. An intervention can also create or enable a new possibility. The model must identify which kind of change its before-and-after comparison represents.

11.8 Philosophical Interpretation of State Sufficiency

The state construction answers a precise question: which distinctions must a description retain for specified updates and observations? This subsection evaluates the philosophical scope of that answer.

A sufficient state is relative to the operations and observations that define the task. It can be minimal for that task while omitting information relevant to another. The relativity is disciplined by an exact criterion and a greatest-property theorem; it permits comparisons between candidate descriptions.

The result leaves open whether the selected task captures what matters about a social situation. A prediction of activity completion can omit the manner of agreement, the participants' reasons, or a dis-

puted history. Those omissions can be acceptable for one calculation and serious for another inquiry. The philosophical argument must state why the chosen observations serve its purpose.

Deterministic state sufficiency also presumes that the operation repertoire captures the relevant way future developments are tested. A test set can exclude interventions that would expose an omitted distinction. Expanding the repertoire can therefore revise the equivalence. The theorem remains correct under either repertoire; the adequacy dispute concerns the choice between them.

This account gives state descriptions a constructive role without treating them as complete inventories of reality. A description organizes possible transitions, supports specified inferences, and exposes its dependence on retained information. Its philosophical value lies in making those commitments available for examination.

11.9 State Use in the Six-Category Framework

The state category coordinates current descriptions with the other analytical roles. This subsection states the connections that a complete application needs to preserve.

Entity identity can continue across changing states. Relations can be represented by conditions or incidence records that influence state transitions. Event tokens identify particular transitions or attempts. Attributes read values at the chosen level. Processes retain sequences and intermediate states. The roles are connected by maps and records, and each can preserve information that another omits.

A successful state model must declare its carrier, observations, operations, and retained equivalence. It must establish representative independence for reduced updates, identify how occupancy is warranted, and distinguish uncertainty from world configuration. Where historical or entity questions exceed predictive sufficiency, it must retain the corresponding records.

The continuation construction that follows provides one exact solution to the dynamic consistency problem. Its proof becomes philosophically useful when read together with these limits and choices, because it states both what the reduced state can do and which further questions remain outside that task.

12 Process Histories and Their Compositional Reductions

This section develops processes from ordered mechanism words and their trajectories. It derives endpoint composition, distinguishes several reductions of a history, and examines segmentation and duration. The philosophical discussion concerns what makes a sequence one process, how actual history differs from possible factorization, and how branching or overlapping activity challenges the sequential model. The aim is to preserve the reasoning carried by a process rather than retain only its final result.

12.1 Ordered Activity and Process Unity

A process description connects occurrences through an order and a criterion of unity. A sequence of operations can produce a result, sustain an arrangement, transform a relation, or end without completing a selected target. The action framework supplies an ordered composition calculus. The interpretation of the sequence as one socially meaningful process requires additional criteria.

A stipulated agreement can be formed through proposal, revision, authorization, and enactment. A record of the final authorization may leave the earlier negotiations invisible. An outcome-oriented calculation may need only the composite effect. A historical explanation may depend on the proposals that failed, the order of revisions, or the interruption of an earlier route. The representations serve different questions even when they concern the same eventual configuration.

A process also need not be goal-directed in its interpretation. An analyst can group a sequence by a recurring pattern, a causal chain, a maintained relation, or a declared interval. A participant's purpose is another possible organizing criterion, requiring its own evidence. The word representation captures order while leaving those interpretations open.

This distinction prevents a sequence's formal availability from being mistaken for an established history. A factorization of a mechanism shows one way its effect can be composed. Evidence is needed to claim that the corresponding operations actually occurred in that order. The process category keeps factorization and occurrence linked through explicit records.

12.2 Words and Trajectories

The construction begins with labeled operations and a seed. This subsection specifies the conventions and derives the trajectory.

Let B be an operation alphabet, with total maps $T_b : S \rightarrow S$. A finite word $w = b_1 \cdots b_k$ is executed from left to right. For a starting configuration $x_0 = x$, define

$$x_i = T_{b_i} x_{i-1} \quad (1 \leq i \leq k). \quad (26)$$

The process presentation retains

$$\text{Proc}(x, w) = (x_0, b_1, x_1, \dots, b_k, x_k).$$

The empty word ϵ has zero steps and leaves the seed unchanged.

Proposition 12.1 (Trajectory determination). *A seed and a finite word determine a unique trajectory under the declared deterministic maps. Its endpoint is $T_w x$, where $T_w = T_{b_k} \circ \cdots \circ T_{b_1}$ and $T_\epsilon = \text{id}_S$.*

Proof. For the empty word, the unique trajectory contains only x . Suppose the first $i - 1$ states have been determined. The total function T_{b_i} assigns exactly one value to x_{i-1} , determining x_i uniquely. Induction constructs the entire finite trajectory.

Repeated substitution into Equation (26) gives $x_k = T_{b_k} \cdots T_{b_1} x$, which is the stated composite-map expression. $\square$

The proof depends on determinism and totality. A partial map requires executability at every prefix. A stochastic transition requires a distribution over possible next states. A nondeterministic relation can yield several trajectories. Those extensions can be developed, but their path objects and claims of uniqueness differ from the present construction.

The labels matter independently of their maps. Two named operations can have the same action on the selected variables. A history of negotiation and a history of imposition can therefore be represented by distinct labels with identical maps when the model deliberately retains their formation names. Interpreting those names requires evidence or stipulation beyond the endpoint formula.

12.3 Words, Products, Maps, and Endpoints

Several reductions connect a process to its outcome. This subsection isolates their different information losses.

When labels specify elements of the base semigroup, a nonempty word has a product

$$e(w) = b_k \cdots b_1.$$

The action sends that product to a transformation of S . Evaluating the transformation at one seed gives an endpoint. The sequence of reductions is

$$w \longmapsto e(w) \longmapsto T_w \longmapsto T_w x \longmapsto q(T_w x). \quad (27)$$

Each arrow answers a narrower question about the process.

Distinct words can have the same product because the semigroup satisfies equations. If $b^2 = b$, one and two applications have the same product. Distinct semigroup elements can have the same map when

the action is nonfaithful. Distinct maps can agree at the selected seed while differing elsewhere. Finally, distinct endpoints can belong to the same quotient class. These losses are logically independent.

A concrete seed-level example uses the identity map and a constant-zero map on $\{0, 1\}$. They agree at seed 0 and differ at seed 1. Knowing the endpoint from 0 therefore fails to identify which transformation was used. Recovering the map would require observations at enough inputs or another source of information.

Nonfaithfulness occurs at another level. A semigroup can contain distinct mechanism labels whose actions coincide on every configuration in the selected carrier. Observing all endpoints within that carrier still leaves the label distinction unrecoverable. The distinction may encode a provenance difference that the dynamics deliberately omits.

The quotient can discard still more. Two histories with different endpoints can agree on a selected outcome, such as whether a joint activity occurred. Their differences in authorization, resources, or recorded formation may remain outside the outcome class. An explanation that requires those differences must retain a richer stage of the reduction sequence.

12.4 Recovery of Process Queries

A process query can concern length, order, duration, participants, or another property of the retained history. This subsection derives a general test for recovering it after compression.

Let H be a declared set of process presentations, let $\rho : H \rightarrow Z$ be a retained record, and let $g : H \rightarrow V$ be a target query. A function $\bar{g} : \rho(H) \rightarrow V$ can recover the target exactly when

$$\rho(h) = \rho(h') \implies g(h) = g(h'). \quad (28)$$

Necessity follows because equal records are equal arguments to $\bar{g}$. For sufficiency, assign each represented record the common query value of its fibre. The condition makes that value well-defined and gives $g = \bar{g} \circ \rho$. This elementary argument gives each proposed reduction a query-specific test.

The test can succeed for one query and fail for another. A composite effect determines the endpoint at a fixed seed. It can fail to determine application count under idempotence. A full word determines its ordered labels, but may fail to determine elapsed time when durations vary. A timestamped record can retain elapsed time while leaving a participant's intention open.

The result is useful because it locates a failed inference precisely. If two candidate histories share the retained record and differ in the target, exact recovery is impossible from that record alone. A richer record can repair the loss, or the inquiry can retain a set of compatible answers. A guessed answer would add information without identifying its source.

The criterion remains model-relative. It assumes that the history domain contains the relevant alternatives and that ρ correctly describes retained information. A query can be formally recoverable in an overly restrictive history domain while remaining empirically uncertain. Adequacy therefore requires scrutiny of both the compression and the candidate histories.

12.5 Segmentation and Composite Events

A process can be divided into stages or compressed into a single event. This subsection derives the endpoint consistency of segmentation and its limits.

For a nonempty word of length k , choose

$$0 = t_0 < t_1 < \dots < t_r = k.$$

The j th block contains positions $t_{j-1}+1, \dots, t_j$. Its subword acts on $x_{t_{j-1}}$ and yields x_{t_j} by the trajectory recursion. A segmentation therefore provides stage boundaries in addition to the underlying word.

Proposition 12.2 (Endpoint consistency of segmentation). *Composing the block transformations of a segmented word gives the same transformation as the entire word. The total transformation generally leaves the segmentation unrecoverable.*

Proof. Each block transformation is the composition of its consecutive operation maps. Composing the blocks concatenates those maps in their original order. Associativity removes the parentheses and yields T_w .

For failure of recovery, a two-letter word bb can be treated as one block or as two. Both segmentations yield T_b^2 , while their boundary records differ. A function of the total transformation alone cannot recover both boundary choices. $\square$

The proposition justifies stage-level calculation without proving that a stage boundary is socially appropriate. A group of operations can be one negotiation phase under one criterion and several events under another. The criteria can concern purpose, institutional procedure, temporal continuity, or a change in the relevant relation.

An event-level description often uses such compression. The overall transition of a meeting can be represented by one composite mechanism while an internal process record retains its steps. Their connection is mathematically clear when the event map is the composite of the process maps. The interpretive claim that these steps form one event remains additional.

Keeping both levels can be useful. The coarse event supports comparison across cases, while the detailed process explains how particular outcomes were reached. The model should identify which conclusions survive compression and which depend on the finer history.

12.6 Duration and the Limits of Algebraic Time

Order alone does not specify duration. This subsection constructs a simple time assignment and tests whether it can survive product compression.

Let $\delta : B \rightarrow \mathbb{R}_{\geq 0}$ assign a duration to each operation label. Under an additive sequential convention,

$$\tau(\epsilon) = 0, \quad \tau(b_1 \cdots b_k) = \sum_{i=1}^k \delta(b_i). \quad (29)$$

The equation treats repeated applications as separate contributions to time.

For a product representation e , duration descends to a function on $e(H)$ exactly when equal products have equal τ -values. The necessity and sufficiency proof is the fibre-constancy argument in Equation (28).

If $e(b) = e(bb)$, descent requires

$$\delta(b) = 2\delta(b),$$

so $\delta(b) = 0$. A positive duration for each application therefore fails to descend through this idempotent product equation. The total effect can forget how long its enactment took.

The assumption of a fixed label duration is itself restrictive. An operation can take different times at different configurations or under different resources. A state-dependent duration map can retain that dependence along the trajectory. Interrupted or overlapping activity needs a schedule rather than a simple sum.

The mathematical distinction corresponds to several temporal questions. Sequence position records order. Duration records elapsed time under a convention. Calendar time locates an occurrence within another temporal frame. Historical periodization groups activities by an interpretive criterion. A natural-number word index supplies the first and leaves the others open.

This prevents the semigroup law from being read as a complete temporal theory. Associativity concerns the grouping of composition. It neither assigns duration nor establishes that every social process admits one fixed sequential time scale.

12.7 Feasibility at Intermediate Stages

A composite effect can be algebraically available while a particular route to it is infeasible. This subsection identifies the role of prefix conditions in the process category.

Let $A(x) \subseteq B$ be the labels admissible at x . A word is feasible from seed x when

$$b_i \in A(x_{i-1}) \quad (1 \leq i \leq k).$$

Every step is tested at the configuration produced by its earlier prefix. A final endpoint test alone can omit a blockage encountered earlier in the process.

For a two-state flip f with $f^2 = \text{id}_S$, allow f only at state 0. From 0, the word ff would return to 0 under the total maps, but its second step is unavailable at state 1. The empty word and ff have the same total effect while differing in prefix feasibility.

The example shows why arbitrary availability domains need not respect equations among earlier total products. A process representation can preserve the words and their prefix tests. Another approach composes the partial maps themselves, so domain information participates in equality. Each approach states explicitly what the product means.

The social interpretation concerns realization conditions. A plan can specify an eventual result while requiring authorization, resources, or cooperation at intermediate stages. Those requirements are added data. The feasible-path paper develops their systematic representation and the consequences of shared conditions.

12.8 Branching, Interruption, and Open-Ended Histories

Finite words provide a clear starting point. This subsection examines cases that require a broader process representation.

Branching can represent several possible continuations from one history. A set of words retains alternative routes; a tree can make shared prefixes visible. Selecting one branch as actual requires occurrence information. A probability distribution or policy would add further assumptions about selection.

Interruption can be represented by a finite prefix that stops before the selected target. Whether it is an incomplete process, a completed attempt, or a separate event depends on the process criterion. The word alone records the performed or proposed prefix and leaves that classification to the inquiry.

Restarting can preserve some conditions while changing others. A new seed may encode retained resources, records, or commitments. Connecting the restarted activity to the earlier process requires a continuity criterion. Merely concatenating labels can conceal a gap whose significance matters historically or institutionally.

Open-ended activity requires a choice of representation. One may retain all finite prefixes, introduce infinite sequences, or use a changing process graph. Results proved for finite composition transfer only when the new structure and its limiting operations are defined. The existence of arbitrarily long finite words alone supplies no completed infinite historical object.

The possibility of generating new operation labels or units raises another issue. A fixed alphabet can encode a range of possible changes in advance. A model that changes its alphabet or carrier during execution needs an explicit comparison across stages. The base framework can motivate such an extension, while its fixed-domain theorems retain their stated scope.

12.9 Historical Actuality and Explanatory Relevance

A process record becomes an explanation through an interpretation of its steps and their evidence. This subsection distinguishes that achievement from a compositional calculation.

A factorization can show how an effect could arise. A historical explanation must support the claim that the relevant sequence occurred and that its steps have the meanings assigned to them. A sequence of labels named negotiation, agreement, and enactment does not establish those meanings without evidence or an explicit stipulated example.

Different histories can support the same final outcome and different explanations. One route may preserve participants' commitments; another may replace them. One may involve repeated failed attempts; another may reach the target immediately. The endpoint calculation can treat these routes alike for one purpose while the historical account distinguishes them.

A process account also selects which intermediate differences matter. A complete physical or documentary record can contain more detail than the inquiry needs. Compression is therefore unavoidable in practice. The question is whether it preserves the distinctions required by the explanatory target, and whether its losses are stated.

The factorization criterion provides a formal version of that question. It can show that a particular query is unrecoverable after a chosen compression. It leaves the importance of the query to philosophical and empirical argument. This combination makes the mathematics useful without turning every retained distinction into an equally important feature.

12.10 Philosophical Role of Process Unity

A sequence becomes one process through a criterion of connection. This subsection considers the philosophical implications of making that criterion explicit.

Causal connection, common purpose, institutional procedure, and historical narrative can each organize activity. They may overlap, but one can hold while another fails. A procedure can continue despite changed intentions. A shared purpose can persist through different procedures. A causal chain can connect events that participants never understood as one undertaking.

The model can represent those criteria through separate predicates or records. It then becomes possible to ask whether one process has several forms of unity, whether its boundary is disputed, or whether a later interpretation reorganizes earlier occurrences. A single operation word leaves those questions open rather than settling them through sequence alone.

A relational account can emphasize how processes form and modify the conditions of later processes. This proposal is compatible with the action framework when the relevant conditions are encoded and updated. The explanatory work lies in specifying those conditions and demonstrating their role. Composition alone guarantees an endpoint calculation, not a general theory of historical generation.

Within ontological suspension, the process category therefore provides a disciplined way to discuss ordered formation and transformation. It can support arguments that give processes explanatory priority in a case. A universal claim that all entities or social facts are ultimately processes requires a further philosophical argument.

12.11 Process Use in the Coordinated Framework

The process category retains the sequence and intermediate manifestations needed to connect the other five roles over time. This subsection states its contribution to a complete representation.

Entity appearances can be tracked along a process under an identity or lineage criterion. Relations can be formed, exercised, or revised by its steps. Event indices identify particular occurrences. Attributes can describe initial conditions, intermediates, or the whole history. States provide the configurations through which the process proceeds.

A successful application declares the seed, operation labels, order convention, feasibility conditions, and occurrence status. It identifies the process-unity criterion and any duration or segmentation records. It then checks which queries survive reduction to products, effects, or endpoints.

The resulting account preserves the insight that a process is more informative than its final formula. The formula records a conditional consequence of the steps. The process record retains the path by which that consequence is calculated, interpreted, and, where evidence permits, situated in history.

13 A Cooperative-Agreement Illustration

This section coordinates the six categories within one finite model. It specifies the complete Boolean action, derives its transformation normal forms, compares order effects and commuting prerequisites, and constructs the observation-relative state quotient. It then examines the additional records required for entities, relations, events, attributes, and processes. The agreement vocabulary is an analytic interpretation of stipulated rules; empirical adequacy remains open.

13.1 Configuration Variables and Their Scope

The example retains authorization, a resource condition, and an activity flag. This subsection defines their roles and explains which questions the small carrier can answer.

Let $S = \{0, 1\}^3$, with $x = (a, r, e)$. Interpret a as a recorded authorization under a cooperative agreement, r as an available resource condition, and e as whether a focal joint activity has occurred at least once. The carrier has eight configurations.

The variables have deliberately limited meanings. Authorization is represented through a recorded condition; its recognition, validity, and practical effect require interpretation. The resource coordinate records persistent availability, rather than a consumable quantity. The activity coordinate records occurrence at least once and omits multiplicity.

Participant identities and the identity of the agreement are additional records. The three bits do not identify which organizations participate or which agreement is being tracked. They describe selected conditions of a case whose referents must be supplied separately.

The complete carrier leaves the initial configuration open. A particular inquiry can select a seed and study its reachable subset. Keeping the full carrier is useful for checking equality of transformations and continuation equivalence across all represented conditions.

13.2 Operations and the Generated Action

The maps specify how the selected variables change. This subsection gives the complete rules and the relation between words and transformations.

Define

$$\begin{aligned} c(a, r, e) &= (1, r, e), \\ k(a, r, e) &= (a, 1, e), \\ j(a, r, e) &= (a, r, e \vee (a \wedge r)). \end{aligned} \tag{30}$$

Operation c records authorization, k supplies the resource condition, and j attempts activation. Let M be the transformation semigroup generated by nonempty words over these labels.

The base equivalence is equality. The selected observation is $o(a, r, e) = e$. Equality of raw configurations is therefore finer than the current activity observation.

The maps are total. When a prerequisite is absent, j leaves the configuration unchanged. An application that distinguishes an unavailable attempt, a refusal, or a failed execution needs an availability or observation enrichment.

The resource condition remains set after activation. A consumable resource would require a different map or a quantity ledger. The current rule is suitable only for the persistent condition interpretation.

Words retain operation order and repetition. Their products in M retain only transformations of the eight configurations. The following derivation characterizes exactly what that effect compression preserves.

13.3 Normal Forms of the Transformations

The generated action contains twelve distinct nonempty-word transformations. This subsection derives that number and supplies a witness for every form.

For a word w , let C, K indicate whether c, k occur anywhere. Since neither coordinate is reset, the final first two values are $a \vee C$ and $r \vee K$.

If j never occurs, the activity flag remains e . If it occurs, let C_j, K_j indicate whether c, k occur before the last j . Successive activation attempts read prerequisite values that can only increase. Their contributions to e therefore are contained in the last contribution.

Consequently, when j occurs,

$$T_w(a, r, e) = (a \vee C, r \vee K, e \vee [(a \vee C_j) \wedge (r \vee K_j)]). \tag{31}$$

Table 4: Transformation normal forms in the cooperative-agreement model

Final flag C	Final flag K	Added activity term $f(a, r)$	Witness word
0	0	$a \wedge r$	j
1	0	0	c
1	0	$a \wedge r$	jc
1	0	r	cj
0	1	0	k
0	1	$a \wedge r$	jk
0	1	a	kj
1	1	0	ck
1	1	$a \wedge r$	jck
1	1	r	cjk
1	1	a	kjc
1	1	1	ckj

When it does not, the last coordinate is simply e . Table 4 lists every possible effect.

Each row denotes $(a, r, e) \mapsto (a \vee C, r \vee K, e \vee f(a, r))$. The witness column contains words read left to right. It is distinct from the left-action product notation used when maps are applied to a configuration.

Proposition 13.1 (Complete finite transformation classification). *The nonempty-word action has exactly twelve transformations, represented by Table 4. The zero-step identity is a thirteenth transformation in the generated monoid.*

Proof. If $(C, K) = (0, 0)$, a nonempty word consists of j operations and has added activity term $a \wedge r$. If $(C, K) = (1, 0)$, the possible terms are zero when there is no activation, $a \wedge r$ when the last activation precedes authorization, and r when it follows authorization. The symmetric case $(0, 1)$ gives zero, $a \wedge r$, and a .

For $(C, K) = (1, 1)$, the last activation can precede both setters, follow only c , follow only k , or follow both. Together with the case containing no activation, the terms are zero, $a \wedge r$, r , a , and one. Thus at most $1 + 3 + 3 + 5 = 12$ maps occur.

The table supplies a word for every form. Different flag pairs have different first or second output functions. Within a flag pair, the listed activity terms differ on an input with $e = 0$. Hence the twelve maps are distinct.

A word containing c or k changes a corresponding zero input bit. A nonempty word containing neither consists of j and changes $(1, 1, 0)$. Thus no nonempty word induces the identity. The empty word adds that map. $\square$

The proof explains the enumeration. It also locates the information lost by the transformation record: many histories share the same setter flags and last-activation effect. Their earlier occurrences can remain distinct in a process record.

13.4 Order Effects and Joint Prerequisites

The model exhibits order sensitivity and joint contribution by commuting operations. This subsection derives both patterns and explains why they should remain distinct.

At $x = (0, 1, 0)$, applying authorization before activation gives

$$jc(x) = (1, 1, 1),$$

while activation before authorization gives

$$cj(x) = (1, 1, 0).$$

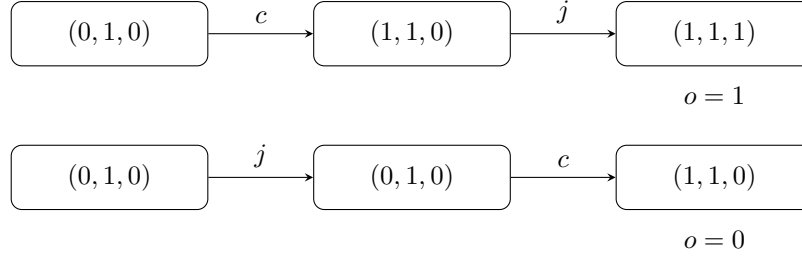


Figure 1: Two operation orders from the same resource-equipped configuration. Authorization followed by attempted activation sets the activity flag; the reverse order leaves it unset.

The expressions use left-action composition, so $jc(x)$ means $j(c(x))$. Figure 1 shows the corresponding word orders.

The order-effect predicate $R_{\text{ord}}(c, j, x)$ therefore holds. The authorization changes the result of a later activation under this stipulated comparison.

Meanwhile, $ck = kc$ on every configuration because the maps set different coordinates. Starting from $(0, 0, 0)$, either setter alone leaves one prerequisite absent. Applying both before j sets the activity flag.

The commuting setters thus jointly contribute to a later operation. Their commutation does not establish independence from every dynamic purpose. It states that their mutual order has the same effect on the chosen carrier.

The converse caution also holds. The order inequality is a mathematical property of the maps. Its interpretation as authorization dependence comes from the declared coding and update rules. Anonymous maps with the same table would satisfy the same predicate.

13.5 Occurrences and Historical Compression

Idempotence makes the difference between effects and occurrences especially clear. This subsection derives the relevant identity and identifies the records needed for repeated activity.

Boolean idempotence gives $j^2 = j$. After one activation attempt, the activity coordinate is $e \vee (a \wedge r)$. Applying the same rule again gives

$$[e \vee (a \wedge r)] \vee (a \wedge r) = e \vee (a \wedge r).$$

The first two coordinates are unchanged, proving equality of maps.

The one-step word j and the two-step word jj therefore have the same product and endpoint. Their occurrence counts differ. A process record can retain that difference through word positions or occurrence identifiers.

The activity flag adds another compression. Once $e = 1$, every later activation leaves the observed value one. A record of whether activity ever occurred cannot recover how often it occurred or how many attempts were made.

A counter extension can retain attempt count, but the extended operation then fails the old idempotence equation. One application adds one to the counter, and two applications add two. Projection to the old variables preserves the old effect. The event section derives that distinction in general.

The example therefore has two separate information losses: word-to-transformation compression and the choice of a Boolean occurrence summary. Identifying their locations makes a proposed repair precise.

13.6 Continuation Refinement

The activity observation merges states that later tests distinguish. This subsection derives the complete refinement and relates it to the common theorem.

Table 5: Continuation partitions for the cooperative-agreement illustration

Horizon	Classes	Distinctions retained
0	2	Current activity flag e
1	3	With $e = 0$, the configuration $(1, 1, 0)$ separates from the other three
2	5	Four singleton classes with $e = 0$; one class containing all configurations with $e = 1$
3	5	The partition is unchanged, certifying stabilization

At horizon zero, o gives two classes: the four configurations with $e = 0$ and the four with $e = 1$. The pair $(0, 1, 0)$ and $(1, 1, 0)$ belongs to the first class, yet j distinguishes it.

At horizon one, $(1, 1, 0)$ separates from the other incomplete configurations. Only this state activates immediately. The other three still give zero under every single operation.

At horizon two, the remaining incomplete states separate. The word kj distinguishes $(1, 0, 0)$ from $(0, 0, 0)$, and cj distinguishes $(0, 1, 0)$ from $(0, 0, 0)$. The word kj also distinguishes $(1, 0, 0)$ from $(0, 1, 0)$.

Every configuration with $e = 1$ continues to report one under every word. These four states therefore form one final class. The four incomplete states are singletons, giving five classes in total.

Table 5 records the horizons and the repeated stable stage. The common theorem makes that stability an exact certificate within the model.

The observation-relative quotient is coarser than the original equality regime. If the inquiry must also retain every original distinction, Equation (12) intersects with equality and keeps eight classes. The two records serve different tasks.

13.7 Coordination of the Six Categories

The same finite inquiry can use all six categories without treating them as disjoint kinds of object. This subsection states the additional commitments associated with each use.

Entity records identify the participants and the agreement across indexed appearances. Their continuity criteria can involve documents, membership, or institutional succession. None is fixed by the three Boolean coordinates.

The relation description concerns authorization and its connection to formation and possible activation. Its historical account requires a record of how the condition was established. Its potential role is represented by the activation comparison.

Event records identify particular attempts or enactments. An occurrence can leave the current state unchanged, so transition difference alone cannot supply its identity. Evidence is needed to establish that the represented attempt occurred.

Attributes select or describe conditions, such as authorization, resource availability, or completion. Their coding rules and domains determine whether they descend to a chosen quotient. An activity-only quotient can omit attribute distinctions that another task requires.

State descriptions retain selected configurational distinctions and support the corresponding updates when compatibility holds. Process records retain ordered operations and their intermediate states. The product normal form can summarize effects while the process record preserves the sequence.

The categories therefore cooperate through linked records. The question is which information each role requires and whether the links preserve it. The example supplies a common setting in which those requirements can be checked.

13.8 Limits and Verification of the Illustration

The construction is deliberately small and its rules are complete. This final subsection states what verification supports and what a substantive application must add.

Finite enumeration corroborates the eight configurations, twelve nonempty transformations, and horizon counts 2, 3, 5, 5. Independent paired-state search gives the same continuation classes. The normal-form proof explains why the transformation count is complete.

The checks concern the stipulated model. They leave the interpretation of authorization, resources, and activity open. A real cooperative arrangement can involve withdrawal, consumption, uncertain responses, and disagreements about recognition.

Adding those features changes the maps or records. The existing proofs remain valid under their displayed assumptions, while the extended model requires renewed checks. Preserving that genealogy prevents a refinement from silently changing the meaning of an earlier result.

The next illustration takes a different constructive route: it derives candidate collective units from a relational graph. Together, the examples show how a common formal base can coordinate category uses while requiring different added interpretations.

14 A Constructive Model of Relational Unit Formation

This section supplies a positive construction of candidate collective units from a declared relational configuration. It derives connected components, examines their dynamics under addition and deletion, and applies the six categories to one finite history. The model also exposes a precise limit: a partition that is sufficient under one operation repertoire becomes insufficient under another. The construction is a stipulated illustration of relational unit formation, with social interpretation and temporal identity kept explicit.

14.1 Elementary Positions and Relational Configurations

The model begins with a finite set of elementary positions and a specified type of relation among them. These are declared inputs. The collective units will be constructed from their configuration, so the derivation concerns those units rather than every object appearing in the model.

Let V be finite and let

$$E_0 = \{\{u, v\} : u, v \in V, u \neq v\}$$

be the possible undirected edges. A configuration is $G \subseteq E_0$, and the carrier is $S = 2^{E_0}$. An edge can be interpreted in a hypothetical application as a specified cooperative connection. The model treats it as undirected and Boolean; direction, intensity, and contested meaning remain outside this illustration.

For every $e \in E_0$, define total maps

$$a_e(G) = G \cup \{e\}, \quad d_e(G) = G \setminus \{e\}. \quad (32)$$

They add and delete the selected edge. Nonempty compositions generate a transformation semigroup acting on S . Full graph equality is the base equivalence.

Both elementary maps are idempotent. Adding an already present edge leaves the graph unchanged; deleting an absent edge does likewise. The equalities concern effects and permit separate occurrence indices for repeated applications. Thus the event distinction developed earlier is already present in this simple system.

The interpretation is intentionally bounded. Elementary positions can stand for participants only if that meaning is warranted. Edges can stand for cooperation only if the selected relation has evidence and a clear criterion. The graph construction derives groups relative to those inputs. It leaves their input justification and a broader account of individuals open.

14.2 Connected Components and Candidate Collective Units

The selected unit criterion is connectedness. This subsection proves that it gives a partition and explains the domain on which the resulting entities are represented.

For $u, v \in V$, define $u \simeq_G v$ when a finite undirected path in G connects them. The empty path connects each position to itself.

Proposition 14.1 (Component partition). *The relation $\simeq_G$ is an equivalence on V . Its classes form a partition $\mathcal{C}(G)$, whose members are candidate collective units under the declared connectedness criterion.*

Proof. The empty path gives reflexivity. Reversing an undirected path gives a path in the opposite direction, proving symmetry. Concatenating a path from u to v with a path from v to w gives a finite walk from u to w ; removing repeated segments if necessary gives a path. Thus the relation is transitive.

Every position belongs to its equivalence class. Two classes that intersect are equal by symmetry and transitivity. The classes therefore cover V and are pairwise disjoint. $\square$

The proof establishes a mathematical partition. Interpreting each component as a socially relevant unit requires the connectedness criterion to be appropriate to the inquiry. A chain of pairwise connections may support one collective task and fail another. The unit interpretation remains a proposal to test.

There are two different equivalences in this construction. The base equivalence on S compares whole graph configurations. The relation $\simeq_G$ compares elementary positions within one configuration. Confusing these domains would identify a state class with an internal collective unit. Their distinction is essential to the six-category argument.

Define the appearance domain

$$I = \{(G, C) : G \in S, C \in \mathcal{C}(G)\}.$$

The map $p(G, C) = G$ connects a component appearance to its configuration. One graph can have several appearances, one for each component. In a time-indexed history, appearances can instead be (t, C) , retaining repeated visits to the same graph configuration.

This is a constructive answer to part of the entity question. The units are derived from relational configurations under a specified criterion. Their identity across configurations still needs a further rule. Deriving a unit at a moment and establishing its historical continuity remain different tasks.

14.3 Component Dynamics under Edge Addition

The component partition can serve as a reduced state for a restricted repertoire. This subsection derives that positive result before examining its limitation.

Let $\pi : S \rightarrow \mathcal{P}_V$ send a graph to its partition $\mathcal{C}(G)$, where $\mathcal{P}_V$ is the set of partitions of V . For an edge $e = \{u, v\}$, consider the blocks containing u and v .

Proposition 14.2 (Addition-compatible partition). *Adding $\{u, v\}$ merges the two blocks containing u and v , with no change if they are already one block. Consequently, $\ker \pi$ is preserved by every addition map a_e .*

Proof. If u and v lie in the same component, a path already connects them. Adding their edge creates no connection between distinct components.

If they lie in distinct components, the new edge connects those components. Any path newly connecting old components must use the added edge, because all its other edges were already present. The new connection therefore joins exactly the two endpoint components and leaves the other blocks separate.

Two graphs with the same old partition have the same blocks containing u and v . The stated merger therefore gives the same new partition in both graphs. This is preservation of $\ker \pi$. $\square$

The partition can consequently be updated deterministically under the addition-only repertoire. Internal edge differences are irrelevant to the next component partition for those operations. The quotient is useful because its sufficiency has been proved for a declared task.

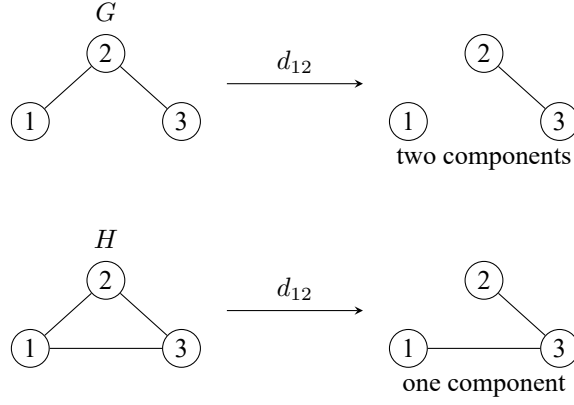


Figure 2: Edge-deletion comparison for graphs with a common initial component partition. The successor partitions expose a difference in internal relational support.

The construction also makes unit merger explicit. Adding a cross-component edge reduces the number of components by one. Adding an internal edge leaves the number fixed. Neither operation adds an elementary position. The change concerns how the fixed positions are grouped under the unit criterion.

An interpretation of merger must still explain why one new connection unifies the groups for the selected purpose. The mathematical component rule supplies that consequence once the criterion is adopted. It does not establish that every social connection should have that grouping effect.

14.4 Deletion and the Loss of Reduced-State Sufficiency

Allowing deletion changes which internal differences matter. This subsection gives a complete counterexample to partition-level sufficiency.

Take $V = \{1, 2, 3\}$ and abbreviate $\{i, j\}$ by ij in edge sets. Let

$$G = \{12, 23\}, \quad H = \{12, 23, 13\}.$$

Both graphs are connected, so $\pi(G) = \pi(H) = \{\{1, 2, 3\}\}$. Deleting 12 gives

$$d_{12}(G) = \{23\}, \quad d_{12}(H) = \{23, 13\}.$$

The first has components $\{1\}$ and $\{2, 3\}$. The second remains connected through the path 1–3–2. Hence their component observations differ.

Figure 2 displays the comparison. The inputs have the same group partition, while one common deletion produces different partitions.

The reduced-update criterion therefore excludes a deterministic function of the old partition alone for this deletion. The problem is located in the omitted internal structure, specifically whether another path maintains connectivity after the edge is removed.

The example gives a conditional account of robustness. The triangle retains connectivity after this deletion, whereas the path loses it. The component partition records current unity and omits this difference in its support. A model of persistence under withdrawal needs a richer description.

The result also explains why entity formation and entity persistence can demand different information. Connectedness identifies a present unit. Predicting whether it remains connected after a change can require the detailed arrangement of its relations. The unit label alone leaves that prediction unresolved.

14.5 Continuation Tests and Internal Structure

The enlarged repertoire can distinguish every graph in this finite model. This subsection derives the exact result and states its testing assumptions.

Table 6: Graph configurations and component observations on three positions

Present edges	Component partition
$\emptyset$	$\{\{1\}, \{2\}, \{3\}\}$
$\{12\}$	$\{\{1, 2\}, \{3\}\}$
$\{13\}$	$\{\{1, 3\}, \{2\}\}$
$\{23\}$	$\{\{2, 3\}, \{1\}\}$
$\{12, 13\}$	$\{\{1, 2, 3\}\}$
$\{12, 23\}$	$\{\{1, 2, 3\}\}$
$\{13, 23\}$	$\{\{1, 2, 3\}\}$
$\{12, 13, 23\}$	$\{\{1, 2, 3\}\}$

Proposition 14.3 (Graph separation by deletion words). *With component observation π and all edge deletions available, distinct graph configurations have different continuation signatures. Their full continuation equivalence is therefore equality.*

Proof. Let $G \neq H$. Choose an edge e that belongs to exactly one of them. Apply a word that deletes every edge in $E_0 \setminus \{e\}$. The word is finite because V is finite.

These deletions leave e unchanged and remove all other edges. One resulting graph is $\{e\}$; the other is empty. The endpoints of e lie in one component only in the first graph. The final component observations therefore differ.

Every distinct pair has such a witness. Equal configurations trivially have equal observations under every word. Thus signature equality is graph equality. $\square$

The theorem concerns exact tests in a stipulated system. It assumes the full deletion repertoire, a fixed edge universe, and exact component observation. Restricted access to operations or observations can leave some internal differences unresolved.

It also leaves the desirability of the tests open. Deleting connections can be a destructive intervention in an interpreted case. The proof uses hypothetical continuations to characterize information; it supplies no recommendation to perform them.

The addition-only and deletion-enriched models therefore give different continuation quotients on the same carrier. The difference follows from their test repertoires. It does not require a different underlying configuration to have appeared.

This demonstrates the constructive and critical roles of the continuation method. It can justify a coarse state when the task permits one, and prove that the same state is inadequate after the task changes.

14.6 Complete Three-Position Enumeration

The three-position case makes the construction fully inspectable. This subsection enumerates the states, observation classes, and transformation structure.

There are three possible edges, so the carrier has $2^3 = 8$ graphs. Table 6 lists every configuration and its component partition. Four graphs share the connected partition, giving five distinct current observations.

Under addition alone, graphs with the same partition remain equivalent under every word by Proposition 14.2. The empty word distinguishes different current partitions, so the full quotient has exactly five classes. Under addition and deletion, Proposition 14.3 gives eight classes.

The appearance domain I has thirteen members in this case. The empty graph contributes three component appearances. Each one-edge graph contributes two, giving six more. Each of the four connected graphs contributes one, giving four more. Thus $3 + 6 + 4 = 13$.

The full nonempty transformation semigroup has twenty-six elements. Each edge coordinate can be left unchanged, set to zero, or set to one. In a word, the last operation on that coordinate determines its constant value. There are 3^3 such coordinate descriptions, including the identity.

A nonempty word sets at least one coordinate and therefore cannot be the all-unchanged identity. Every other description is realized by setting its constant coordinates once. The number is therefore $3^3 - 1 = 26$. The empty word remains available in the word calculus as the zero-step identity map.

A finite verification program enumerates these objects and independently compares paired configurations under the operation repertoires. Its results agree with the five-class and eight-class calculations. The checks concern the stipulated model; the displayed proofs provide the general conditional reasoning.

14.7 Formation, Merger, Fission, and Reappearance

A selected history gives the components temporal appearances. This subsection uses one history to distinguish group formation from identity across changes.

Begin with the empty graph and apply

$$a_{12}, a_{23}, a_{13}, d_{12}, d_{13}, a_{12}.$$

The component history starts with three singletons, joins 1 and 2, then joins all three positions. Adding 13 changes internal structure without changing the component partition.

Deleting 12 from the triangle leaves all three positions connected through 13 and 23. Deleting 13 then leaves $\{1\}$ and $\{2, 3\}$. Adding 12 reunites all three positions. The graph trajectory records formation, internal reinforcement, fission, and reappearance under the connectedness criterion.

Whether the final three-position group is the same entity as the earlier one requires another criterion. Equality of membership sets would identify them. A criterion requiring uninterrupted connected continuity would distinguish the two intervals. Both can be represented, and the graph history makes their disagreement explicit.

A directed lineage can link components at consecutive steps when their vertex sets overlap. At fission, one component overlaps two successors. At merger, a successor overlaps several predecessors. The lineage therefore preserves directional continuation without satisfying the identity equivalence implied by treating all such links as sameness.

This history shows why a derived unit criterion does not settle every entity question. It constructs components at each configuration. The interpretation of their temporal continuity remains an additional commitment that can be tested against the purpose of the inquiry.

14.8 Coordination of the Six Categories

The same graph case can represent all six analytical roles. This subsection makes their connections and differences explicit.

The entity representation consists of component appearances and a declared continuity criterion. The relation representation includes direct edges and derived path connectivity. These differ: two positions can be connected without a direct edge between them.

Events are indexed applications of additions and deletions. Adding a present edge can be a silent graph transition and still remain one application in the history. Exact event count requires retaining positions or a counter, because the update maps are idempotent.

Attributes can concern edge presence, component size, or connectivity of a selected pair. Edge presence fails to descend to the component partition, because connected graphs can have different edges. Pairwise connectivity does descend: it is determined by whether the pair occupies one block.

States can be full graphs or reduced partitions when the repertoire permits. The addition-only case proves that a reduced state can be exact for a task. The deletion counterexample proves its limit.

Processes are operation words and graph trajectories. Their products can forget repeated applications and intermediate partitions. The selected history therefore retains information needed for formation and continuity questions that the final graph omits.

These coordinated uses establish representability under declared commitments. They leave open whether the social meaning of a graph component is appropriate in a particular case. The formal clarity helps locate that remaining question.

14.9 Philosophical Interpretation and Alternative Unit Criteria

Connectedness is one possible way to derive a collective unit. This subsection evaluates its strengths and limitations without giving it universal authority.

The criterion has a clear construction and definite consequences. It can identify groups from relations among supplied positions, explain mergers through new links, and describe fission through loss of connections. It also supports precise questions about which internal differences matter to persistence.

A social unit can require more than connectedness. A chain may connect participants whose commitments are incompatible. A single weak link may join otherwise independent groups. An institution may recognize a group despite sparse interaction, or participants may coordinate across a structure that this graph omits.

Alternative criteria can therefore use direction, thresholds, shared procedures, repeated interaction, or institutional records. Each criterion changes the unit construction and its evidence requirements. A thresholded graph, for example, needs a scale and a threshold whose meaning must be argued.

The model's philosophical contribution is narrower and constructive. It shows how a relational configuration can support derived units without taking those collective units as initial carrier elements. It also shows that the chosen relation structure and criterion are indispensable inputs.

This leaves ontology suspended. The result concerns a representational and explanatory strategy. An argument that all entities are ultimately relational formations would need to extend the reasoning beyond the fixed elementary positions and this specific grouping criterion.

The case nevertheless advances the original question beyond an inventory of limitations. It provides a workable construction, proves its dynamical adequacy under stated conditions, and demonstrates exactly where a richer representation becomes necessary.

15 Coordination of the Six Category Records

This section develops a joint representation in which the six categories can cooperate without losing their different roles. It specifies the domains for appearances, occurrences, states, and processes; derives the coherence of event chains with the action; and examines which quotient maps should commute. A central countercondition is that a continuing entity can occupy different states. Requiring every record to descend through every quotient would therefore impose an unjustified identity criterion.

15.1 Distinct Domains in a Common Inquiry

The category treatments introduced several kinds of record. Their coordination begins by preserving their domains. This subsection collects the necessary maps and explains why a single undifferentiated carrier of descriptions would obscure their responsibilities.

Retain a configuration carrier S , labeled operations T_b , and an action-compatible quotient $q : S \rightarrow Q$. Add an appearance domain I with anchor $p : I \rightarrow S$. The anchor associates an appearance with its represented configuration. Let E_{id} be a separate equivalence on I , with quotient $e : I \rightarrow I/E_{\text{id}}$.

Add an occurrence domain Ω with source, target, and label maps

$$\text{src}, \text{tgt} : \Omega \rightarrow S, \quad \text{lab} : \Omega \rightarrow B.$$

These maps describe how an occurrence record relates to the operation model. The occurrence itself remains independently indexed.

A process record can retain a seed and an ordered chain of occurrences. Attributes can be attached to configurations, appearances, occurrences, or processes according to the question. Incidence relations likewise need a declared participant domain and context.

The collection is an enriched construction. An inquiry can use only the components it needs. The purpose is to expose their relationships, rather than to require a maximal record for every social fact.

15.2 Occurrence Coherence with the Action

An occurrence described as an instance of a modeled transition should agree with that transition's source and target. This subsection states the condition and its evidential status.

For the relevant transition occurrences, require

$$\text{tgt}(\omega) = T_{\text{lab}(\omega)}(\text{src}(\omega)). \quad (33)$$

The equation links the occurrence description to the action model. It is a coherence assumption about the representation.

The assumption does not establish that the occurrence happened. A hypothetical event record can satisfy it, and an actual record can fail it because the model, coding, or evidence is incomplete. The failure needs interpretation before it becomes a historical conclusion.

An attempted operation can leave source and target equal. The equation permits such a record when the map fixes the source. A blocked attempt requires an availability or failure convention. An unattempted operation is a possibility in the model, rather than an occurrence record by default.

Several occurrence tokens can share the same source, label, and target. Repeated idempotent attempts supply a direct example. The coherence equation therefore need not make the description map injective.

The distinction protects occurrence identity from effect compression. A record system can retain the token, its evidential status, and its transition description separately. Each component can be revised without silently replacing the others.

15.3 Event Chains and Process Endpoints

Compatible occurrence records can be assembled into a process chain. This subsection derives the relation between the chain's endpoint and the composite action of its labels.

Let a process record be $h = (x, \omega_1, \dots, \omega_k)$, where the seed satisfies $\text{src}(\omega_1) = x$ when $k > 0$, and adjacent occurrences satisfy

$$\text{tgt}(\omega_i) = \text{src}(\omega_{i+1}).$$

The empty chain retains its seed explicitly.

Proposition 15.1 (Coherence of process endpoints). *If every occurrence in the chain satisfies Equation (33), then its final endpoint is*

$$T_{\text{lab}(\omega_k)} \cdots T_{\text{lab}(\omega_1)} x.$$

Applying q gives the same outcome as successively applying the induced quotient operations to $q(x)$.

Proof. For one occurrence, the endpoint formula is exactly the coherence equation. Suppose it holds for the first i occurrences. Adjacency identifies their endpoint with the source of ω_{i+1} . The next coherence equation applies the next labeled map to that endpoint. Induction proves the first statement.

For the quotient statement, action compatibility gives $q(T_b z) = \bar{T}_b(q(z))$ at every step. Substituting this equality successively through the endpoint formula gives the quotient trajectory. The empty chain has endpoint x and quotient endpoint $q(x)$. $\square$

The proof identifies the exact role of chain completeness. If two recorded occurrences have mismatched endpoints, an unrecorded intermediate step may be needed. Silently joining them would assert a transition that the record has not represented.

A process can also have a coarser segmentation whose stages contain several occurrences. Associativity preserves the endpoint when the underlying chain is coherent. It leaves the segment boundaries and their meaning to a separate record.

The result coordinates event, process, and state descriptions while preserving their different information. A shared endpoint does not recover the chain, and a coherent chain does not establish actual occurrence without evidence.

15.4 Entity Identity and State Anchors

Entity identity and state equivalence occupy different domains. This subsection examines their relationship and proves why state constancy should not be imposed on identity by default.

The relation E_{id} is a subset of $I \times I$. The state relation $\ker q$ is a subset of $S \times S$. A direct intersection has no intended role in this typed construction. A set-theoretic operation on their encodings would not supply the missing domain alignment.

The anchor permits a pullback:

$$uE_{\text{state}}v \iff q(p(u)) = q(p(v)).$$

This is an equivalence on appearances. Its intersection with E_{id} is mathematically defined, but it adds state agreement to the identity criterion.

Proposition 15.2 (State assignment to an entity class). *The map $q \circ p : I \rightarrow Q$ factors through the entity quotient e if and only if*

$$uE_{\text{id}}v \implies q(p(u)) = q(p(v)). \quad (34)$$

Proof. If $q \circ p = \bar{p} \circ e$, equal entity classes receive the same value under $\bar{p}$, proving necessity. Conversely, if the displayed condition holds, assign to each entity class the common state class of its appearances. The value is representative-independent and gives the required factorization. $\square$

A continuing organization can have appearances before and after authorization, resource acquisition, or activity. Those appearances can belong to different state classes. The factorization condition then fails even when the identity criterion is appropriate.

The failure is not a defect in tracking change. It shows that current state is generally a property of an entity appearance or an entity at a context, rather than a timeless function of entity identity alone.

The pair record

$$u \mapsto (e(u), q(p(u)))$$

retains both answers on the appearance domain. One entity can appear in several states. Repeated appearances with the same pair can still require an occurrence or time index. The pair is not automatically a complete appearance identifier.

15.5 Attributes and Incidence across Records

Attributes and relations can be transferred between record levels only under their own descent conditions. This subsection coordinates those conditions with identity and state.

A state attribute $a : S \rightarrow V$ descends through q precisely when it is constant on state classes. An appearance attribute descends through e precisely when it is constant on identity classes. The two tests concern different domains and can have different answers.

A changing attribute can be retained on appearances or on entity-context pairs. Requiring it to be constant over the entity's entire history would replace a dynamic attribute with a timeless one. That change needs an explicit research purpose.

For incidence at a configuration, let $I_s = p^{-1}(s)$ and $R_s \subseteq I_s^n$. A relation on reidentified participants requires saturation under the restricted identity criterion in each coordinate. The relation section derives that condition.

Context matters because the same participants can enter or leave a relation over time. A timeless relation on their global identity classes can merge those changes unless the context is retained in the predicate or its domain.

The compatibility tests should therefore follow the question. Some maps should commute because they report the same target through different records. Other maps should remain distinct because their targets change through time or belong to different analytical roles.

15.6 Shared Subjects and Category Overlap

One subject can support several category uses in a single inquiry. This subsection explains that overlap without reducing all roles to one identity criterion.

An agreement can be tracked as a continuing referent, described through obligations, located in signing or revision occurrences, assigned attributes, reported in a state, and reconstructed through a process. The records answer different questions about the same inquiry.

A process can itself become a referent in another question. An investigator can track a negotiation across stages while also using its internal occurrences to explain an outcome. The entity-like tracking role and the process role can coexist through different maps.

A relation can likewise be named, revised, or entered into another relation. A finite model can add records of those relation instances to an appropriate participant domain. The enrichment should state the new roles and their identity criteria.

Such overlap creates no requirement that every object occupy one exclusive category. An exclusive classification would be an additional theoretical proposal whose usefulness needs argument.

The six-category repertoire also remains open to challenge. A research question can expose a missing role or an inadequate distinction within a current role. The appropriate response is to specify the missing information and test a revised construction.

15.7 A Compatibility Audit for a Joint Model

A joint model can be checked through a small set of explicit obligations. This subsection organizes them by the map whose use is being claimed.

For operation-to-state transfer, verify that the state equivalence is preserved. For occurrence-to-transition transfer, verify the source, label, and target equation. For process construction, verify adjacency or state the convention for unrecorded gaps.

For attributes and incidence, check the appropriate fibre constancy and saturation conditions. For historical queries, check the record factorization criterion. For entity tracking, state the continuity criterion independently of the dynamic observation quotient.

The audit should also identify conditions that are intentionally not imposed. A continuing entity need not occupy one state. A quotient-silent event need not disappear from the occurrence record. A process with the same endpoint need not share a formation history.

A failed coherence check has several possible causes. The model can omit a relevant variable, the records can refer to different contexts, or the proposed transfer can be inappropriate for the question. The failure should be investigated at that level.

This method provides a constructive alternative to a single grand equation for all six categories. It specifies the maps and conditions that coordinate their particular uses, while preserving the distinctions that the inquiry needs.

15.8 Representation and Factual Warrant

The coherence results concern a model and its records. This final subsection states the additional requirement for a claim about social facts.

An internally coherent occurrence chain can be hypothetical or incorrectly recorded. An entity criterion can be consistently applied and still poorly suited to the practice under study. An attribute can descend through a quotient while its coding rule remains disputed.

Factual warrant requires a connection between the representation and the case. Sources, observation, interpretation, and comparison design supply that connection. The formal audit identifies where they enter.

The distinction permits a model to remain useful during disagreement. Several candidate histories or identity criteria can be represented and their consequences compared. The research state can preserve the alternatives until evidence or argument supports a refinement.

The coordinated category account therefore combines mathematical discipline with epistemic openness. Its maps make commitments explicit, its proofs establish conditional consequences, and its records retain the questions that those consequences leave unresolved.

16 Philosophical Scope and Factual Warrant

This section examines the philosophical commitments of the category representations and the evidence needed for their use in social inquiry. It develops ontological suspension, distinguishes formation from description and classificatory uptake, and considers historical explanation, identity, and evaluation. The aim is to preserve the questions that the formal constructions clarify without allowing their mathematical consistency to decide those questions by itself.

16.1 Ontological Suspension and Analytical Commitment

The six categories serve analytical roles in an inquiry about social facts. This subsection explains what ontological suspension permits and what it leaves the modeler responsible for.

The framework can ask how an entity is reidentified, how a relation is formed, how an occurrence is recorded, how an attribute is assigned, how a state supports updates, and how a process is reconstructed while leaving ultimate ontological priority unsettled. The questions concern representational and explanatory commitments that can be investigated within a declared scope.

Suspension does not make every modeling choice neutral. Selecting a carrier, identity criterion, or observation can privilege some distinctions and obscure others. Those choices can shape the conclusions of an inquiry even when the author withholds a metaphysical claim about what fundamentally exists.

The appropriate response is to make the choices inspectable. The entity section states its continuity criterion. The attribute section states its coding and scale. The state section states its observation and update responsibilities. The process section retains the relation between a sequence and its compressed effect.

The inherited series title connects this paper to Part I. The present subtitle and argument specify the narrower task of modeling six categories of social facts. A successful construction establishes an analytical possibility under its assumptions. An argument about the ultimate nature of social reality would require further premises.

The suspension therefore has a positive methodological role. It allows competing representations to be developed and criticized without deciding their metaphysical hierarchy in advance. Their adequacy can first be tested through the questions they answer and the distinctions they preserve.

16.2 Formation, Description, and Uptake

A pattern can form, become distinguishable in a description, and enter subsequent action through recognition or classification. This subsection separates those achievements and examines their possible interaction.

Formation concerns the development of the practice, organization, or dependence pattern being studied. Description concerns a record through which an investigator or participant distinguishes it. Uptake concerns the use of that description in further activity.

These achievements can occur in different orders. A practice can persist before an official category recognizes it. A proposed category can precede the organization it is intended to support. A description can become consequential when an institution uses it to allocate access or select a procedure.

The graph illustration derives a candidate unit from a declared connectivity rule. That construction identifies a partition of a relational configuration. A claim that participants recognize those components or act through them requires an additional uptake model and evidence.

The distinction prevents two opposite shortcuts. A new description need not create a new social organization. A description can nevertheless contribute to later formation when participants respond to it. The response must be represented explicitly before its effects are inferred.

A comprehensive account can therefore connect the three roles without identifying them. It can record a formation history, a descriptive rule, and a response process, then examine their dependencies and disagreements. The next subsection gives one conditional representation of the response process.

16.3 Classification-Mediated Updates

A classification can enter dynamics through a rule selecting subsequent operations. This subsection defines that enrichment and derives its compatibility with a state quotient.

Let $o : S \rightarrow Y$ be a classification or observation, and let $\pi : Y \rightarrow B$ be a deterministic response rule. Define

$$U_\pi(x) = T_{\pi(o(x))}(x). \quad (35)$$

The rule selects an operation from the observed value and applies it to the configuration.

Proposition 16.1 (Compatibility of an observation-mediated response). *If an equivalence E is preserved by every T_b and $E \subseteq \ker o$, then E is preserved by U_π . The response therefore has a well-defined induced update on S/E .*

Proof. If xEy , containment in $\ker o$ gives $o(x) = o(y)$. The response rule selects the same label $b = \pi(o(x)) = \pi(o(y))$. Preservation by that map gives $T_b x E T_b y$, which is the required relation between $U_\pi(x)$ and $U_\pi(y)$. $\square$

The theorem is conditional on uptake. The classification o and response rule π are additional inputs. A description's existence does not establish that an actor uses it, nor does a formal response rule establish its legitimacy or empirical accuracy.

The response map can lie outside the original semigroup generated by the primitive maps. On $\{0, 1\}$, let the primitives be identity and flip. Observe the state, choose flip at zero and identity at one. The response sends both states to one, while every old primitive composition is a bijection.

The example shows that conditional selection can introduce a noninvertible global update even when every selected primitive is invertible. The selection rule is part of the enlarged mechanism. The result supplies no contradiction to an earlier theorem about one fixed semigroup.

This construction provides a limited formal account of classificatory uptake. A substantive study must identify the classification, the responding actors, their information, and the actions through which the description becomes consequential.

16.4 Social Facts and Assertions About Them

A representation can encode a claim about a social condition without establishing that condition. This subsection separates the record, its asserted content, and the warrant for that content.

A database can contain an authorization entry. That is a fact about the record under a suitable evidential description. The proposition that authorization holds under the relevant practice is another claim. A disagreement can concern the entry's accuracy, the coding rule, or the governing criterion.

The same separation applies to events. An occurrence record can assert that an agreement was signed or an activity attempted. Its presence in the model supplies a representation of that claim. Evidence is needed to connect it to an actual episode.

Identity claims can also be recorded without being settled. Two sources can disagree about whether an organization continues through a change. The model can preserve both criteria and compare their consequences. A single identifier should not silently resolve the dispute.

PROV-DM offers a useful comparison through its separate treatment of entities, activities, and derivation records [6, Sections 2.1 and 5.2.1]. The present framework has its own category commitments. A translation between the schemes requires an account of the meanings and information preserved.

Huang's conformance specification likewise emphasizes declared encodings and partitions in reconstructing generative structure [4]. The category audit adds a complementary responsibility: identify the base data, added records, and interpretive premises on which each claim depends.

16.5 Historical Explanation and Predictive Sufficiency

A record sufficient for future observation can remain inadequate for historical explanation. This subsection examines why both uses can be legitimate and how they should be coordinated.

Two histories can reach the same configuration and agree under all selected future observations. A query about formation mode, contribution, or an earlier alternative can still distinguish them. Predictive equivalence therefore leaves historical questions open unless their information is included in the task.

A historical explanation can seek why the present arrangement arose, which processes sustained it, or how participants understood a transition. Those questions can require records that a compressed predictor omits. Their relevance needs a substantive argument beyond future-output difference.

Conversely, a detailed history can contain distinctions irrelevant to a selected forecast. The framework permits a smaller operational record while preserving the larger history in an archive. The relation between the records should remain traceable.

If a later operation consults a historical field, the predictive problem changes. The sufficient state must retain that field or equivalent information. The historical-relations companion paper develops this dynamic coordination in a finite example.

The philosophical lesson is that historicity has several roles. A process can produce a present condition, a record can preserve its formation, and later activity can respond to that record. The roles can interact without becoming one unqualified claim that history determines every future difference.

16.6 Identity, Behavior, and the Boundary of Selfhood

The entity construction tracks appearances through a declared equivalence or lineage. This subsection states what such tracking can contribute to identity questions and where further philosophical work is needed.

Behavioral equality can hold between separately identified entities. A continuing entity can also change its behavior. The continuation quotient therefore cannot replace every identity criterion. Its equivalence concerns a selected observation task.

A lineage records directional continuity through change and can branch through fission or merge through combination. Turning that relation into numerical identity requires additional choices. Undirected connectivity can join branches that the inquiry needs to distinguish.

The same caution applies to a model of a self. Embodied continuity, first-person appropriation, social recognition, and practical responsibility can disagree. A causal chain or an observation profile can contribute to an account of those questions while leaving their resolution open.

A commitment to a future self can matter precisely because future preferences or abilities may change. A model that identifies identity with unchanged behavior would miss that possibility. The relevant continuity criterion needs an independent philosophical argument.

The present metamodel supplies formal preparation for such an inquiry. It distinguishes appearance, state, history, and lineage and identifies their information requirements. An account of lived selfhood remains a further research project.

16.7 Evaluation and the Selection of Descriptions

Descriptive choices can influence what an inquiry evaluates, while the formal results leave evaluative standards open. This subsection examines that relation.

An observation of whether activity occurred differs from an observation of its distribution, burden, or meaning. A quotient preserving the first can merge distinctions needed by the others. Selecting the observation therefore shapes the questions the model can answer.

The choice can be justified by a bounded purpose. A completion flag can be sufficient for one technical calculation. Its use becomes problematic when the result is extended to a broader judgment without retaining the missing dimensions.

Generative and reproductive roles also remain evaluatively open. Formation of a new route can support or undermine participants' aims. Maintenance of an existing relation can preserve cooperation or sustain an unwanted dependency. The formal role alone supplies no fixed ranking.

A plural description can retain several observations and their joint continuation requirements. That construction preserves the selected information but does not decide how conflicting values should be weighed.

The framework therefore separates a formal measure from the reasons for using it. Cardinality, robustness, and predictive sufficiency can inform an evaluation when their relevance is argued. They should remain linked to the standards and participants that give the comparison its meaning.

16.8 Adequacy through Explicit Revision

The metamodel's adequacy depends on its capacity to support and survive criticism in a specified inquiry. This final subsection identifies the kinds of revision that the preceding distinctions permit.

A failed update prediction can expose an omitted condition or an overly coarse state. A disputed identity can expose an inadequate continuity criterion. A history collision can expose insufficient records for a target query. These failures belong to different parts of the representation.

A revised construction should retain the earlier formulation and explain the change. Adding a variable, refining a quotient, or changing an operation repertoire can repair one problem while introducing new assumptions. Those assumptions should remain visible.

The category framework is useful when it makes the repair more precise. It should help identify which record needs enrichment, which proof needs rechecking, and which interpretive claim remains unsupported.

Its philosophical contribution is therefore a disciplined coordination of questions. The six categories organize different responsibilities within social inquiry, and the formal constructions show how some of those responsibilities can be met under declared assumptions. The remaining tensions continue to guide research.

17 Changes of Regime and Extensions of the Dynamics

This section examines changes to the mathematical setting that the six category representations may require. The discussion first treats transitions between declared regimes of description. It then introduces a finite probabilistic extension and derives its consistency condition. Partial availability, concurrent activity, and history-dependent responses complete the comparison. Each extension identifies its additional inputs and the conclusions that survive under those inputs.

The fixed action supplies a controlled starting point. A more detailed inquiry can reveal a changing classification, a restricted operation domain, uncertain outcomes, or several activities occurring during an overlapping interval. These discoveries pose different mathematical problems. Treating them all through a larger carrier can sometimes produce a formally unified representation. The substantive value of that representation depends on the interpretation of its new coordinates and on the evidence available for their updates.

17.1 A Declared Family of Descriptive Regimes

A regime can specify a vocabulary of observations, a criterion for following entities, or an operation repertoire. To represent anticipated changes among such regimes, the model must retain which regime is active and how a transition alters the represented situation. The following construction makes those commitments explicit.

Let $\mathcal{R}$ be a set of regime labels. For each $r \in \mathcal{R}$, let S_r be a carrier and let E_r be an equivalence on that carrier. The tagged union

$$X = \coprod_{r \in \mathcal{R}} \{r\} \times S_r$$

keeps the regime and its state together. A point (r, x) records a state under a specified regime. Define

$$(r, x) E (s, y) \iff r = s \text{ and } x E_r y.$$

This definition compares states within a common regime. It leaves cross-regime alignments to an additional construction.

A primitive operation can now be a partial map $U_b : X \rightharpoonup X$. Whenever it is defined, write

$$U_b(r, x) = (\kappa_b(r, x), F_b(r, x)), \quad F_b(r, x) \in S_{\kappa_b(r, x)}.$$

The map κ_b chooses the successor regime, while F_b supplies the successor state in its corresponding carrier. This typing matters: a state code valid under one regime may lack a declared interpretation under another. Even when the carriers happen to use the same numerical codes, their alignment requires a modeling decision.

Proposition 17.1 (Descent across declared regimes). *A partial operation U_b induces a partial operation on X/E if and only if E -related inputs have the same definedness and, whenever their outputs are defined, the outputs have equal regime tags and equivalent state coordinates under that common successor regime.*

Proof. A partial quotient operation needs a domain consisting of whole E -classes. Thus definedness must agree for related representatives. For an admitted class, its successor class must also be independent of the representative. By the definition of E , this requires equality of the successor regime tags and equivalence of the successor state coordinates under that tag. These requirements are necessary. Conversely, their satisfaction defines an unambiguous domain on X/E and one successor class for each admitted input class. The resulting partial map is the required quotient operation. $\square$

Internal consistency of each regime supplies only part of this condition. A regime can preserve its own state distinctions under its ordinary operations while a transition to another regime separates states previously identified. The transition therefore deserves its own proof. This distinction is relevant whenever a modeling account proposes to accommodate changing institutions simply by allowing its parameters to change. The parameter change can alter the very distinctions needed to determine a successor.

17.2 Refinement of Descriptions and Recovery of Omitted Information

A common-carrier example isolates the cost of moving to a finer description. Suppose $S_c = S_f = S$, where c and f name coarse and fine regimes. Let a regime switch preserve the raw coordinate:

$$U(c, x) = (f, x).$$

Related inputs in the coarse regime have $x E_c y$. Their outputs are related in the fine regime exactly when $x E_f y$. Consequently the switch descends precisely under

$$E_c \subseteq E_f.$$

The direction of this inclusion deserves attention. A finer description has smaller equivalence classes, so its equivalence typically satisfies $E_f \subseteq E_c$. A strict refinement therefore generally violates the condition needed to compute the switch from a coarse class alone.

For example, take $S = \{0, 1\}$. The coarse regime identifies both points, while the fine regime distinguishes them. The unique coarse class supplies insufficient information to choose between the two fine classes. A raw-state model retaining x can execute the switch without difficulty. Its ability comes from information retained in that model. The coarse record cannot acquire the missing bit merely through a change of notation.

This example separates three tasks in the interpretation of social change. A descriptive revision can introduce a distinction already represented in the raw data. An evidential revision can recover a distinction through additional observation or historical records. A substantive transition can change the represented situation itself. The three tasks can coincide in a case, yet their justification differs. A revised institutional classification may use archived information, require new documentation, and affect future action. Each component should receive its own update or evidential account.

The construction also clarifies the status of anticipated novelty. A regime catalog can include several possible successor vocabularies and explicit transitions between them. Such a catalog models revision within its declared possibilities. An account of how investigators or participants develop a previously unrepresented vocabulary requires additional work. Including a symbol named *new regime* leaves the content and interpretation of that regime to be supplied.

Cross-regime entity identity raises a related issue. The tagged equivalence above keeps regimes separate because it serves a state description. An organization may remain the same organization under two descriptions. A cross-regime alignment can represent that continuity through matched entity appearances, documentary succession, or another declared criterion. Merging regime tags in the state quotient would conflate this identity question with the consistency of the descriptive dynamics.

17.3 Finite Probabilistic Transitions

Some inquiries seek probabilities of successor states. A stochastic transition model adds information beyond deterministic reachability. The finite case permits a complete statement of the exact aggregation condition and a direct comparison with the quotient action.

Let S be finite and let K be a stochastic matrix, with $K(x, y) \geq 0$ and $\sum_{y \in S} K(x, y) = 1$. Let $q : S \rightarrow Q$ be a surjective description. For a quotient value $C \in Q$, its block is $q^{-1}(C)$. The probability of entering this block from x is

$$K(x, q^{-1}(C)) = \sum_{y: q(y)=C} K(x, y).$$

An exact quotient transition must assign the same block probability to every state receiving the same current description.

This requirement is the classical finite-state lumpability condition. Kemeny and Snell give the block-sum criterion and its proof in Theorem 6.3.2 of *Finite Markov Chains* [5]. The result is used here as an established antecedent. Its role is to specify the added consistency obligation when a category description is asked to support probabilistic prediction.

Proposition 17.2 (Exact finite probabilistic aggregation). *A representative-independent quotient transition matrix exists for every initial distribution if and only if*

$$q(x) = q(x') \implies \sum_{y: q(y)=C} K(x, y) = \sum_{y: q(y)=C} K(x', y) \quad (C \in Q).$$

Under this condition,

$$\bar{K}(q(x), C) = \sum_{y: q(y)=C} K(x, y).$$

For a distribution μ on S , write $\mu_q(C) = \sum_{q(x)=C} \mu(x)$. Then

$$(\mu K)_q = \mu_q \bar{K}.$$

Proof. Initial distributions concentrated at two representatives of the same block induce the same quotient initial distribution. A common quotient transition therefore requires equal probabilities into each

target block, proving necessity. Under the stated condition, the displayed formula defines $\bar{K}$ independently of the representative. Its entries are nonnegative, and its row sums equal one because the blocks partition S . Finally,

$$\begin{aligned} (\mu K)_q(C) &= \sum_x \mu(x) \sum_{y:q(y)=C} K(x, y) \\ &= \sum_{A \in Q} \sum_{x:q(x)=A} \mu(x) \bar{K}(A, C) \\ &= \sum_{A \in Q} \mu_q(A) \bar{K}(A, C). \end{aligned}$$

This is the required commutation identity. □

The hypothesis concerns every initial distribution. A successful aggregation under one selected distribution can depend on how probability is distributed among hidden states inside each block. The proposition gives a stronger guarantee: the quotient transition remains valid independently of that hidden mixture. A project using a weaker guarantee should state the relevant initial distributions and the resulting restriction of scope.

17.4 Possible Successors and Their Probability Distributions

A small counterexample shows why possible successor classes give insufficient information for probabilistic aggregation. Let $S = \{0, 1, 2\}$, with blocks $A = \{0, 1\}$ and $B = \{2\}$, and set

$$K = \begin{pmatrix} 4/5 & 0 & 1/5 \\ 1/5 & 0 & 4/5 \\ 0 & 0 & 1 \end{pmatrix}.$$

From either state 0 or state 1, the possible successor blocks are A and B . Their probabilities of reaching B are $1/5$ and $4/5$, respectively. The current description A therefore preserves the set of possible successor blocks while omitting a distinction essential to the next-step probability.

For social interpretation, the difference concerns the claim made by a model. An account can state that two arrangements permit the same classes of outcome. An account can also state that those outcomes have equal probabilities under specified conditions. The second statement demands further information. The deterministic architecture supplies a disciplined language for the first kind of possibility comparison; it leaves the probability assignment to an enrichment with its own evidence.

A family of primitive kernels K_b supports finite operation sequences when the same block condition holds for each primitive. Applying the commutation identity successively gives the quotient distribution after any declared finite word. This sequential result presumes that the kernel assigned to the next primitive is appropriate to the represented state. If unrecorded history changes the next-step probabilities, the chosen state is incomplete for that model.

The deterministic case appears as a special instance. For a map $T_b : S \rightarrow S$, define

$$K_b(x, y) = \mathbf{1}_{\{T_b(x)=y\}}.$$

Its mass in a target block is one exactly when the deterministic successor belongs to that block. Equality of block masses is therefore equivalent to equality of successor descriptions. The probabilistic condition recovers preservation of $\ker q$. This relation explains the common structural role of the two criteria while keeping their empirical inputs distinct.

17.5 Partial Availability and the Meaning of Failed Operations

An operation may be executable only on part of the state space. Representing it by a partial map makes availability visible in the domain. A quotient partial operation then requires both constant definedness

within each class and a common successor class whenever execution is available. These are distinct requirements. Equivalent outcomes following successful execution leave open whether execution is possible at each representative.

A total-map representation can encode failure by adding failure states or by declaring that an unavailable operation leaves the state unchanged. Either convention can be useful for a specified question. Their meanings differ. A failed attempt can consume resources, produce a documentary record, alter expectations, or trigger a sanction. A self-loop on the original state can represent such an attempt only if those consequences are outside the declared state question or are retained elsewhere.

The occurrence record is particularly valuable here. Two situations may have the same resulting state while differing in whether an operation was attempted. An event ledger can retain that difference even when the state dynamics compress it. If future responses depend on the attempt, its record must also enter the information available to subsequent updates. The category architecture identifies the relevant coordination obligation without forcing every event detail into every state description.

The feasible-path model developed separately in the paper portfolio supplies more detailed conditions involving permissions, resources, and execution prefixes. The present partial-map discussion establishes the simpler point: availability has a formal location, and that location must be preserved by the description when the description is used to determine executable operations.

17.6 Concurrent Activity and Temporal Resolution

A word orders its primitive occurrences. That order can represent an observed sequence, an imposed scheduling convention, or an approximation at a chosen temporal resolution. Concurrent activity requires an interpretation of how the sequential description relates to overlapping occurrences.

Commutation of two state transformations gives a limited result. If $T_a T_b = T_b T_a$, reversing their order preserves their composite effect on every state in the declared carrier. The equation concerns transformation effects. Occurrence times, exposure intervals, participants' knowledge, and temporary resource conflicts may still differ. A representation retaining only the composite transformation omits those questions.

Conversely, two operations can overlap in clock time while their effective dependencies impose an order on particular subevents. An authorization may be issued during a resource transfer yet become usable only after a confirmation step. A sequential approximation must specify which events it treats as atomic and which timing relations it suppresses. The choice is assessed against the outcome or historical query.

A fuller concurrent model would need an explicit account of compatible occurrences, temporal constraints, and composition. The present architecture supplies two starting resources: occurrence-level records preserve individual executions, and process-level records preserve their organization. The exact relation between such records and the semigroup action remains a research obligation when concurrency carries substantive content.

17.7 History-Dependent Responses and Expanded State

History dependence can be represented by storing a record that determines the information used in the next update. The resulting state might include the current condition, the most recent formation mode, and a selected event summary. This strategy is useful when the retained record has an explicit update rule and a clear relation to the questions being answered.

The cost of the strategy appears in the size and interpretation of the record. A history coordinate can contain an entire sequence, yielding an exact but large state space. A finite summary needs a preservation proof for the selected future and historical readouts. Sufficiency for one query gives a local achievement. Adding another query or operation can require refinement.

The same issue affects institutional learning or classificatory uptake. A response may depend on a participant's remembered experience, an institution's archive, or an investigator's classification. Each

is a different information source. A response function $\pi(o(x))$ gives a compact model only when the observation o contains the information actually used by the response rule.

These extensions support a common methodological conclusion. Enlarging the mathematics should make an additional commitment explicit and testable. A larger carrier, a stochastic kernel, or a regime tag gains explanatory value through its interpretation and the questions it helps answer. The six-category architecture organizes those commitments while leaving the choice of an adequate empirical model open.

18 Approximate Descriptions and Finite-Horizon Error

This section develops a quantified alternative to exact descriptive consistency. It introduces a metric on reduced descriptions, derives trajectory error from local approximation residuals, and examines the consequences for threshold classifications. It then explains why numerical tolerance differs from an equivalence relation and identifies the evidence needed to use an approximate model.

The exact quotient construction provides a clear baseline: equivalent configurations have equivalent successors under each declared operation. An empirical or computational model can aim for a controlled approximation when exact preservation is too demanding. That change requires a measure of discrepancy and a statement of how error propagates. A phrase such as *approximately the same state* leaves these commitments unresolved.

18.1 Metric Structure and the Meaning of Approximation

A metric supplies a nonnegative distance between descriptions, with zero distance identifying equal points, symmetry, and the triangle inequality. The triangle inequality will perform the main work in the derivation. The interpretation of the distance remains an additional modeling choice.

For numerical attributes, a distance can compare values expressed in declared units. For a vector of attributes, a distance can also combine differences across coordinates. Such a combination needs justification. Changing a unit or a coordinate weight can alter which descriptions count as close. The attribute analysis therefore remains relevant to the approximate state model.

A categorical difference may resist a useful graded interpretation. An occurrence can be correctly or incorrectly identified; two organizations can be assigned different identity claims. A numerical distance between their codes has meaning only through an independently specified comparison. The introduction of a metric should follow the question it is meant to answer.

Girard and Pappas develop a broader framework of metric comparisons and approximate simulation and bisimulation for transition systems [2]. That work provides an established antecedent for quantified approximation. The present section proves a narrower elementary bound for two deterministic evolutions following the same label sequence. It leaves a full comparison with the broader framework to later investigation.

18.2 Reduced Updates and Local Residuals

Let $T_b : S \rightarrow S$ be the declared operations of the detailed model. Let $q : S \rightarrow Q$ be a description map, where (Q, d) is a metric space. Proposed reduced updates are maps $\hat{T}_b : Q \rightarrow Q$. Surjectivity of q is unnecessary for the following bound.

The local residual compares two ways of taking one step: update the detailed configuration and then describe it, or describe the configuration and apply the reduced update. Assume that, for each b ,

$$d(q(T_b x), \hat{T}_b(qx)) \leq \delta_b \quad (x \in S). \quad (36)$$

The parameter $\delta_b \geq 0$ is a uniform upper bound on that discrepancy.

Also assume that the reduced update satisfies

$$d(\hat{T}_b u, \hat{T}_b v) \leq \lambda_b d(u, v) \quad (u, v \in Q), \quad (37)$$

with $\lambda_b \geq 0$. This condition controls how a discrepancy already present in the reduced description can change under the next operation.

The two assumptions address different sources of error. The residual is introduced even when the reduced input exactly equals qx . The Lipschitz factor governs propagation of an earlier discrepancy. Their combination gives a bound over a finite sequence.

Both assumptions require support. A fitted model can have small residuals on observed configurations while behaving differently elsewhere. The displayed uniform inequalities are mathematical premises. A finite sample becomes evidence for them only through a separate estimation, coverage, or structural argument.

18.3 Derivation of the Trajectory Bound

Fix a word $w = b_1 \cdots b_n$. Let the detailed and reduced evolutions be

$$x_i = T_{b_i} x_{i-1}, \quad z_i = \hat{T}_{b_i} z_{i-1}.$$

Their discrepancy at step i is

$$e_i = d(qx_i, z_i).$$

The initialization can itself be approximate, so e_0 is retained in the general result.

Proposition 18.1 (Propagation of local description error). *Under (36) and (37),*

$$e_n \leq \left(\prod_{j=1}^n \lambda_{b_j} \right) e_0 + \sum_{i=1}^n \delta_{b_i} \prod_{j=i+1}^n \lambda_{b_j}. \quad (38)$$

An empty product equals one.

Proof. Insert $\hat{T}_{b_i}(qx_{i-1})$ between the two descriptions being compared. The triangle inequality gives

$$\begin{aligned} e_i &\leq d(q(T_{b_i} x_{i-1}), \hat{T}_{b_i}(qx_{i-1})) \\ &\quad + d(\hat{T}_{b_i}(qx_{i-1}), \hat{T}_{b_i}(z_{i-1})) \\ &\leq \delta_{b_i} + \lambda_{b_i} e_{i-1}. \end{aligned}$$

The first term uses the local residual bound and the second uses the Lipschitz condition. For one step, the product-sum formula is exactly this inequality. Suppose the formula holds at $n-1$. Substituting it into $e_n \leq \delta_{b_n} + \lambda_{b_n} e_{n-1}$ extends every earlier product by λ_{b_n} and adds the new residual with an empty trailing product. Induction proves (38). $\square$

The formula retains the order of the operations. A residual introduced early is multiplied by the sensitivity factors of subsequent updates. A large factor late in the word can amplify several earlier errors. A small residual bound for each primitive therefore has to be interpreted together with the continuation.

The initialization term matters for uncertain current records. Even an exact reduced update can propagate initial discrepancy. Conversely, exact initialization removes that term while leaving the residuals introduced by the approximation itself.

18.4 Uniform Bounds and Horizon Dependence

Suppose $\delta_b \leq \delta$, $\lambda_b \leq \lambda$, and $e_0 = 0$. Since all factors are nonnegative, the general bound yields

$$e_n \leq \delta \sum_{k=0}^{n-1} \lambda^k.$$

The finite sum has the explicit form

$$e_n \leq \begin{cases} n\delta, & \lambda = 1, \\ \delta \frac{\lambda^n - 1}{\lambda - 1}, & \lambda \neq 1. \end{cases} \quad (39)$$

For $\lambda \neq 1$, multiplication of the finite sum by $\lambda - 1$ cancels its intermediate powers and leaves $\lambda^n - 1$. For $\lambda = 1$, the sum has n unit terms.

When $0 \leq \lambda < 1$, the bound is at most $\delta/(1 - \lambda)$. The reduced dynamics contract earlier discrepancies enough to bound their accumulation. When $\lambda = 1$, the bound grows linearly with the number of steps. When $\lambda > 1$, the bound can grow geometrically.

These are properties of the declared metric model. A claim of approximation adequate for one horizon therefore needs to specify the horizon. A longer continuation can cross the permissible error threshold even when each individual update appears accurate.

The distinction also affects model comparison. One reduced model can have smaller local residuals and larger sensitivity factors than another. The preferred model for a long horizon can differ from the preferred model for a single step. The derived expression makes the tradeoff available for inspection.

18.5 An Example Attaining the Growth Bound

The geometric expression can be attained exactly. Let $S = Q = \mathbb{R}$, let q be the identity, and consider one operation with

$$T(x) = \lambda x, \quad \hat{T}(z) = \lambda z + \delta, \quad x_0 = z_0 = 0,$$

where $\lambda, \delta \geq 0$. The local residual is exactly δ , and the reduced map has Lipschitz factor λ .

The detailed trajectory remains zero. The reduced trajectory satisfies

$$z_n = \delta + \lambda z_{n-1} = \delta \sum_{k=0}^{n-1} \lambda^k.$$

Since the values are nonnegative, the distance between the trajectories equals the displayed expression. The upper bound is therefore exact for this model.

With $\lambda = 2$, the error is $\delta(2^n - 1)$. A small constant local discrepancy can exceed any fixed tolerance if the horizon is sufficiently long. The example shows that a general inference from small one-step residual to uniform long-run accuracy needs additional assumptions.

The social interpretation remains limited. The scalar state is a mathematical counterexample to an approximation claim. It supplies no empirical assertion that a particular social process has the displayed sensitivity. A substantive application would need to justify the variables, update, and metric.

18.6 Threshold Decisions and Margins

An inquiry can use a numerical description to make a categorical decision. Let $g : Q \rightarrow \mathbb{R}$ satisfy

$$|g(u) - g(v)| \leq K_g d(u, v).$$

If $e_n \leq B_n$, then

$$|g(qx_n) - g(z_n)| \leq K_g B_n.$$

The metric bound therefore transfers to the readout.

Consider the classification $\mathbf{1}_{\{g \geq \theta\}}$. If

$$|g(z_n) - \theta| > K_g B_n, \quad (40)$$

the detailed and reduced classifications agree. When the predicted value exceeds $\theta + K_g B_n$, the detailed value remains strictly above the threshold. When it lies below $\theta - K_g B_n$, the detailed value remains strictly below it.

The margin condition is sufficient. Its failure leaves the classification unresolved by this bound. The actual classifications can still agree. Reporting agreement as guaranteed requires the appropriate margin or a stronger argument.

A small numerical discrepancy can matter near a threshold. If the classification enters a response rule, the two models can then select different subsequent operations. The same-word error theorem ceases to apply directly after that selection difference. A feedback approximation needs a further argument about policy stability or operation mismatch.

This observation connects approximation to social uptake. A descriptive error can remain small in its own metric while crossing a decision boundary that changes access, priority, or the next institutional step. The importance of the error depends on the use of the description.

18.7 Tolerance Relations and Transitive Closure

A common attempt to represent approximation identifies descriptions whenever $d(u, v) \leq \varepsilon$. The resulting relation is reflexive and symmetric, yet it can fail transitivity. Three real values 0, 0.6, 1.2 with tolerance 0.7 give an elementary example.

Taking a transitive closure repairs transitivity at a further cost. On $\mathbb{R}$ with any $\varepsilon > 0$, the closure identifies every pair of points. For given x, y , choose a positive integer n with $|x - y|/n \leq \varepsilon$. The sequence

$$x, \quad x + \frac{y - x}{n}, \quad \dots, \quad x + n \frac{y - x}{n} = y$$

joins them through ε -close pairs. Thus every pair belongs to the transitive closure.

A finite chain can exhibit the same problem at its own scale. Neighboring descriptions may each be within tolerance while the chain's endpoints are far apart. Closing the neighbor relation transitively merges the entire chain.

Consequently an equivalence constructed by closure can lose the original error guarantee. A partition into declared bins supplies another possible description, with its own boundary and diameter properties. The partition should be assessed through those properties, rather than through the tolerance relation it replaced.

This matters particularly for entity modeling. Approximate similarity of attributes can support a comparison among appearances. It leaves historical identity and continuity criteria to be supplied. An approximate state metric should not silently become an entity equivalence.

18.8 Evidence and the Scope of Approximate Adequacy

An approximate application should specify the distance, permitted discrepancy, residual bounds, update sensitivity, and horizon. It should also state which decisions depend on the description and whether those decisions have adequate margins.

Observation uncertainty and approximation residual are different inputs. A record may inaccurately measure the detailed state. A reduced update can also deviate from the detailed model even when that state is known. The general bound retains an initialization discrepancy, while a full measurement model may need uncertainty at several stages.

The theorem compares a fixed label sequence. Partial availability, adaptive choices, changing regimes, and stochastic transitions can require additional reasoning. These extensions should preserve the distinction between a mathematically derived bound and evidence that its assumptions hold in the application.

Approximation therefore broadens the architecture through quantified commitments. It permits a reduced description to be evaluated at a specified precision and horizon while retaining the exact construction as a reference point. The meaning and adequacy of that precision remain questions for the substantive inquiry.

19 Research Design and Revision Obligations

This section develops a proposed research design for evaluating the category architecture in a bounded social inquiry. It first specifies the unit of investigation and the competing representations. It then states evidence requirements for the six categories, explains how failures should be diagnosed, and identifies the conditions for extending or revising the model. The design remains prospective.

The conditional results establish relations among declared mathematical objects. Their use in social inquiry requires interpretations for those objects and evidence relevant to the resulting claims. A satisfactory study should make both levels visible. It should also retain discrepancies that reveal a poor representation, since those discrepancies can identify precisely which enrichment the inquiry needs.

19.1 A Bounded Arrangement and Its Comparison Questions

A first empirical study could follow one cooperative arrangement through formation, activation, and revision. The study would declare a temporal interval, a focal activity, and the organizations or other participants whose appearances enter the record. The cooperative example in Section 13 supplies a set of logical contrasts for designing such a study. Its binary states would be assessed and revised in light of the actual case.

The initial question should specify what the model is expected to recover. An inquiry into whether implementation occurred needs different evidence from an inquiry into the source of an authorization. An inquiry into the continuing identity of an organization needs a different comparison from one into the conditions enabling a particular action. These questions can share a case while retaining distinct success criteria.

A useful first comparison would involve three descriptions. A current-condition description would retain the variables initially proposed to determine the next operation. A history-enriched description would add the formation and occurrence information needed for selected historical questions. A feasibility-enriched description would add conditions that restrict execution. The comparison would assess each description against its declared task, including tasks for which its limitations are expected.

This design avoids rewarding a model simply for retaining more data. An expanded record gains value when the added information supports a stated question or repairs a demonstrated failure. It also carries costs: more demanding evidence, additional ambiguities, and a larger space of possible records. Those costs should be reported alongside gains in recoverability or consistency.

19.2 Evidence for Entity Continuity

Entity reidentification requires a declared relation among appearances and evidence relevant to that relation. The study would retain dated names or identifiers, membership records, succession claims, and the criteria used by the investigation. Participant assertions of continuity would receive their own record, allowing agreement and disagreement among sources to remain visible.

A discriminating case would compare unchanged membership after an interruption with changed membership under an uninterrupted institutional procedure. A membership-based criterion and a procedural-continuity criterion may give different answers. The inquiry should state which answer serves its focal question and what consequences follow from the alternative.

Fission and merger deserve explicit treatment. If one organization divides into two successors, a directed lineage record can preserve both successor relations. A single equivalence-based identity criterion must explain whether it selects one successor, regards both as new entities, or identifies a different continuing unit. The record should retain the lineage even when the focal identity criterion selects only part of it.

Research Obligation 19.1 (Identity criteria and evidential scope). State the appearances being compared, the criterion under which they count as the same entity, and the evidence supporting each required continuity relation. Preserve alternative criteria when they alter a substantive inference.

The resulting obligation concerns the adequacy of a representation for a question. It permits several well-specified identity criteria within one study. It also permits a criterion to be rejected when its consequences conflict with the distinctions the study has committed to retain.

19.3 Evidence for Relations and Their Formation

A relation record should specify its participants, roles, scope, and formation information. For a cooperative arrangement, these might include which party can request performance, which party bears a stated obligation, and which recorded procedure established the arrangement. The coding would remain a proposed representation until checked against the relevant case evidence.

A particularly useful comparison involves two candidate histories leading to the same current-condition record. If the historical question gives different answers under those histories, the current record is insufficient for that question. Documentary evidence may narrow the compatible history set. Its effect should be reported as a reduction of uncertainty, with remaining alternatives identified.

Prospective effects require another comparison. The study must specify the continuation or intervention under which a relational difference is expected to matter. Two arrangements may have the same currently observed implementation status while differing in their response to a request, review, or withdrawal. The model should declare which aspects of the comparison are held fixed and which change.

Research Obligation 19.2 (Historical warrant and continuation comparisons). Separate evidence that a relation was formed through a specified history from evidence that a selected continuation distinguishes its potential effects. State the comparison assumptions needed to attribute that difference to the relation.

This separation protects a central insight of the paper. A relation can remain relevant through unrealized continuations, while its actual formation remains a historical question. Evidence supporting one side can leave the other unresolved.

19.4 Evidence for Occurrences and Attempts

Occurrence records require a method for distinguishing individual events even when their transition descriptions coincide. The study would retain source identifiers, date or interval information, operation labels, and links among reports believed to concern the same occurrence. Conflicting reports would remain separate until their relation is assessed.

Repeated applications of an idempotent operation provide a decisive test of the chosen record. If two authorization attempts leave the represented condition unchanged after the first, a state-only record may merge them. A question about frequency, administrative burden, or procedural compliance requires the occurrence distinction.

Attempts and completions also need separate coding. A request can be submitted, rejected, withdrawn, or left unresolved. The relevant event boundary depends on the question. A record that counts only completed implementation can be adequate for an outcome query while being inadequate for an inquiry into barriers to implementation.

Temporal precision should match the inference. An ordered record can establish that one reported event precedes another without establishing the duration between them. Coarse dates may leave order unresolved. The study should retain those limits through partial temporal information or explicit uncertainty, rather than assign a convenient total order without support.

19.5 Evidence for Attributes and Measurement Choices

Attribute coding requires the subject, value domain, interpretation, and definedness conditions to be declared together. An investigator should be able to identify whether a missing value reflects unavailable evidence, an inapplicable predicate, a disputed classification, or a measurement that has yet to be performed.

When an attribute is used at a quotient level, the study should examine representatives within each proposed class. A value that changes among those representatives fails the descent criterion. The response can involve refining the quotient, changing the attribute question, or retaining the attribute at a more detailed level. The choice should reflect the inquiry's purpose.

Numerical attributes require particular care. A score can provide an ordered comparison without supporting arithmetic differences or averages. If a conclusion changes under a recoding permitted by the stated scale, the conclusion relies on more structure than the measurement convention supplies. A proposed aggregation should therefore state why the relevant operations on values have a substantive interpretation.

A coding rule can also have consequences when it is used by participants or an institution. The study should distinguish the classification recorded by the investigator from the classification that enters an actual decision rule. The feedback construction in Section 16 describes one way to represent uptake after its presence and form are established.

19.6 Evidence for State Sufficiency

A state description supports a declared future task when it retains the distinctions needed for the selected observations under the relevant operations. Empirical assessment therefore requires an operation repertoire, an observation scheme, and a comparison horizon.

The study should seek pairs of cases or modeled configurations with the same current record and different responses to a specified continuation. Such a pair can reveal hidden memory or an omitted condition. The interpretation depends on the evidence: the discrepancy could also reflect recording error, changed operation meaning, or an incorrect deterministic assumption.

Research Obligation 19.3 (Observation and operation design). Specify which social question fixes the observation map, the operation repertoire, and the comparison horizon. Identify distinctions that these choices omit and test whether the omissions affect the intended use.

The continuation quotient gives an exact mathematical response once these inputs are fixed. A finite empirical sample typically supports a more restricted assessment. Successful agreement on observed continuations should be reported at that scope. The extension to unobserved continuations remains an assumption or a further test.

19.7 Evidence for Process Organization

A process record links occurrences through an organizing criterion. The study should state whether the process is bounded by a common objective, a procedural sequence, a continuing institutional activity, or another relation. The criterion determines which occurrences belong to the process and how interruptions are interpreted.

Intermediate conditions can be decisive. A final resource balance can conceal an unavailable intermediate step. A final implementation state can conceal failed attempts, temporary suspensions, or repeated reversals. A process account therefore needs the record required by its path-sensitive questions, even when its endpoint is well established.

Segmentation should also be justified. Grouping several operations into one phase can support explanation at a useful scale. A phase boundary can mark a change of authority, a resource threshold, or a shift in the relevant participants. Alternative segmentations should be compared when they alter the explanation of the process.

Duration introduces an additional evidence requirement. A word supplies an order of labels; durations require temporal data and a rule relating those data to the process. The idempotence counterexample shows why a composite effect alone generally cannot recover elapsed time or occurrence count.

19.8 Diagnosis of Discrepancies and Revision Order

A discrepancy between a model and a record should initiate diagnosis at the narrowest relevant level. The first task is to check that the observed record and the formal prediction refer to the same subject, time, operation meaning, and question. A mismatch of those references can produce an apparent contradiction.

The next task is to examine the representation's assumptions. An attribute may fail to descend, an identity criterion may collapse distinct participants, or a state may omit a formation variable. The mathematical counterexamples in the paper identify possible failure forms. Case evidence is needed to determine which form, if any, explains the discrepancy.

Revision should preserve a traceable account of what changes. Refining a state can alter its continuation classes and the records that depend on those classes. Revising entity identity can alter relation incidence and attributed properties. A coordinated model therefore needs to revisit dependent records after a material change.

The study should also record successful preservation results. A reduced description that retains the intended readouts under tested operations can justify a useful compression at that scope. Demonstrating such success makes the eventual limitations more informative: the inquiry can identify which new question or operation first requires additional structure.

19.9 Extension and Abandonment Conditions

The proposed architecture warrants further development when its distinctions organize evidence, reveal otherwise hidden assumptions, or produce comparisons that can discriminate among candidate models. A new enrichment should name the failure it repairs and the evidence supporting its added inputs.

Abandonment or substantial revision is appropriate when the proposed categories repeatedly obscure the case's central distinctions, when the required identity criteria cannot be defended for the focal question, or when the operation model suppresses the very changes the inquiry aims to explain. These are substantive limitations of an application, even when the conditional algebra remains correct.

The research program can therefore advance through both successful applications and informative failures. Its immediate task is to connect precise constructions to bounded questions while retaining uncertainty about the evidence, the interpretation, and the eventual theoretical reach of the framework.

20 Conclusion

This paper has developed a coordinated representation of six categories of social facts from an abstract action framework and explicitly declared enrichments. The construction assigns entity reidentification, historically formed dependence, event occurrence, attribute description, state comparison, and process history different tasks within a common inquiry. The conclusion draws together the formal results, their philosophical interpretation, and the research obligations that follow.

The central result is an architecture of representational commitments. A common dynamics can support several category-specific records, yet the data required by those records differ. A carrier and an action supply configurations and transformations. A preserved equivalence supports quotient dynamics. Occurrence indices, historical records, attribute interpretations, and identity criteria supply further information. Keeping their roles explicit permits an account to be evaluated at the level where its substantive commitments enter.

The entity construction begins from indexed appearances and a declared reidentification relation. Its quotient groups appearances according to that criterion. Forward reachability alone leaves the criterion underdetermined, and directed lineage retains information that an identity equivalence can obscure. The relational-unit example shows how a membership criterion can generate collective units from configurations while also showing how a change of operation repertoire can defeat a compressed description of those units.

The relation construction distinguishes participant incidence, historical formation, and prospective comparison. Incidence descends through a participant quotient only when it is invariant under the rel-

evant representative choices. Formation questions depend on the historical information retained by the record. Potential effects concern distinctions under declared continuations, including continuations that remain unrealized. A causal interpretation requires further comparison assumptions and evidence.

The event construction preserves occurrence identity separately from transition effects. Repeated idempotent operations can produce the same composite transformation while contributing different numbers of occurrences. Attempts, completions, and quotient-silent events therefore require records suited to the question. The event category supplies a location for those distinctions within the common dynamics.

The attribute construction specifies subjects, value domains, interpretations, and definedness. Its quotient criterion makes clear when a value is well-defined at a reduced descriptive level. Numerical interpretation adds measurement commitments beyond the existence of a value map. Classification can also enter subsequent dynamics through a declared response rule, creating a further distinction between description and social uptake.

The state construction asks which information supports selected current and future readouts. Equality of continuation observations gives a dynamically preserved equivalence under the declared operations. This established behavioral construction supplies a precise sufficiency criterion. Historical questions and entity identity can require distinctions that predictive sufficiency omits.

The process construction retains ordered occurrences and their intermediate states. Its reductions to products, transformations, and endpoints preserve different information. Segmentation, duration, interruption, and prefix feasibility require additional records when they form part of the inquiry. The usefulness of a composite effect is assessed together with the process questions it leaves unresolved.

The formal developments also repair several tempting shortcuts. The exact composition of collapse sets involves a preimage of the later collapse, which prevents an unjustified additive count. Positive-future equivalence differs from a comparison that includes the present observation. A family of internally consistent descriptive regimes can fail to preserve distinctions across a regime switch. A probabilistic quotient requires equal transition masses into each block, extending the deterministic preservation condition through an established finite-state result.

The cooperative-agreement illustration connects the categories in a complete finite action. Its twelve nonempty-word transformations are derived through their normal forms. The example distinguishes order effects, jointly enabling operations, historical alternatives, and several choices of observation. The graph illustration adds a different test: a component-level description supports edge additions while edge deletions can expose distinctions it omits. Together the examples show how a representation's adequacy depends on the operations and questions attached to it.

The philosophical position remains one of ontological suspension. The six categories are developed through the analytical work they perform and the commitments required to sustain that work. This position permits substantive criticism of a model's identity criteria, historical omissions, classification choices, and explanatory reach. It keeps the broader ontological question available for separate investigation.

A bounded application is the next evidential task. Such an application should declare its focal arrangement, category records, operation meanings, observations, and comparison questions. It should examine competing representations and preserve the discrepancies that force revision. The conditional results provide a framework for that assessment; the evidence determines which proposed interpretations survive.

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